# CSE 4303
# Data Structures

## Topic: Deletion on Binary Search Trees

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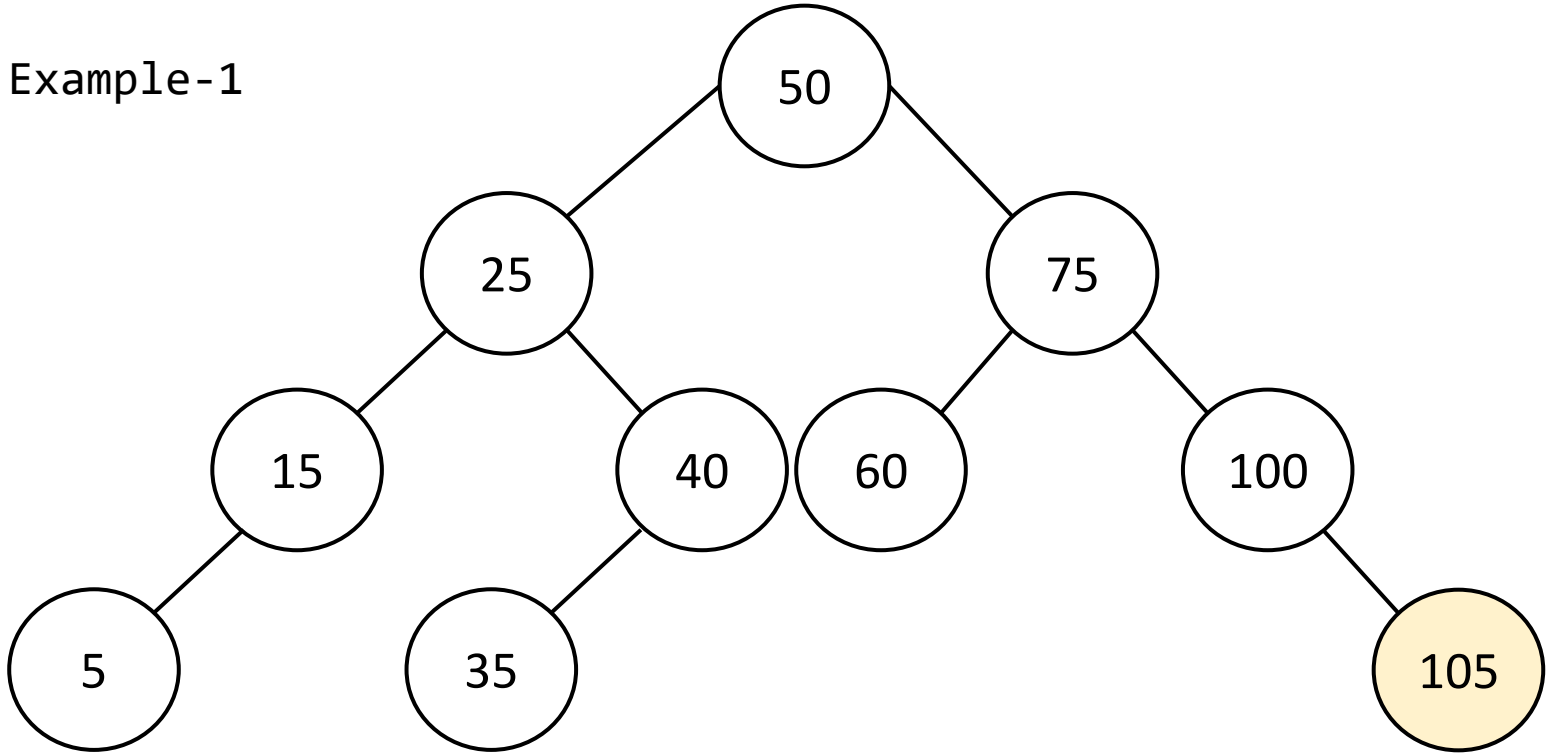
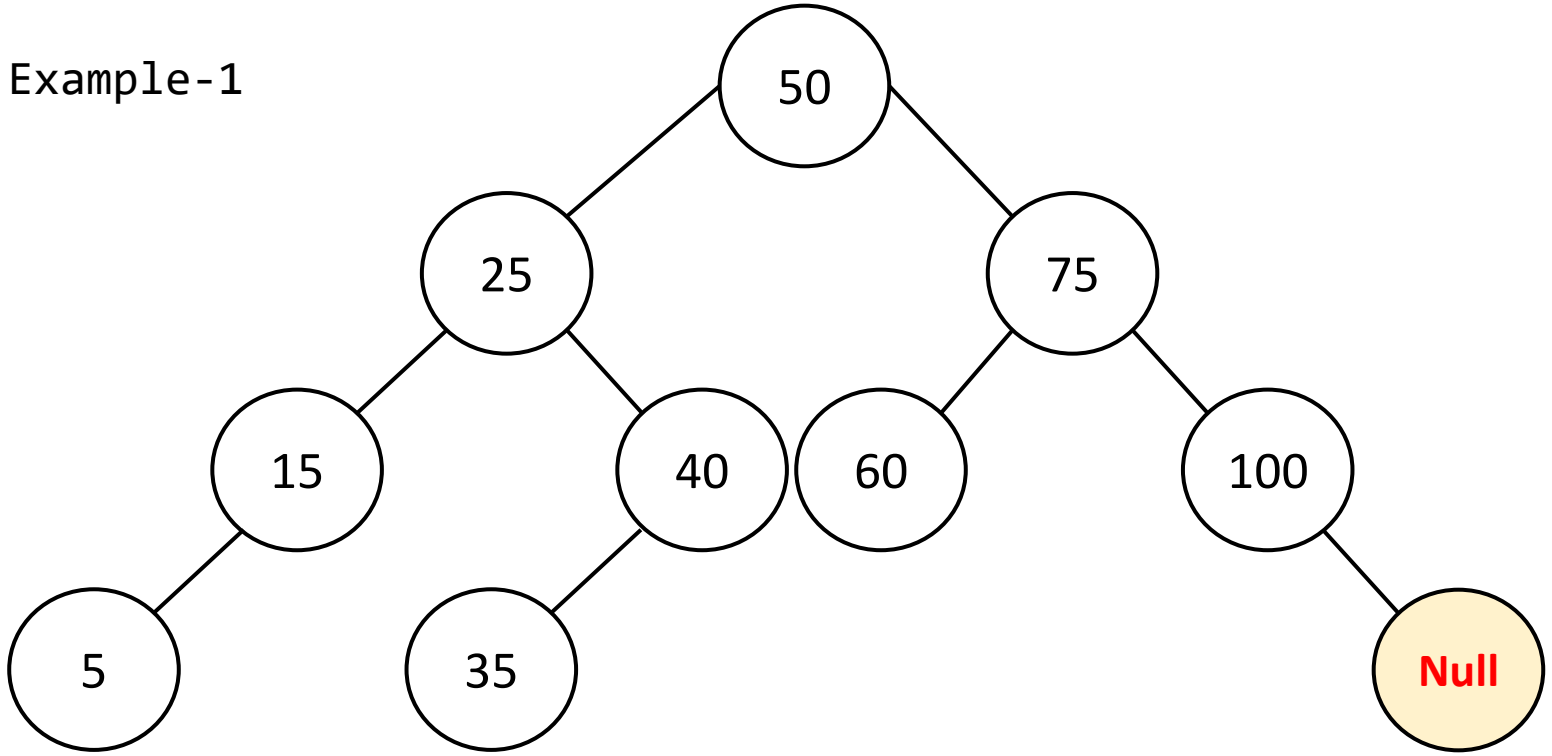
Example-1




## Delete Leaf

Delete(105)

Example-1

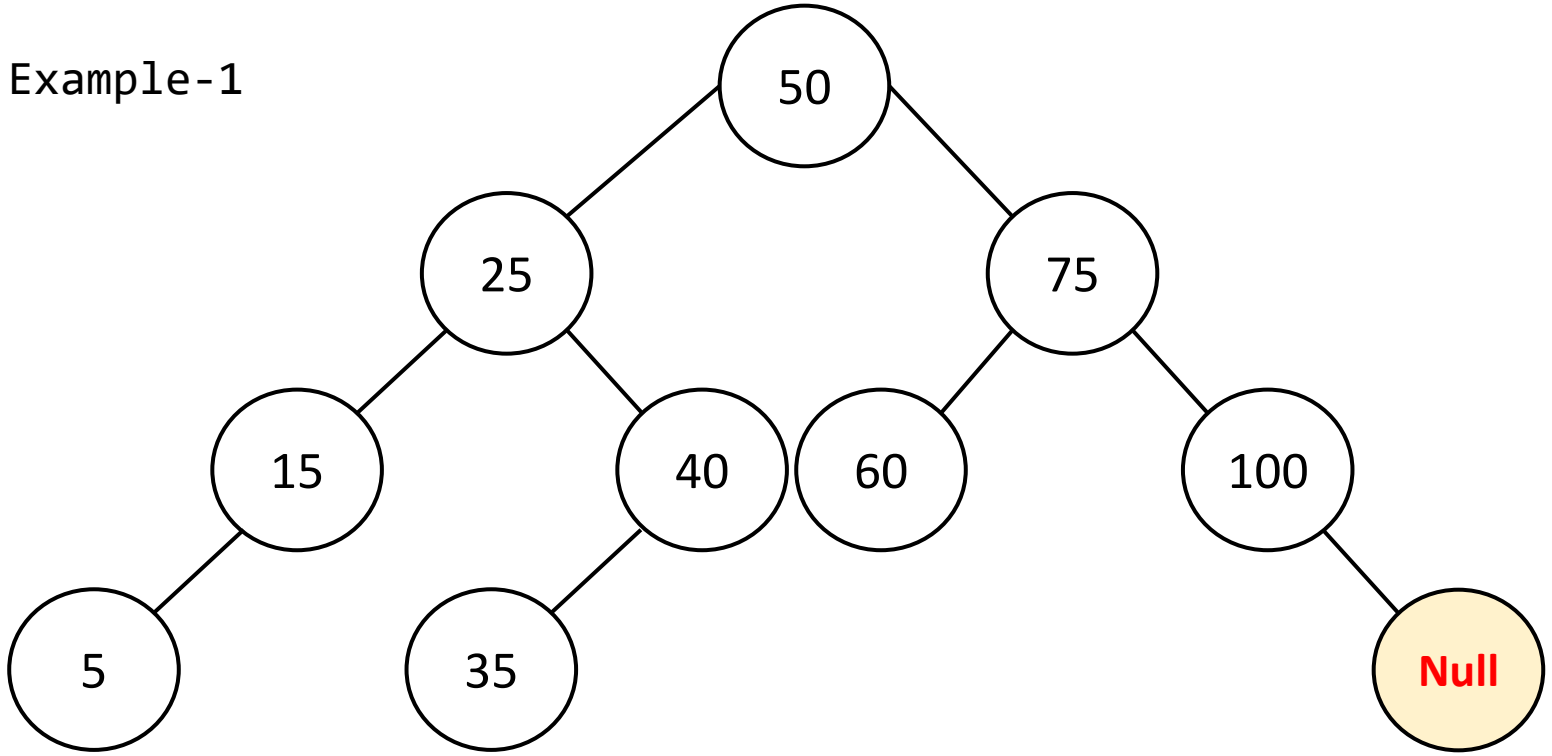



## Delete Leaf

Delete(105)

- 105 has No child (leaf)
- Can be replaced by Null!

Example-1



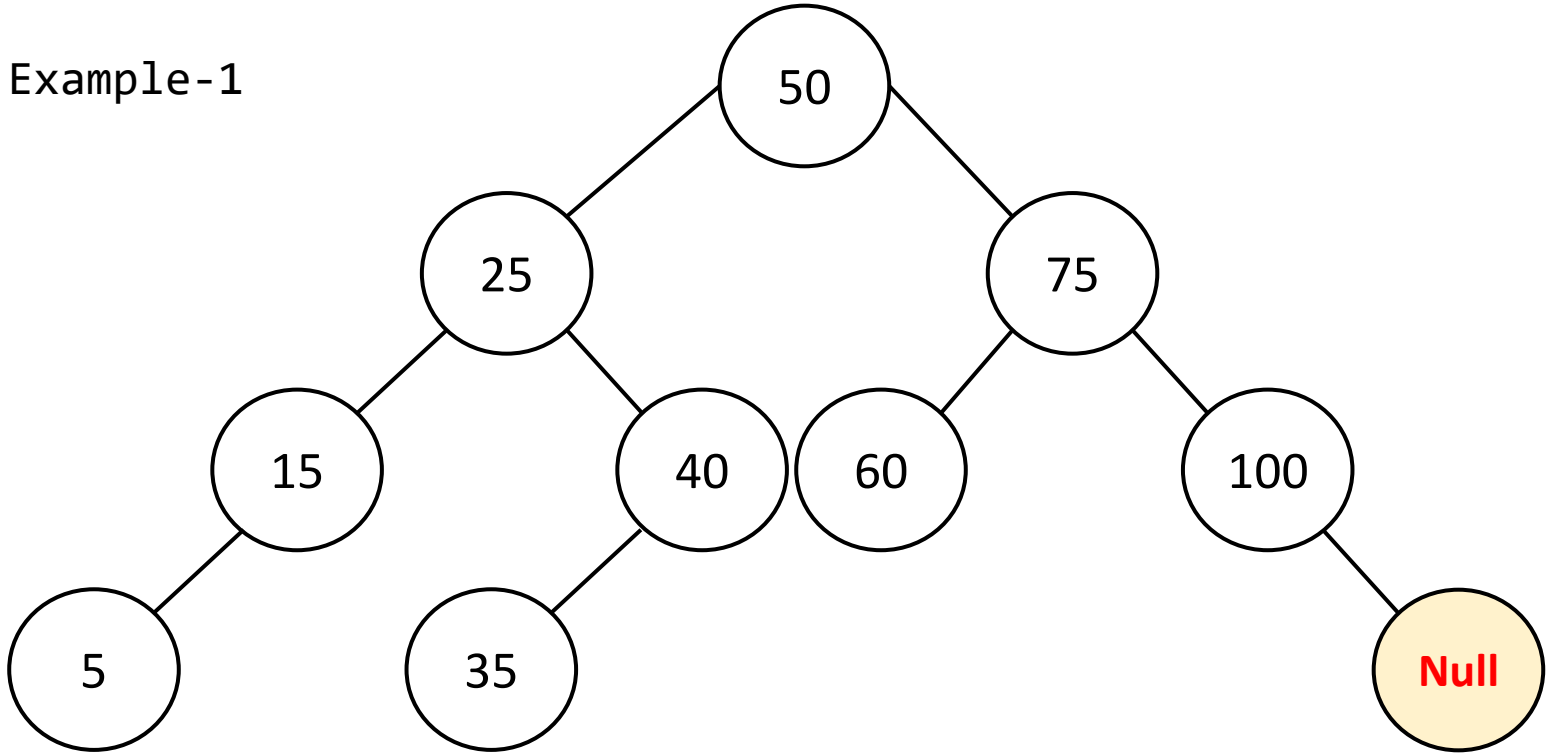

## Delete Leaf

Delete(105)

- 105 has No child (leaf)
- Can be replaced by Null!

Let, z = 105 (z is the node to be deleted)

Example-1




## Delete Leaf

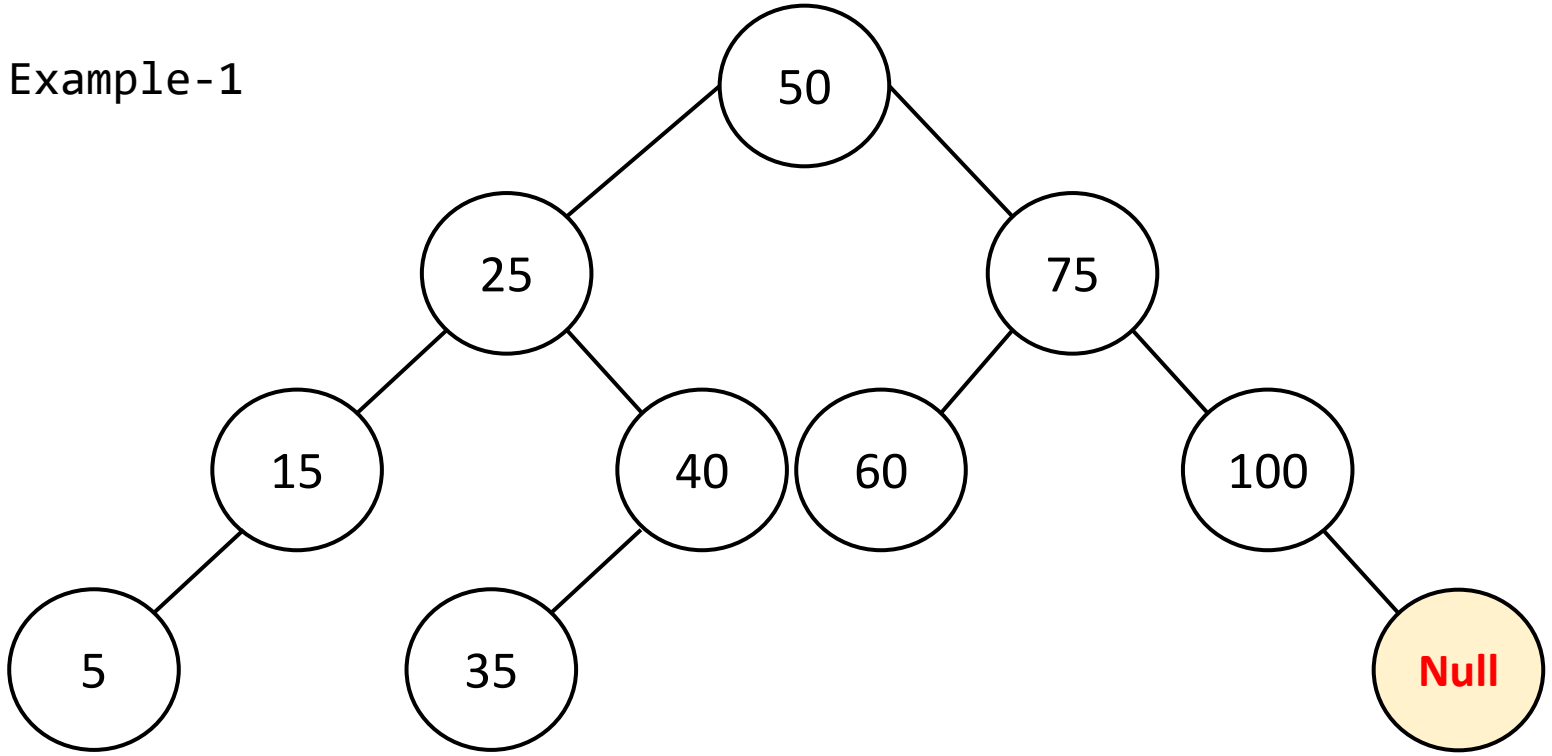
Delete(105)

- 105 has No child (leaf)
- Can be replaced by Null!

Let, z = 105 (z is the node to be deleted)

If z is the right of its parent,
z.p.right = Null

Example-1




## Delete Leaf

Delete(105)

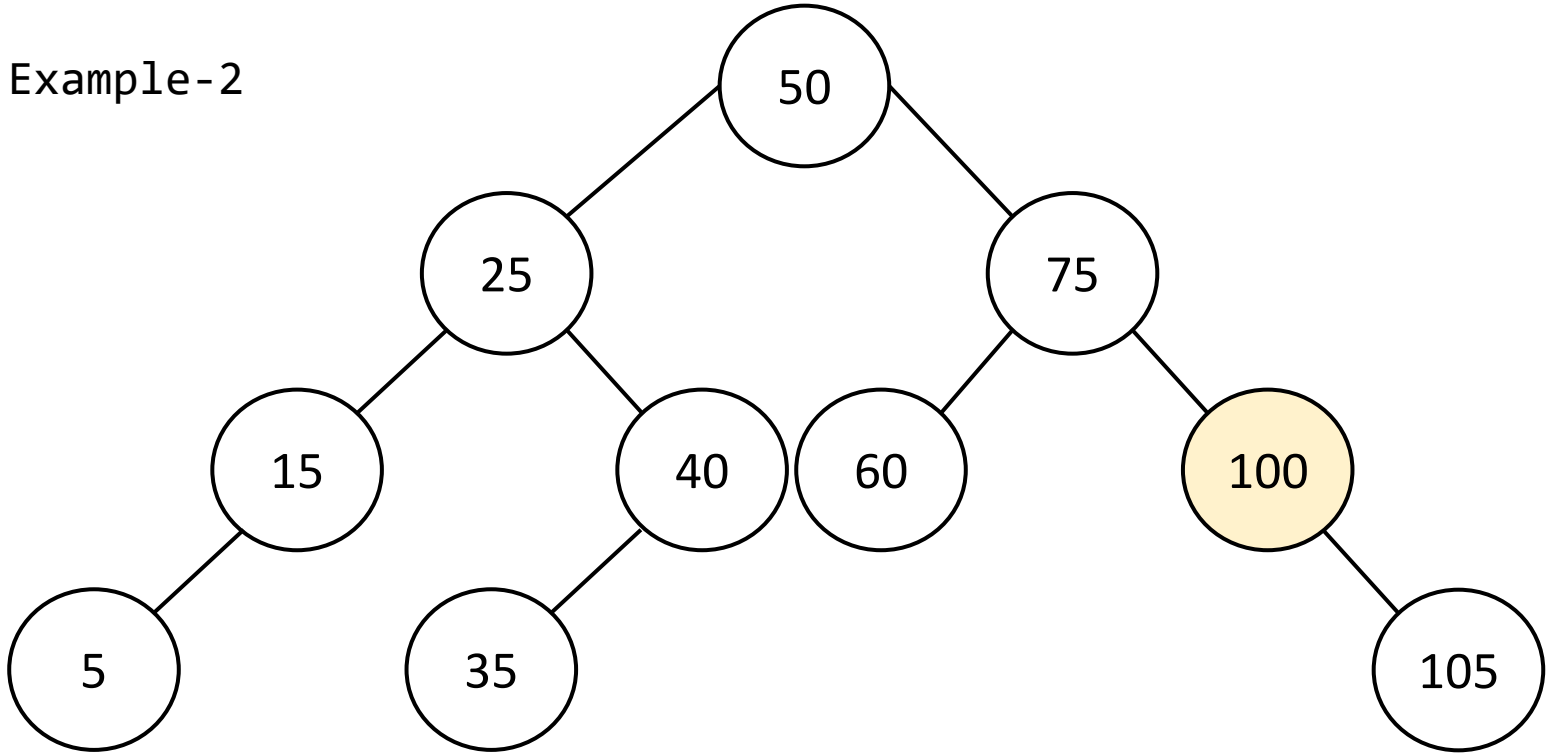
- 105 has No child (leaf)
- Can be replaced by Null!

Let, z = 105 (z is the node to be deleted)

If z is the right of its parent,
z.p.right = Null

Else z.p.left = Null

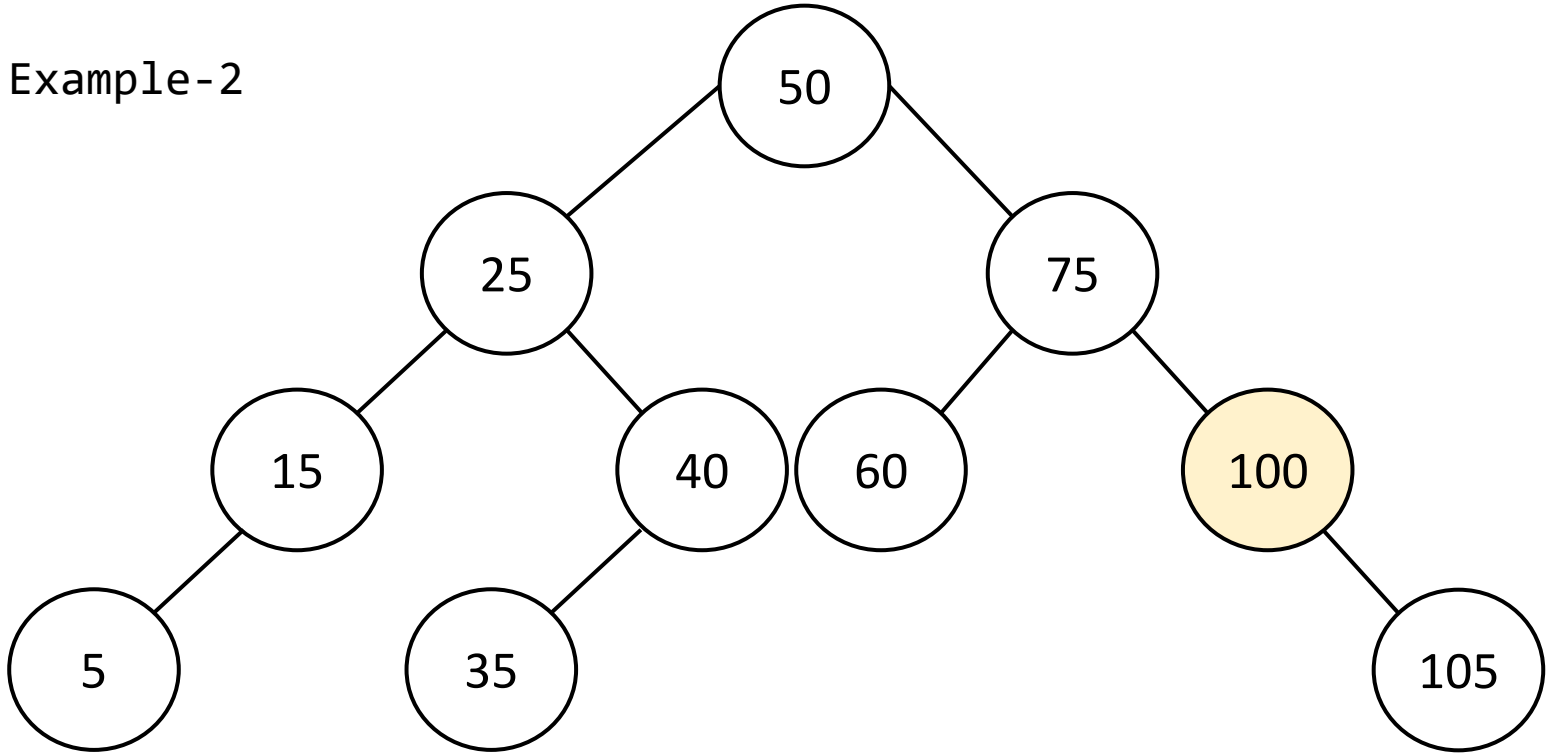
Example-2




## Delete Node with 1 child

Delete(100)

Example-2

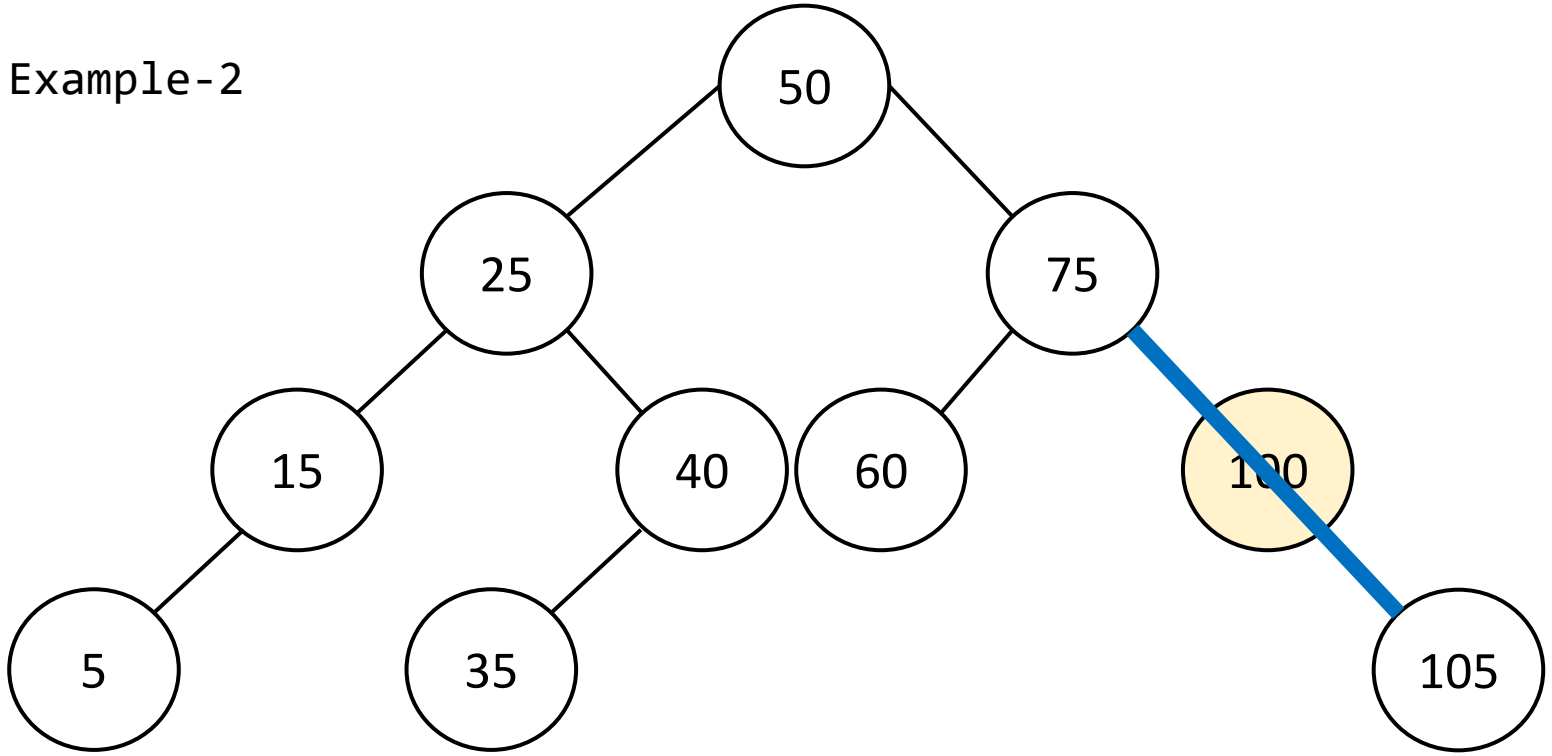



## Delete Node with 1 child

Delete(100)

- z=100, has 1 subtree
- z.right in not Null, but z.left is Null

Example-2



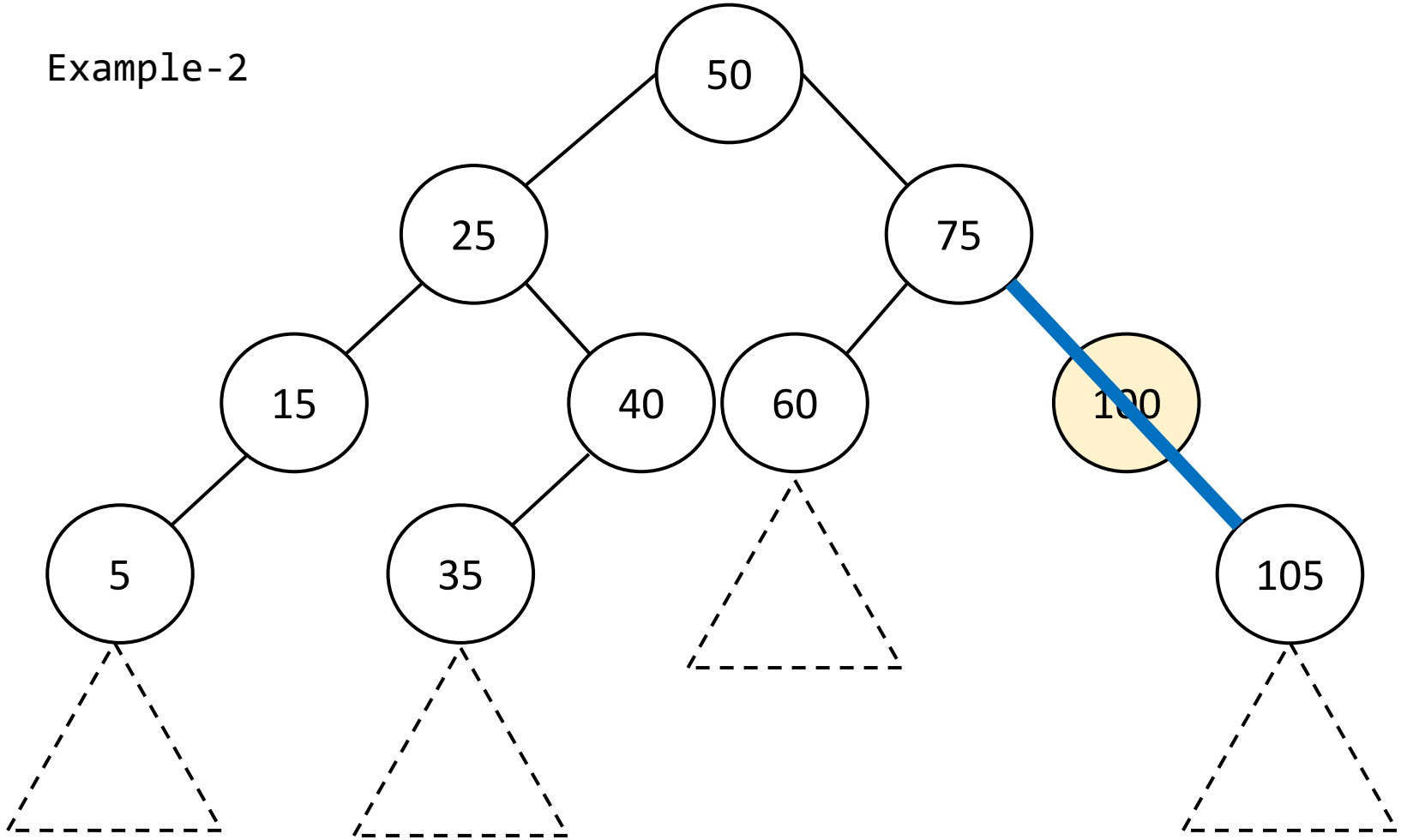

## Delete Node with 1 child

Delete(100)

- z=100 has 1 subtree
- z.right in not Null, but z.left is Null

So z.right can be connected with z.p

Example-2




## Delete Node with 1 child

Delete(100)

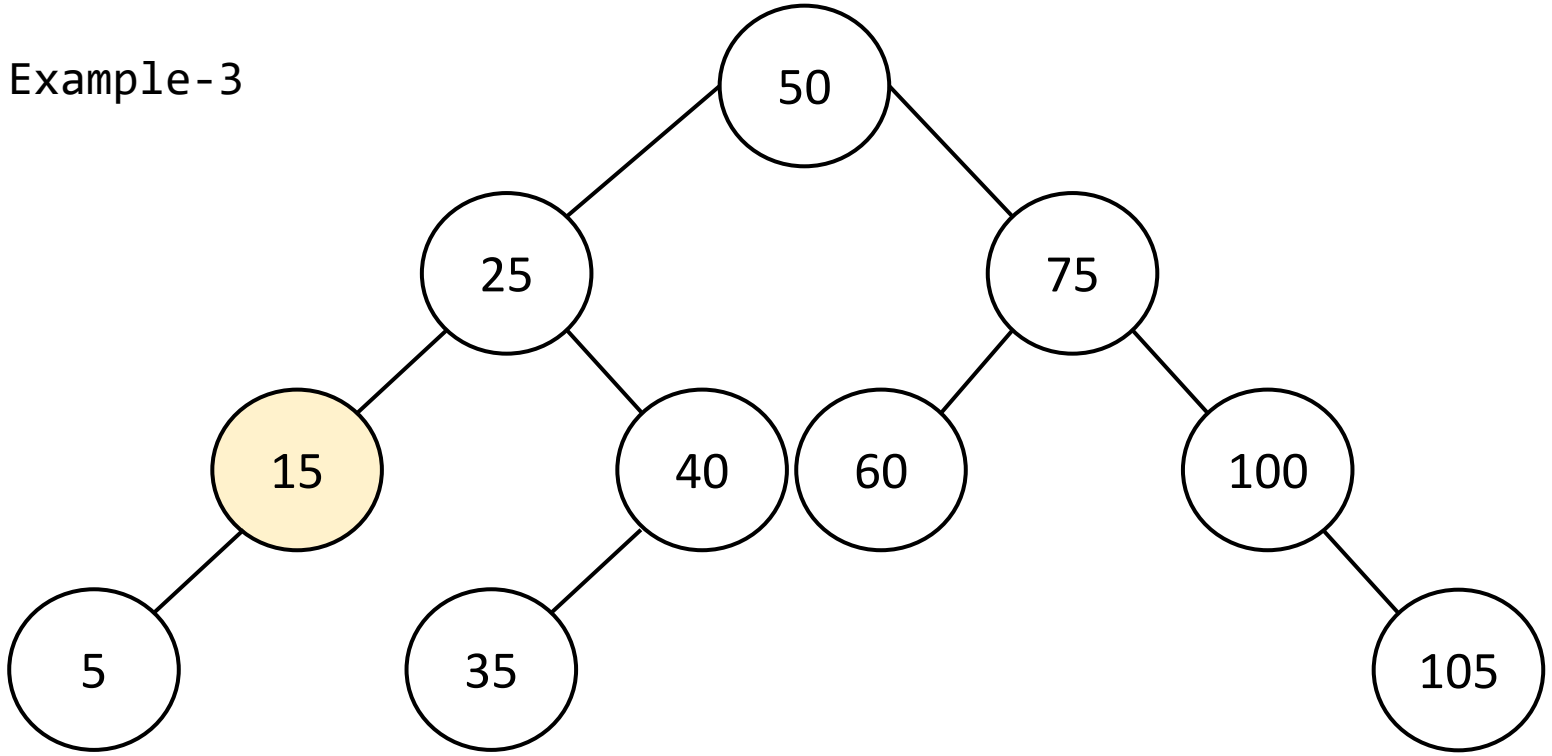
- z=100 has 1 subtree
- z.right in not Null, but z.left is Null

So z.right can be connected with z.p

*Assume, each node has a subtree under it.*

*One a node is connected, the entire subtree gets connected*

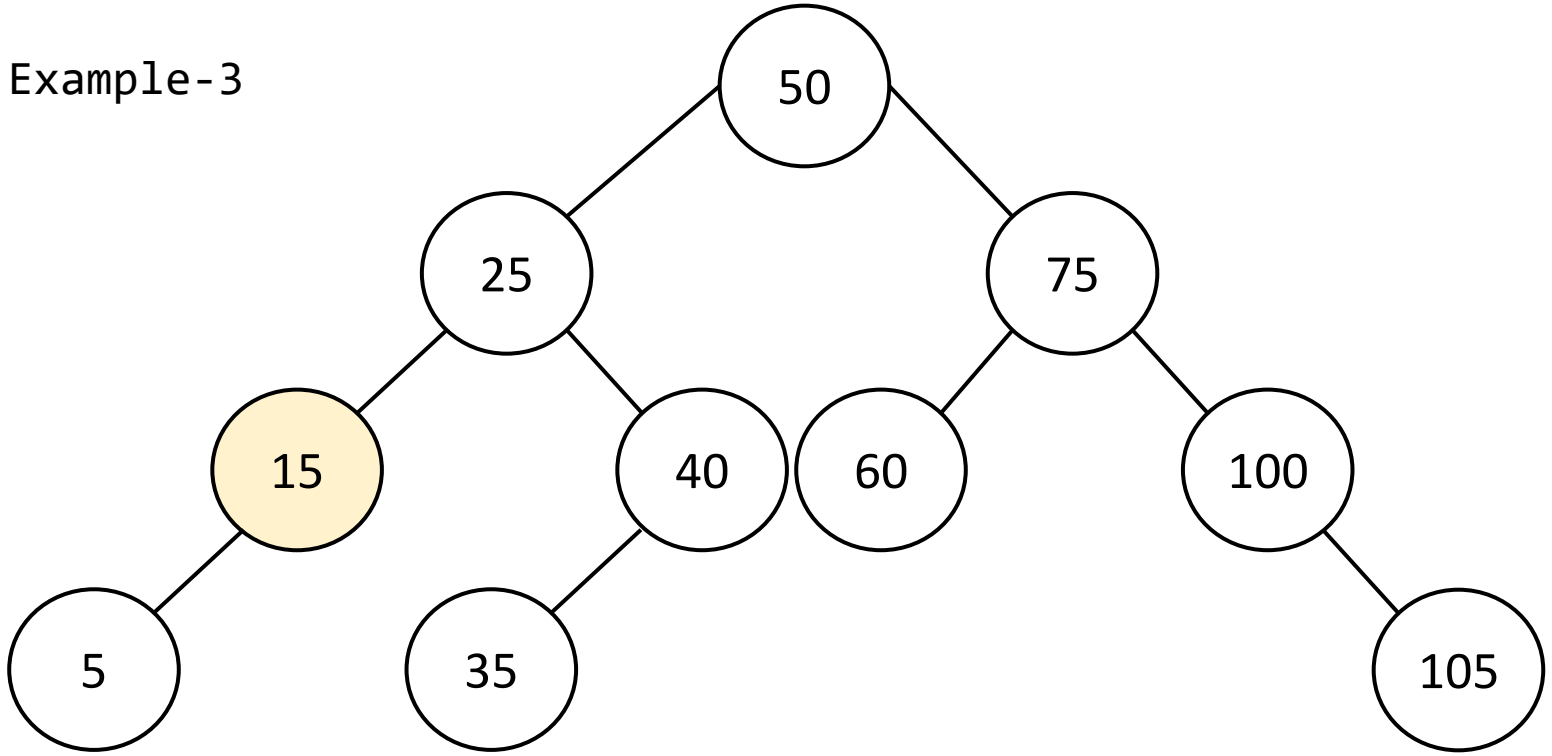
Example-3




## Delete Node with 1 child

Delete(15)

Example-3

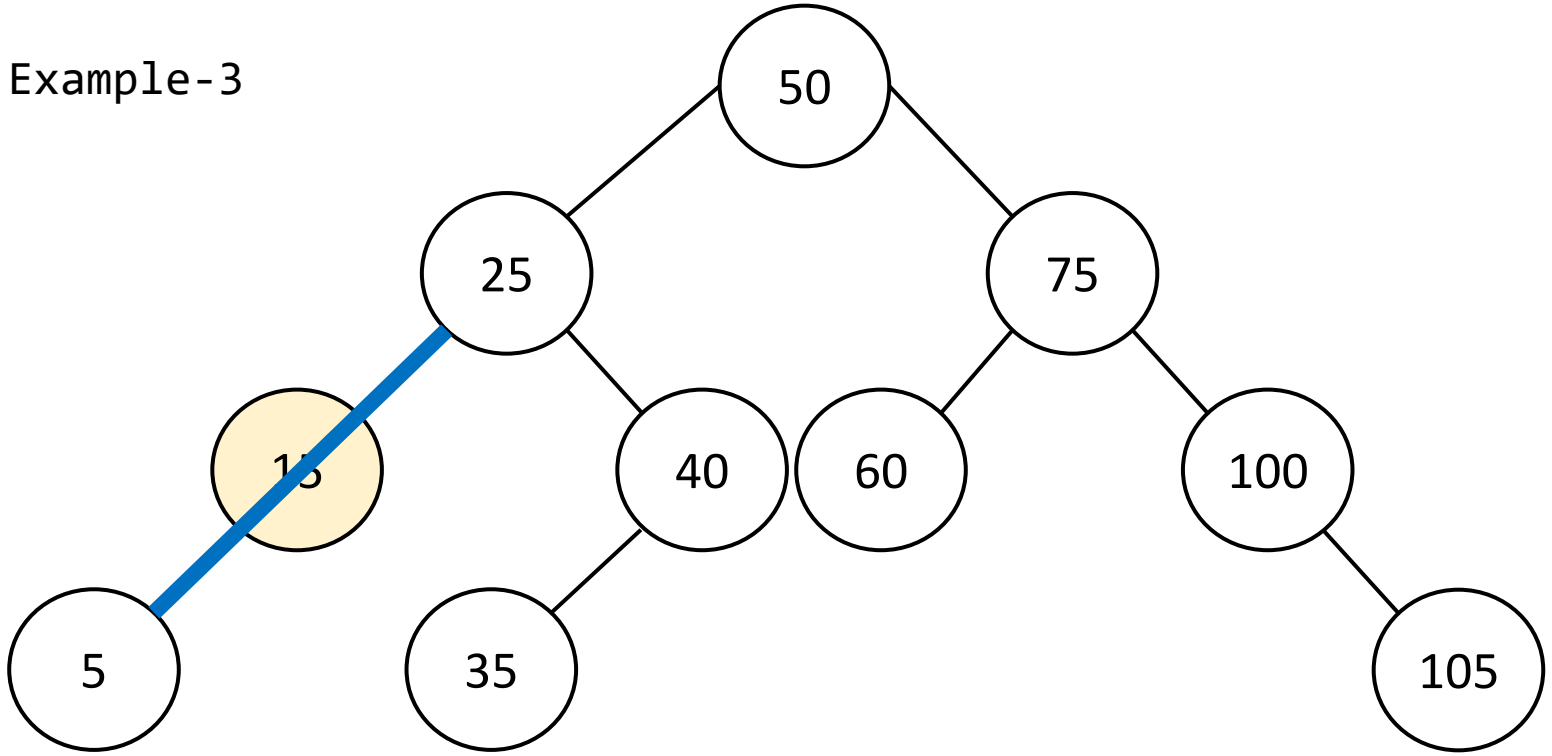



## Delete Node with 1 child

Delete(15)

- z=15 has 1 subtree
- z.left is not Null, but z.right is Null

Example-3



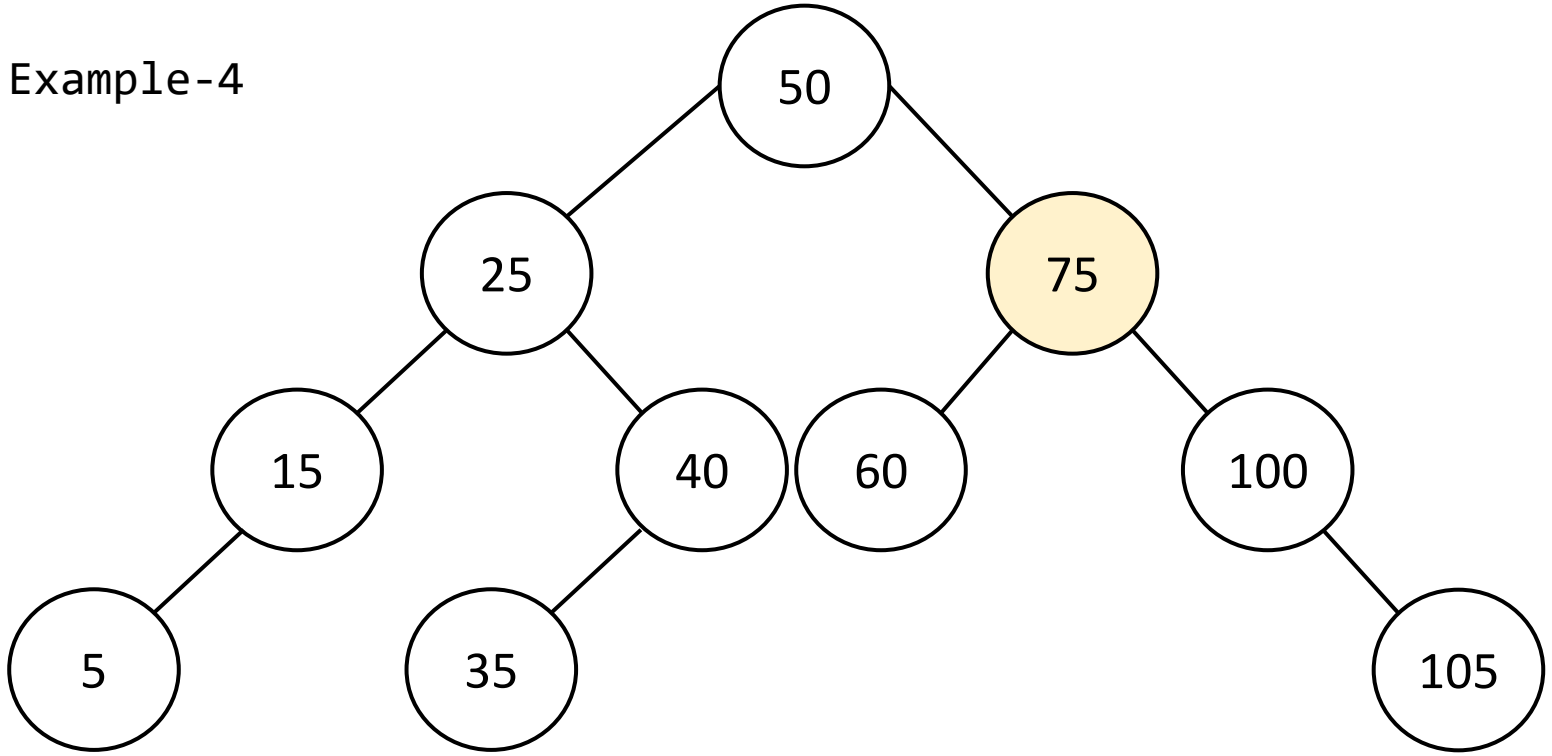

## Delete Node with 1 child

Delete(15)

- z=15 has 1 subtree
- z.left is not Null, but z.right is Null

So z.left can be connected with z.p

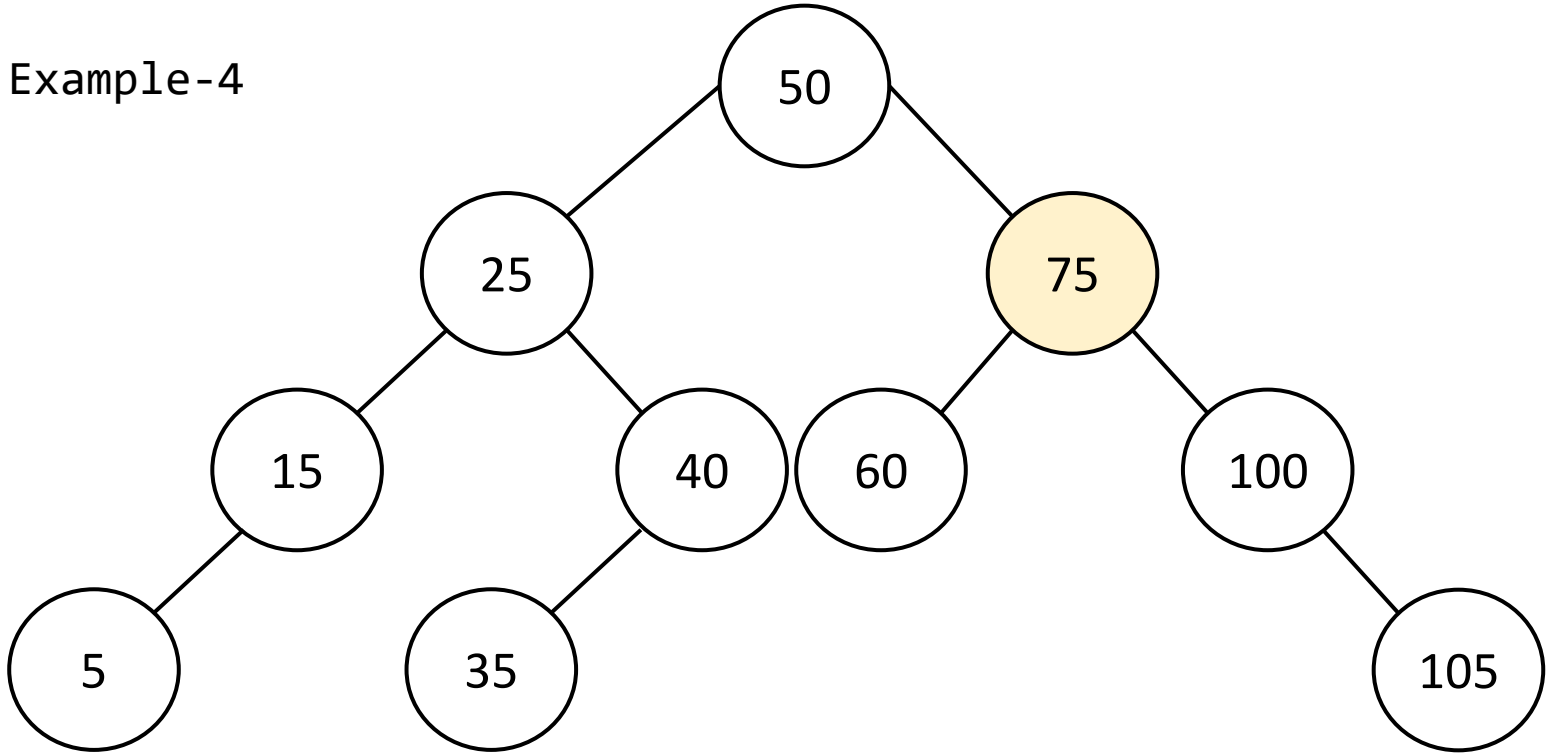
Example-4




## Delete Node with 2 child

Delete(75)

Example-4

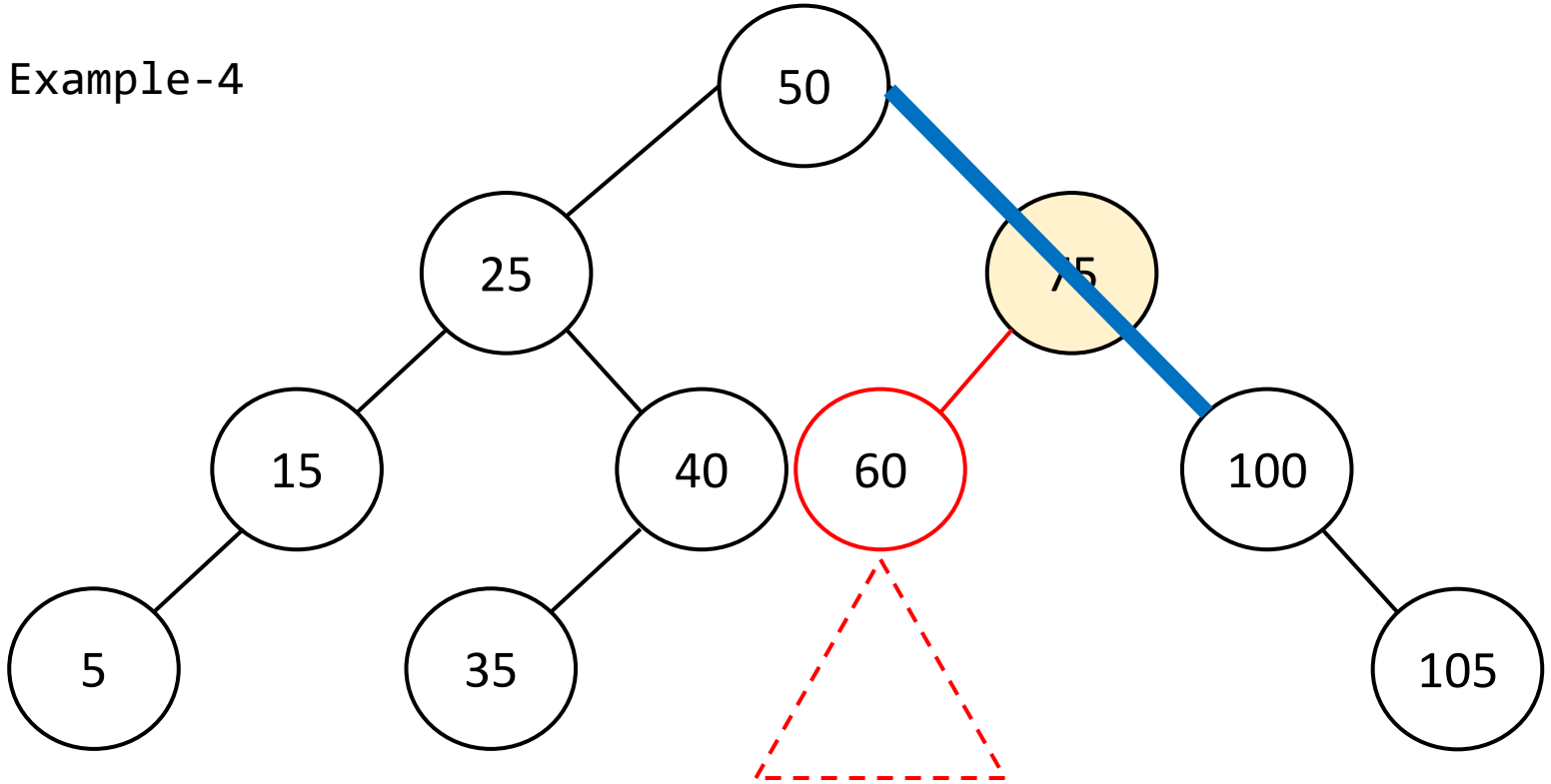



## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

Example-4



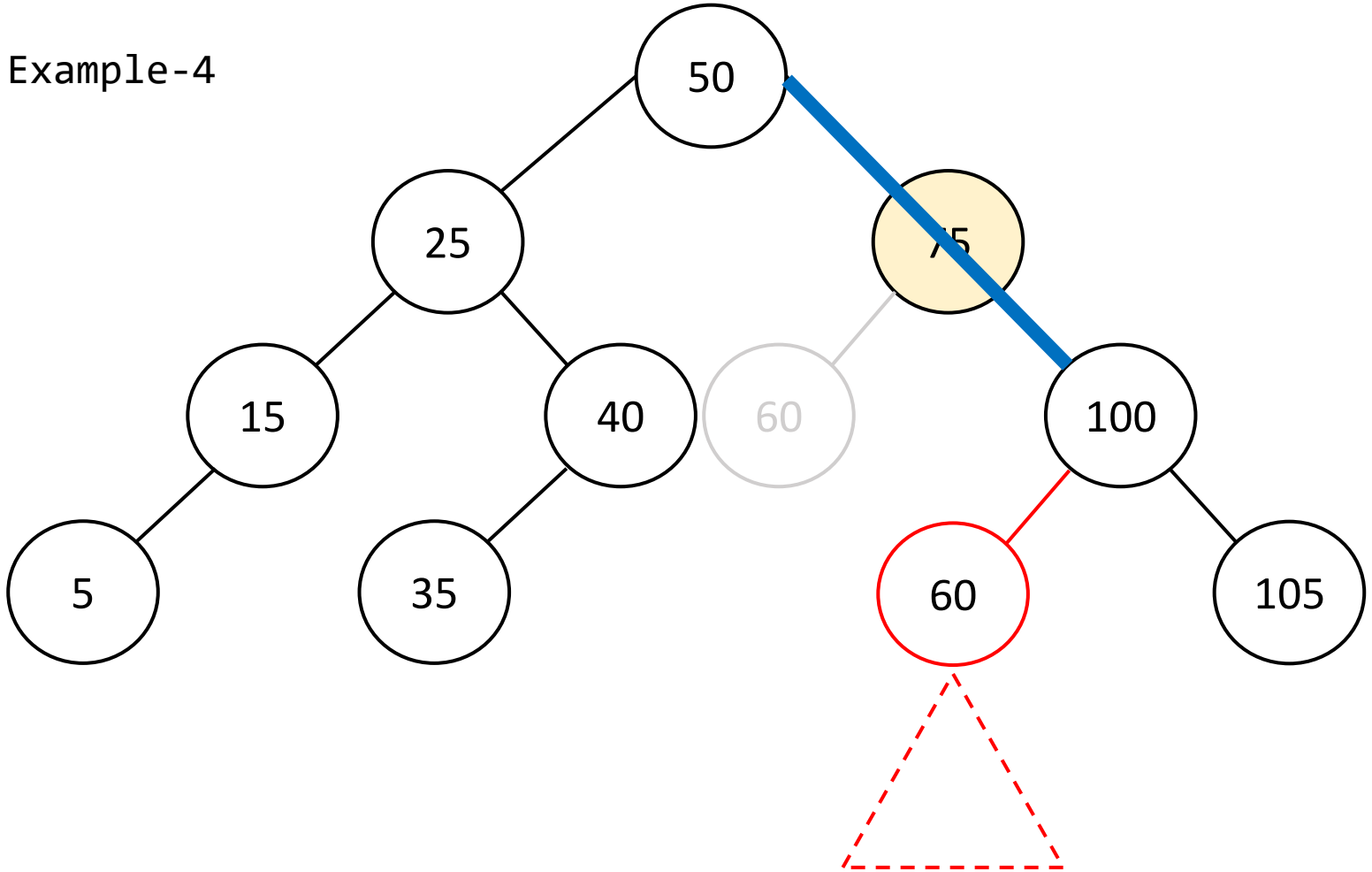

## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

- z.p can be connected with z.right
- But what about z.left subtree?

Example-4



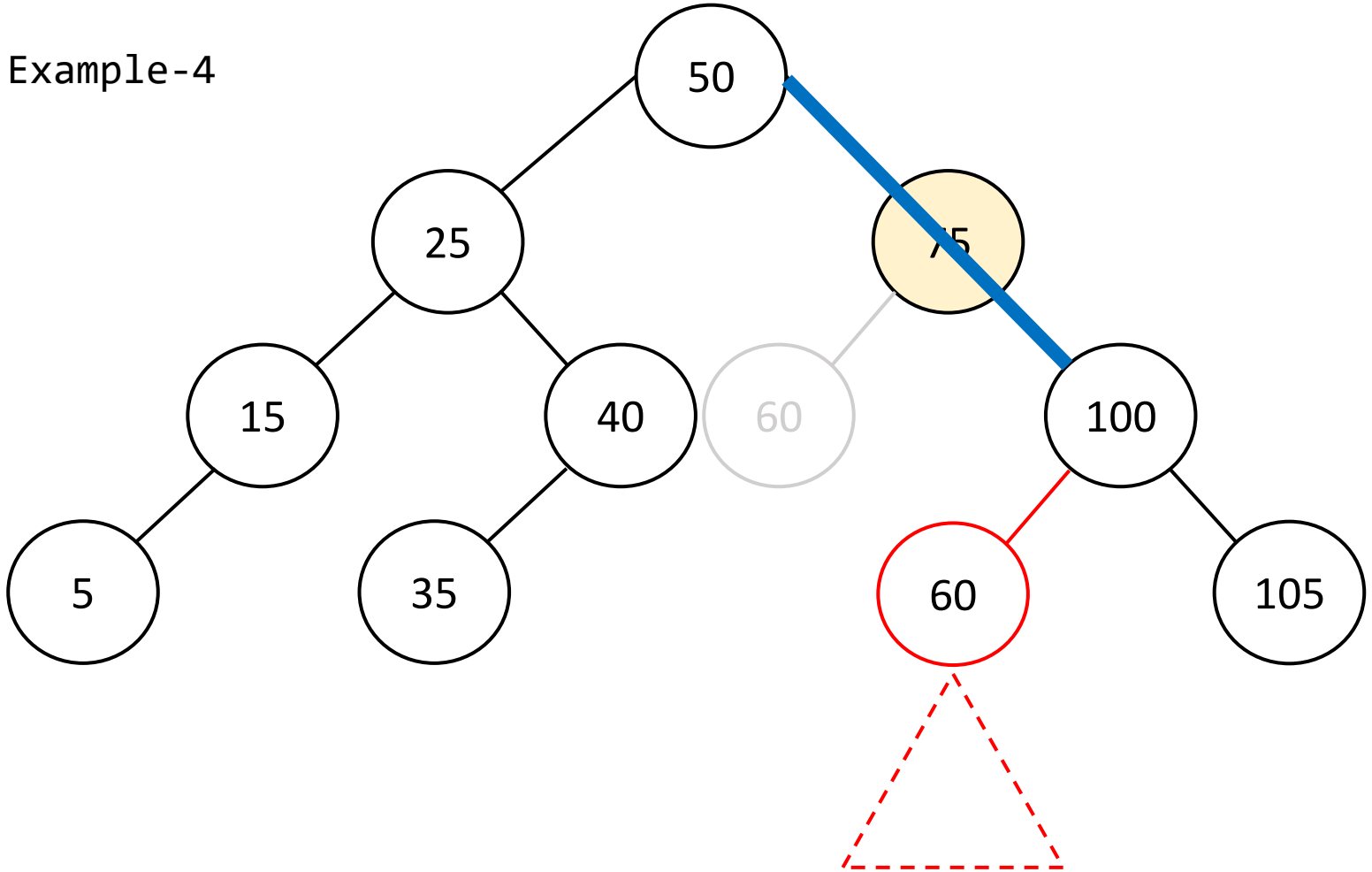

## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

- z.p can be connected with z.right
- But what about z.left subtree?
- **Seems it can be connected with 100.left**

Example-4



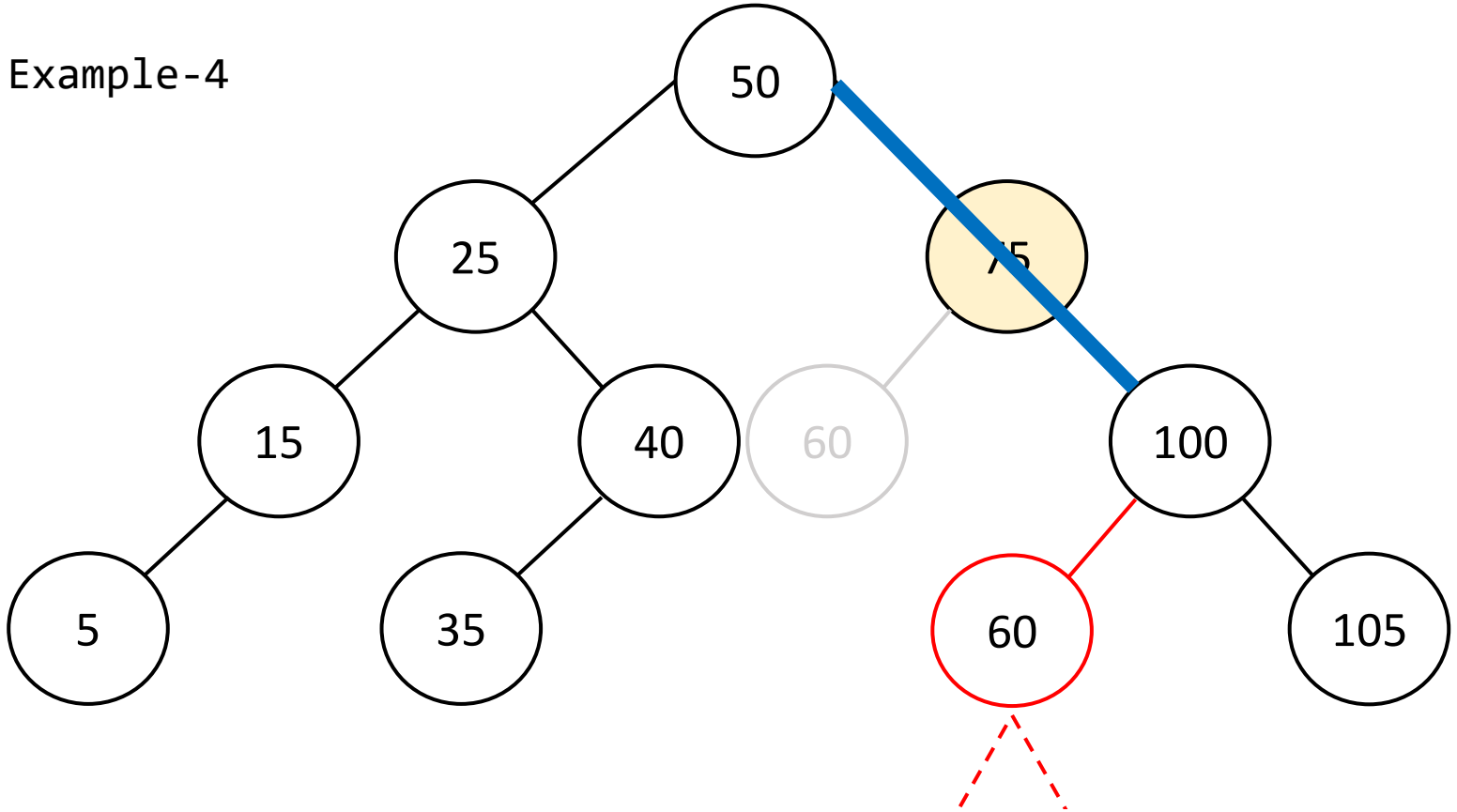

## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

- z.p can be connected with z.right
- But what about z.left subtree?
- **Seems it can be connected with 100.left-** but is it always possible?

Example-4




*Can't guarantee that 100's (z.right's) left subtree is always free so that z.left can go there!*

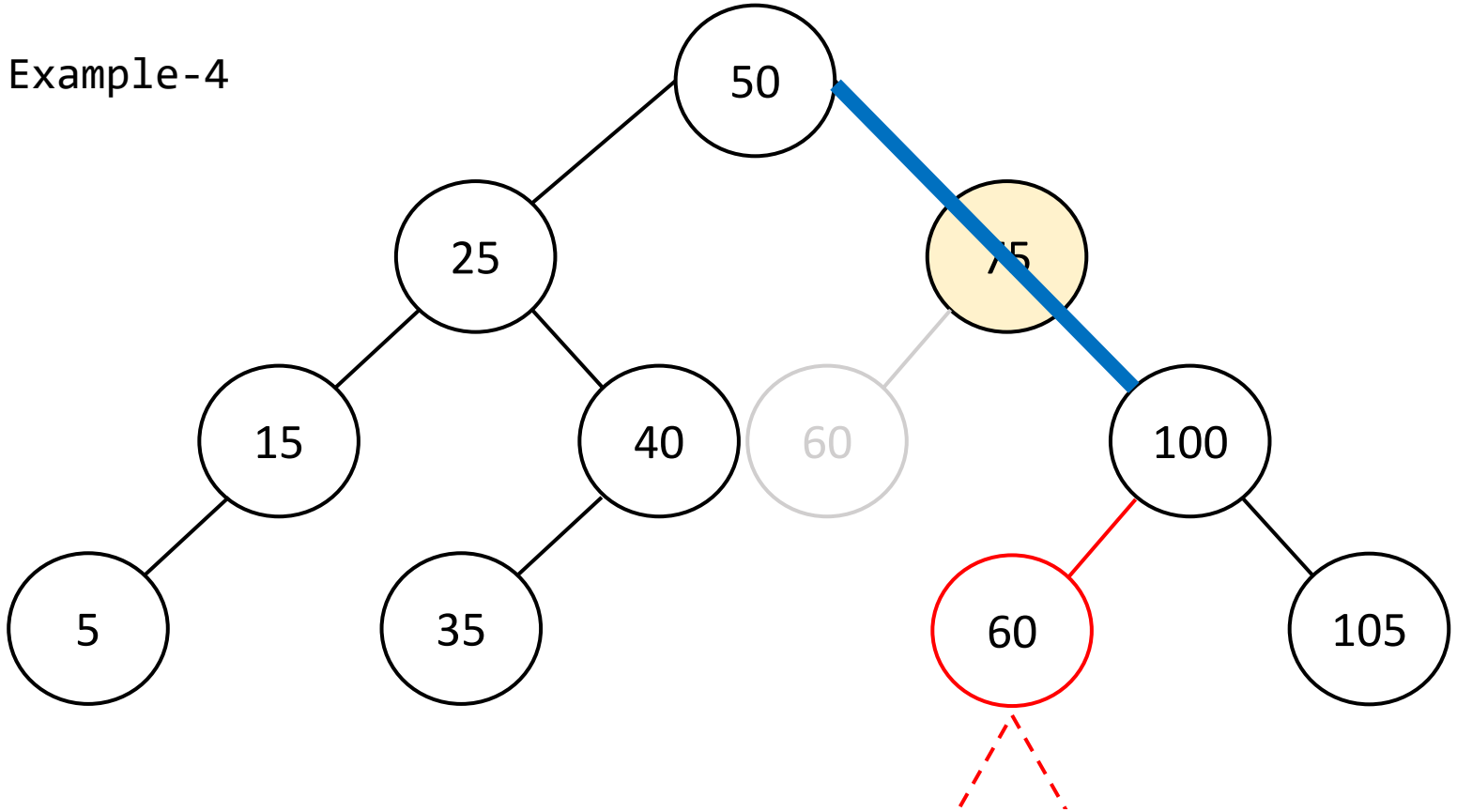
## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

- z.p can be connected with z.right
- But what about z.left subtree?
- **Seems it can be connected with 100.left-** but is it always possible?

Example-4




*Can't guarantee that 100's (z.right's) left subtree is always free so that z.left can go there!*

*Which node provides such a guarantee?*

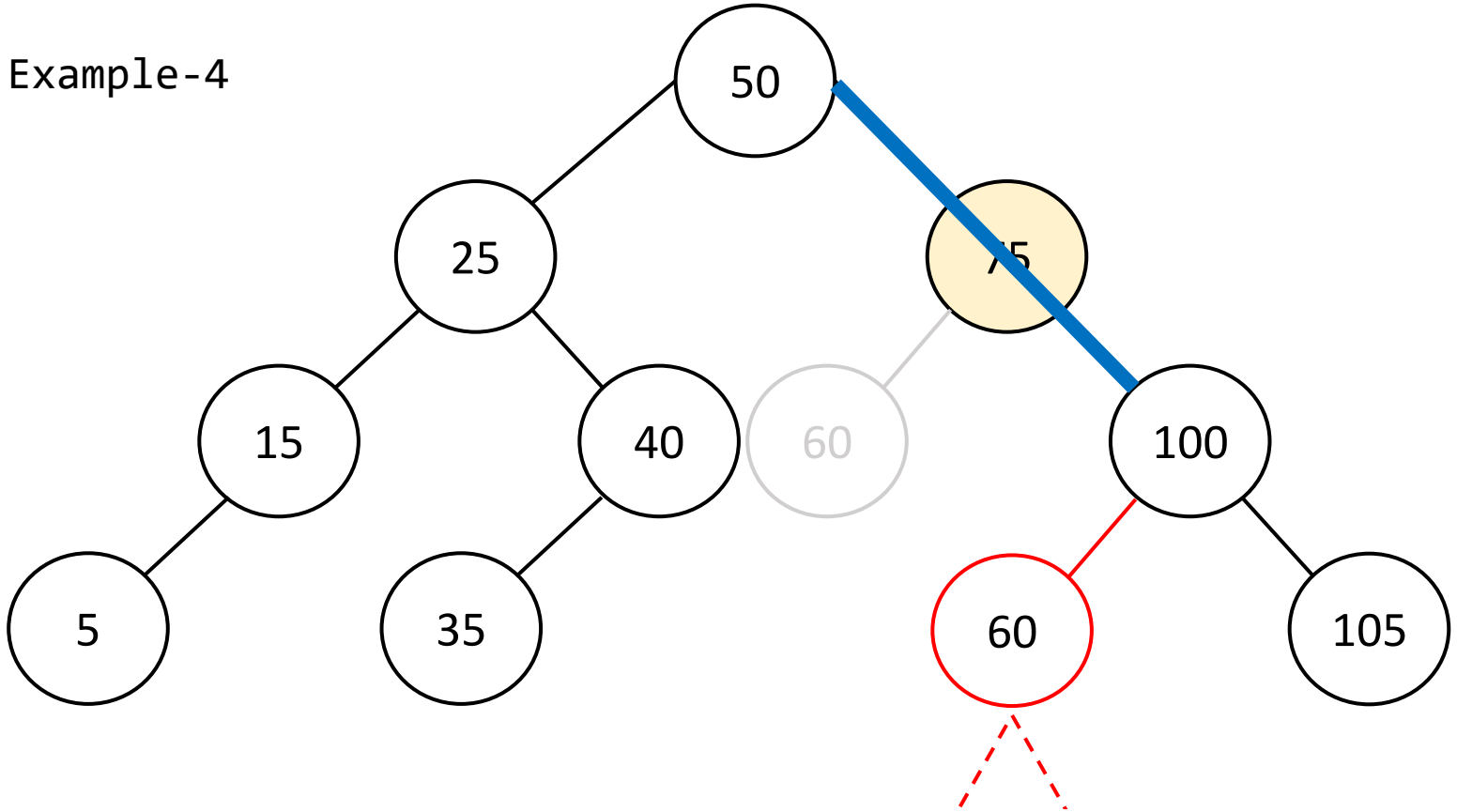
## Delete Node with 2 child

Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

- z.p can be connected with z.right
- But what about z.left subtree?
- **Seems it can be connected with 100.left-** but is it always possible?

Example-4




*Can't guarantee that 100's (z.right's) left subtree is always free so that z.left can go there!*

*Which node provides such a guarantee?*

Successor(z)

## Delete Node with 2 child

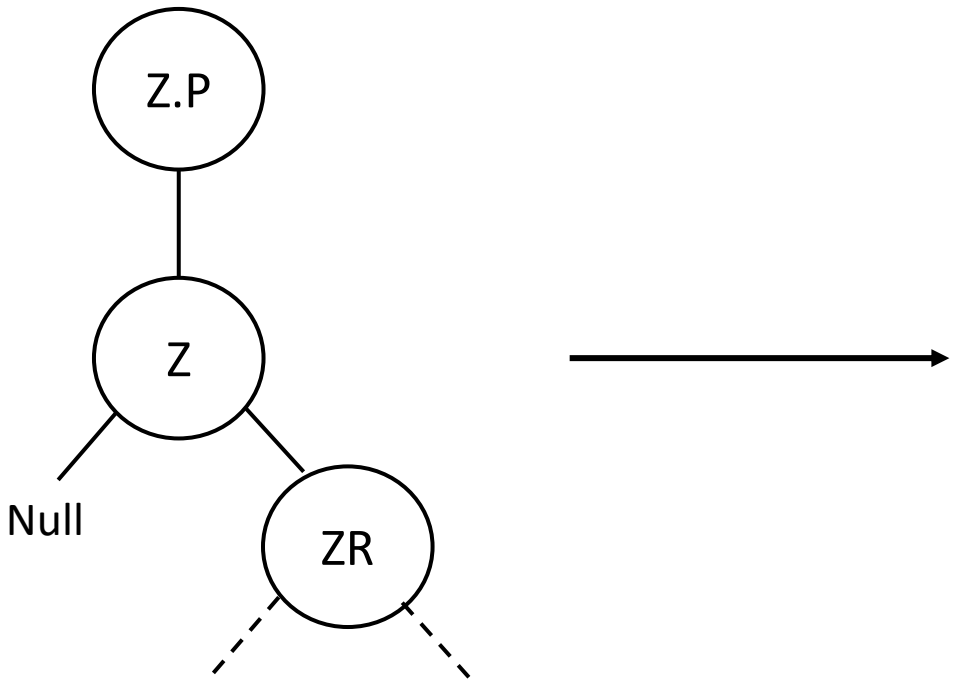
Delete(75)

- z=75 has 2 subtrees
- z.left and z.right both exist

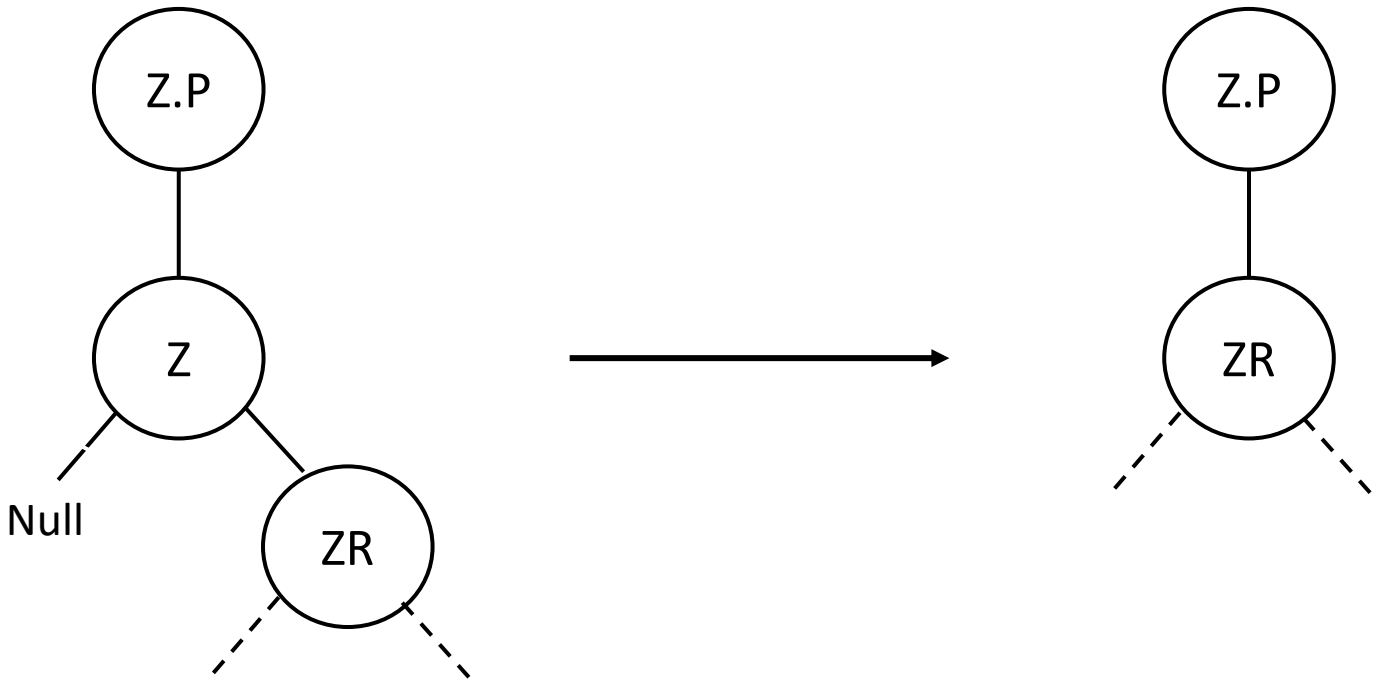
- z.p can be connected with z.right
- But what about z.left subtree?
- **Seems it can be connected with 100.left-** but is it always possible?

*Let’s summarize the different scenarios…*

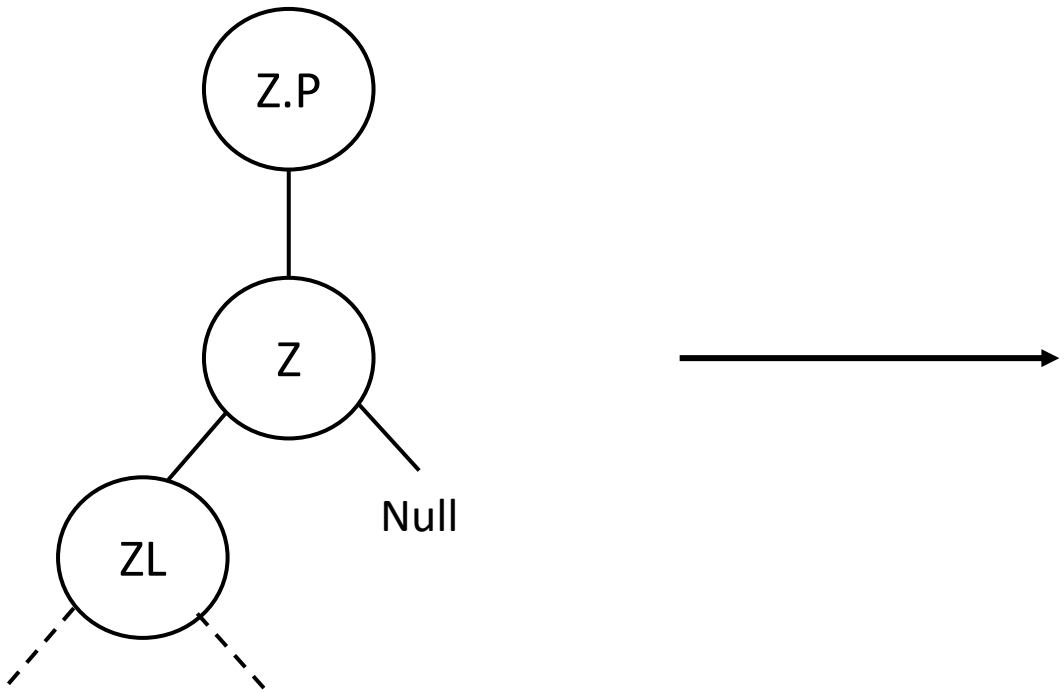
# Case-A



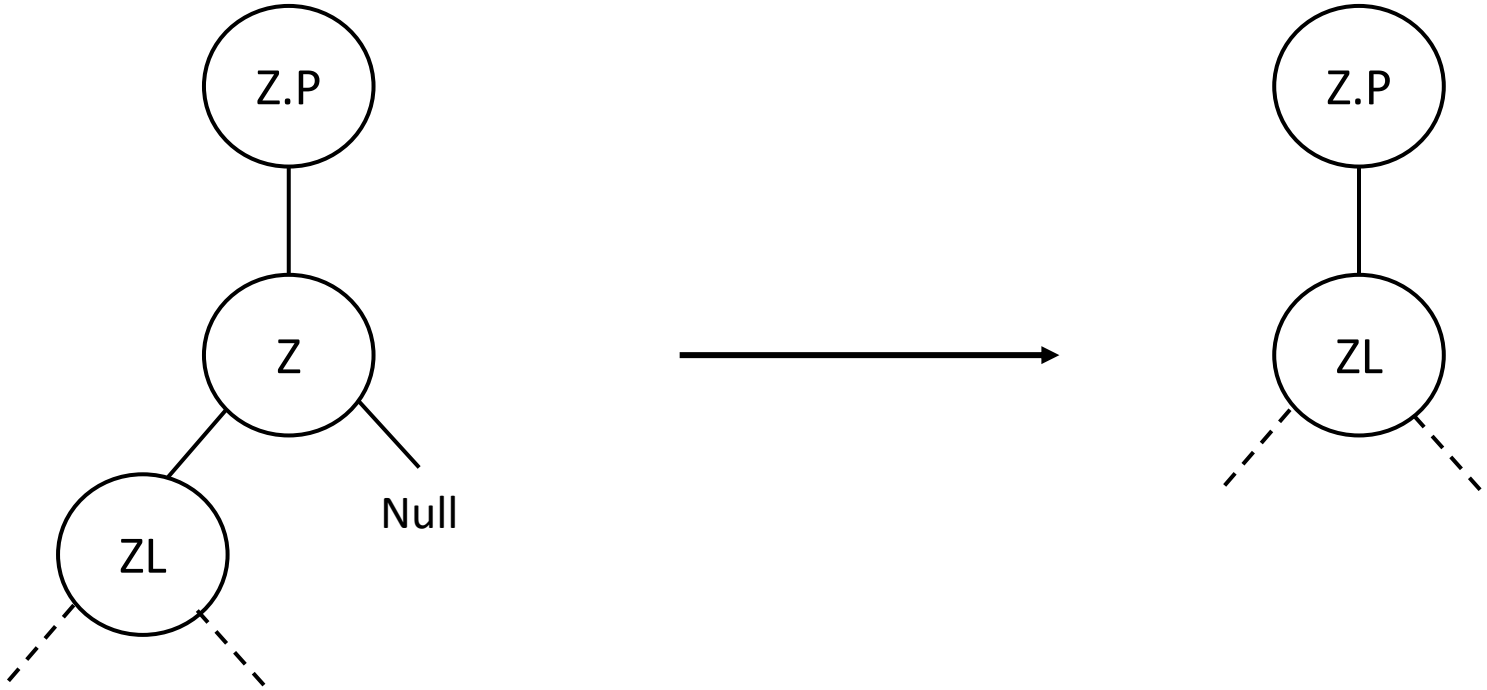
# Case-A



# Case-B



# Case-B



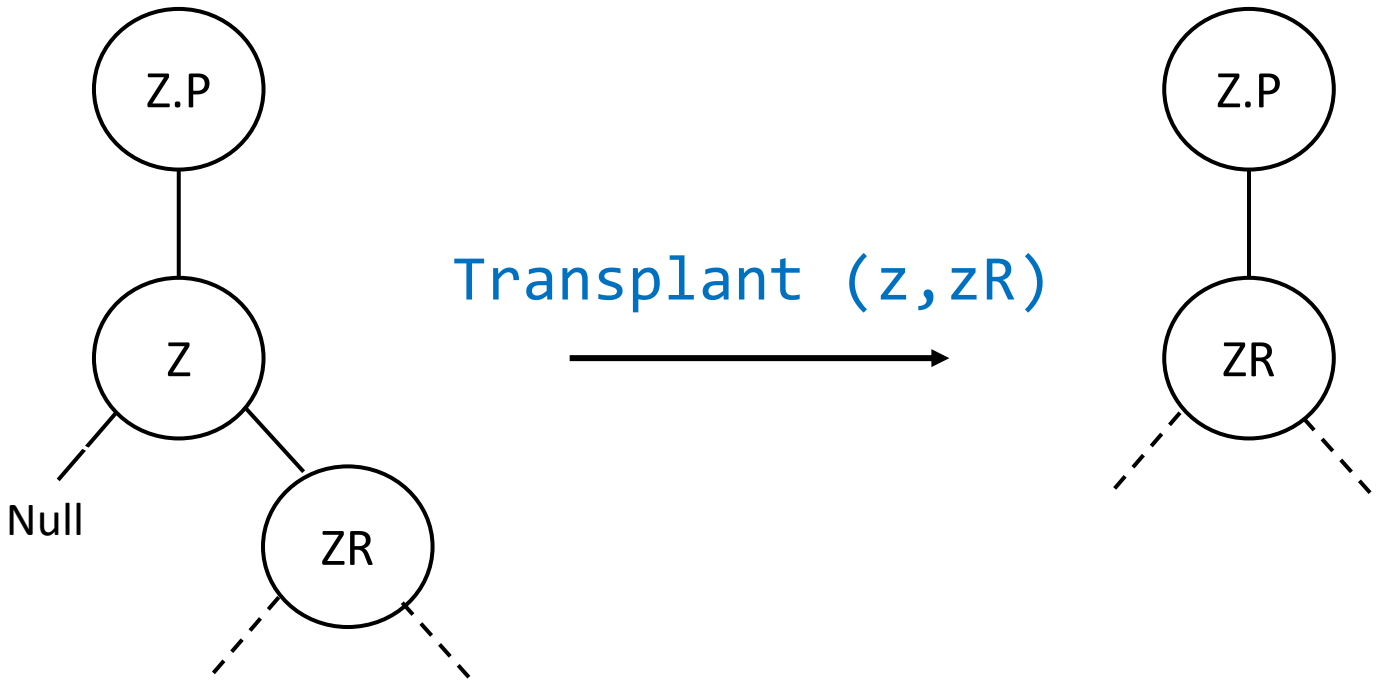
```
transplant (u,v){
    // connects u.p with v as appropriate child

}
```

```
transplant (u,v){
    // connects u.p with v as appropriate child

    if (u.p == Null):
        root = v

}
```

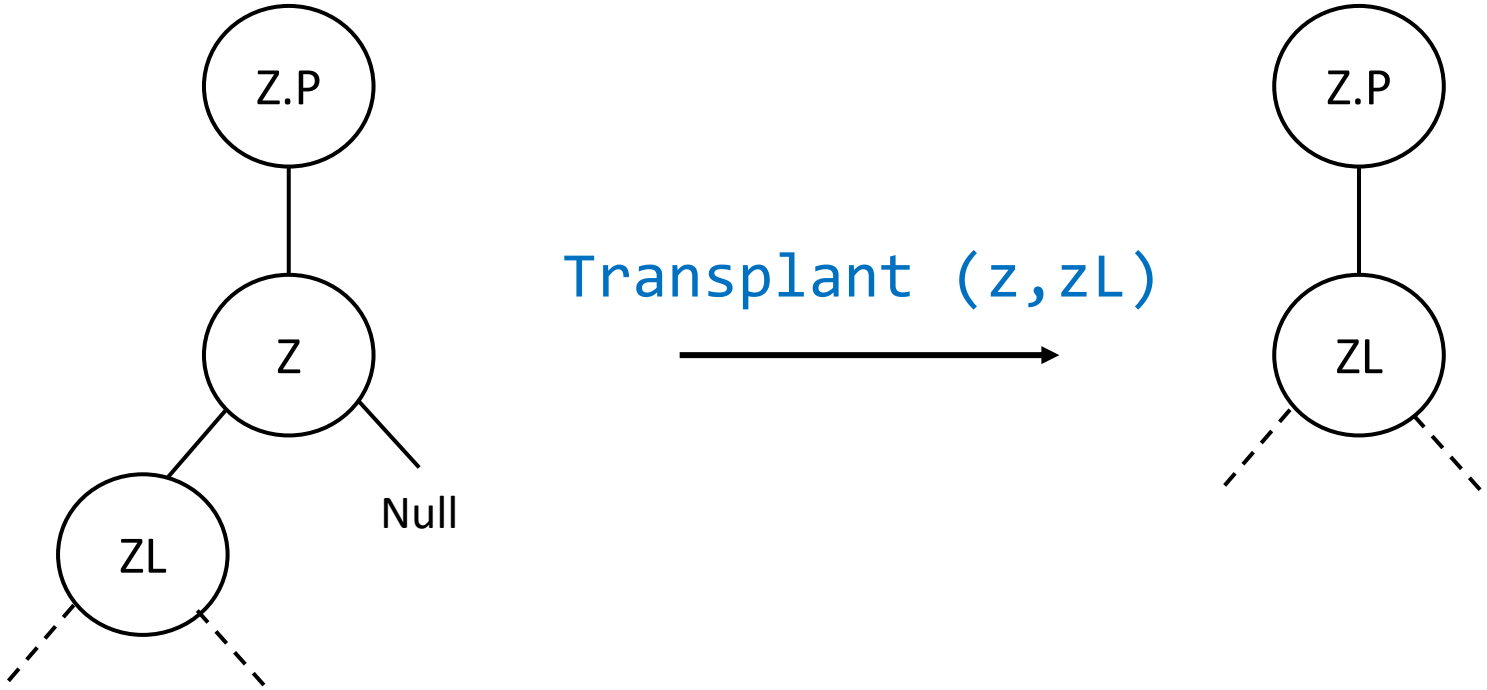
```
transplant (u,v){
    // connects u.p with v as appropriate child

    if (u.p == Null):
        root = v
    else if (u == u.p.left):
        u.p.left = v
    else
        u.p.right = v


}
```

```
transplant (u,v){
    // connects u.p with v as appropriate child

    if (u.p == Null):
        root = v
    else if (u == u.p.left):
        u.p.left = v
    else
        u.p.right = v
    if (v!=Null):
        v.p = u.p
}
```

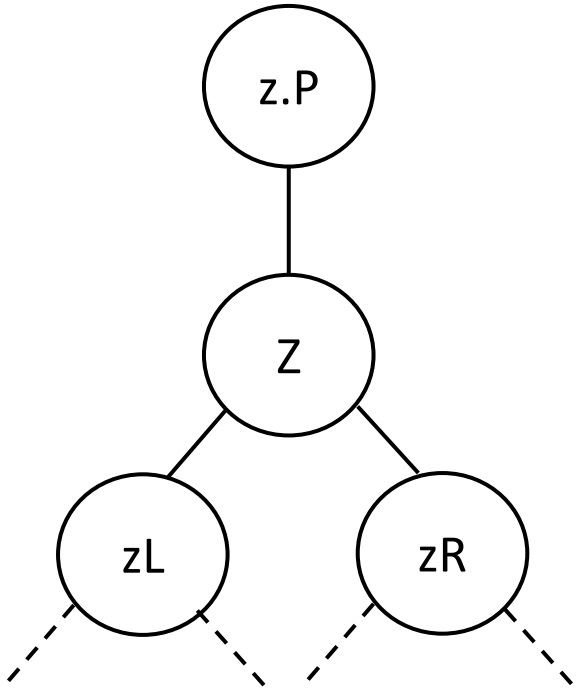
# Case-A



Case-B

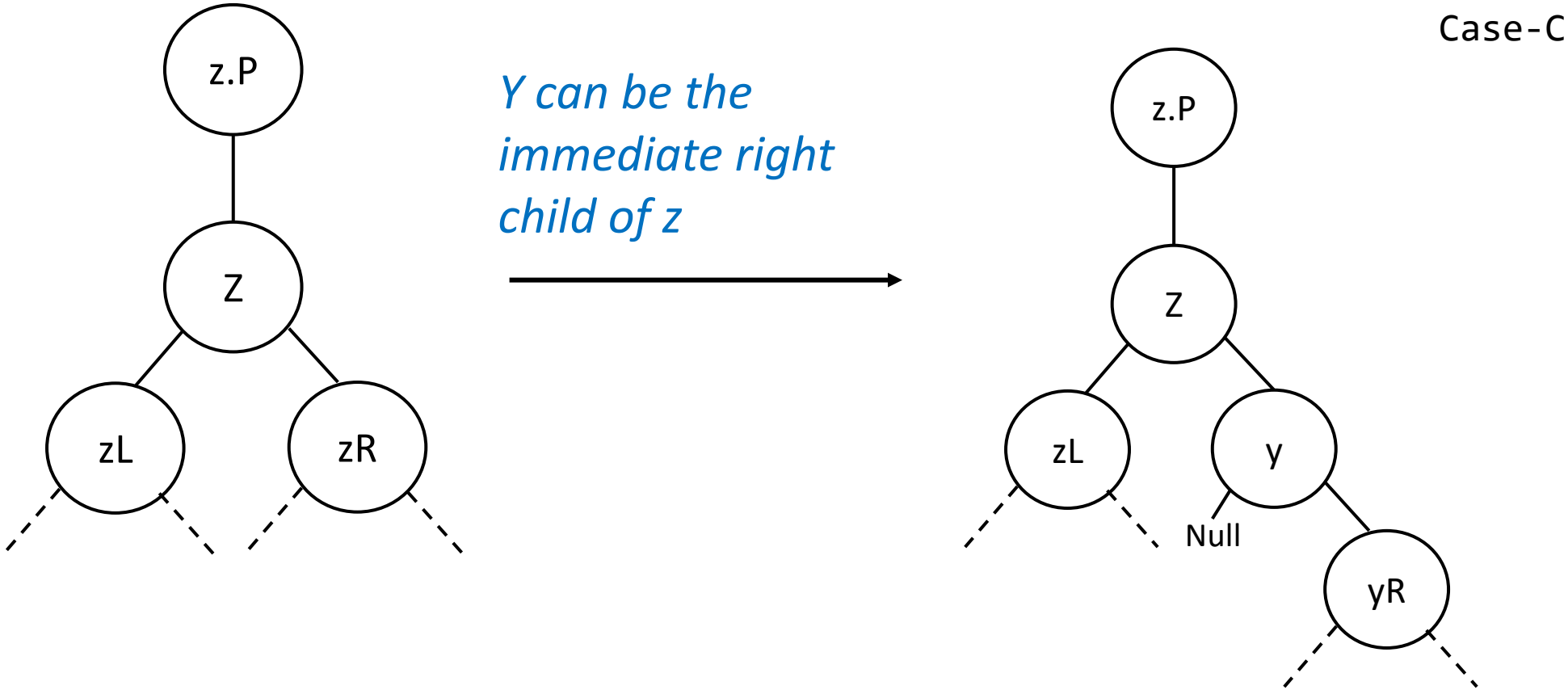


*Z has two subtrees.*
*So lets find y = successor(z)*

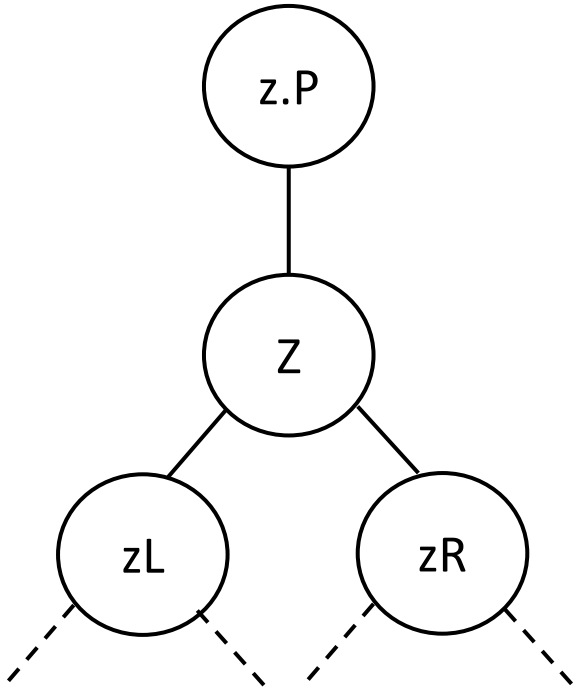


*Z has two subtrees.*
*So lets find y = successor(z)*

Case-C



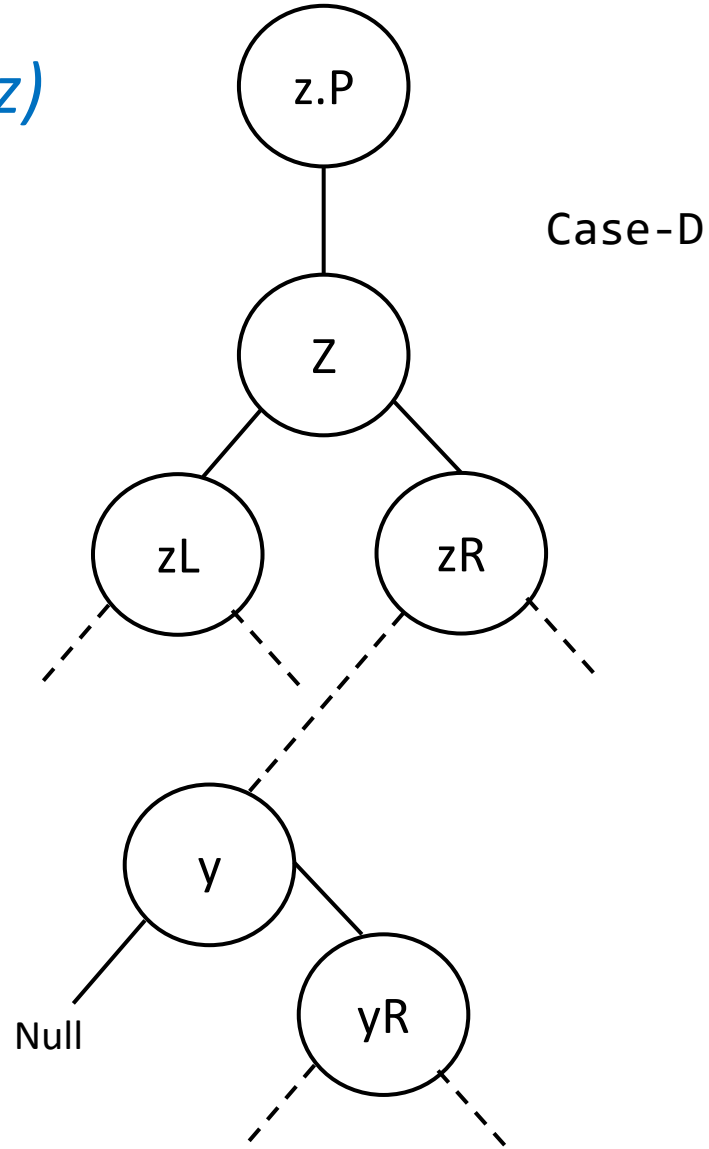
*Z has two subtrees.*
*So lets find y = successor(z)*



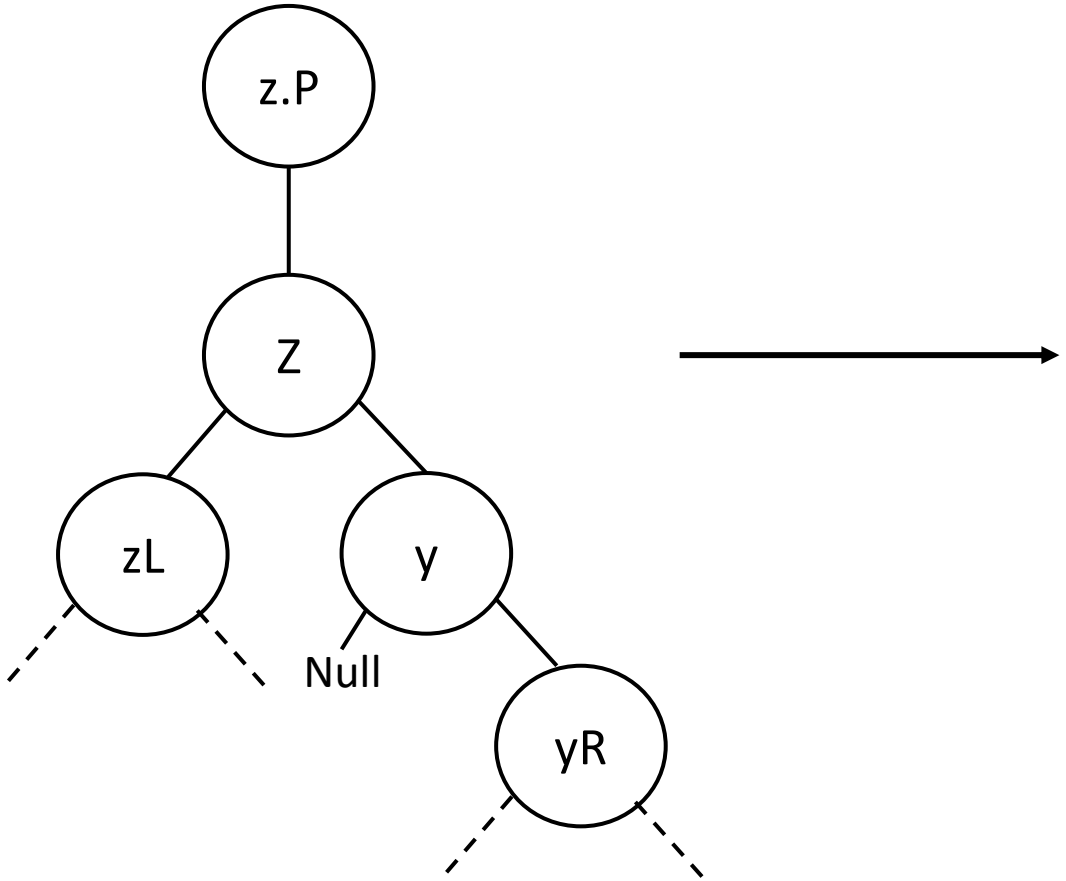

*Y can be the immediate right child of z*

*Otherwise, y will be minimum(z.right)*

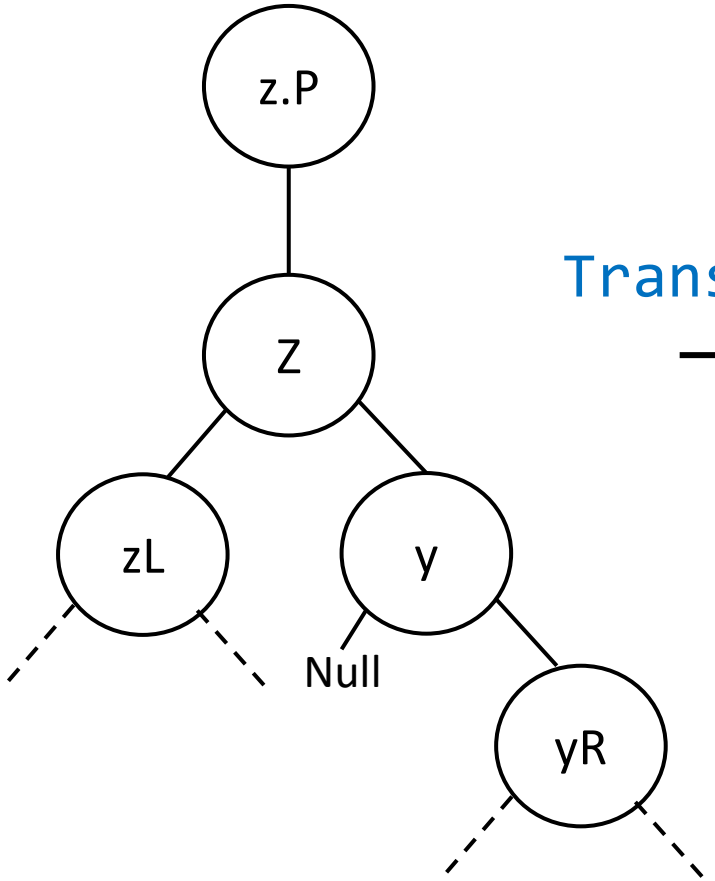
Case-D



# Case-C

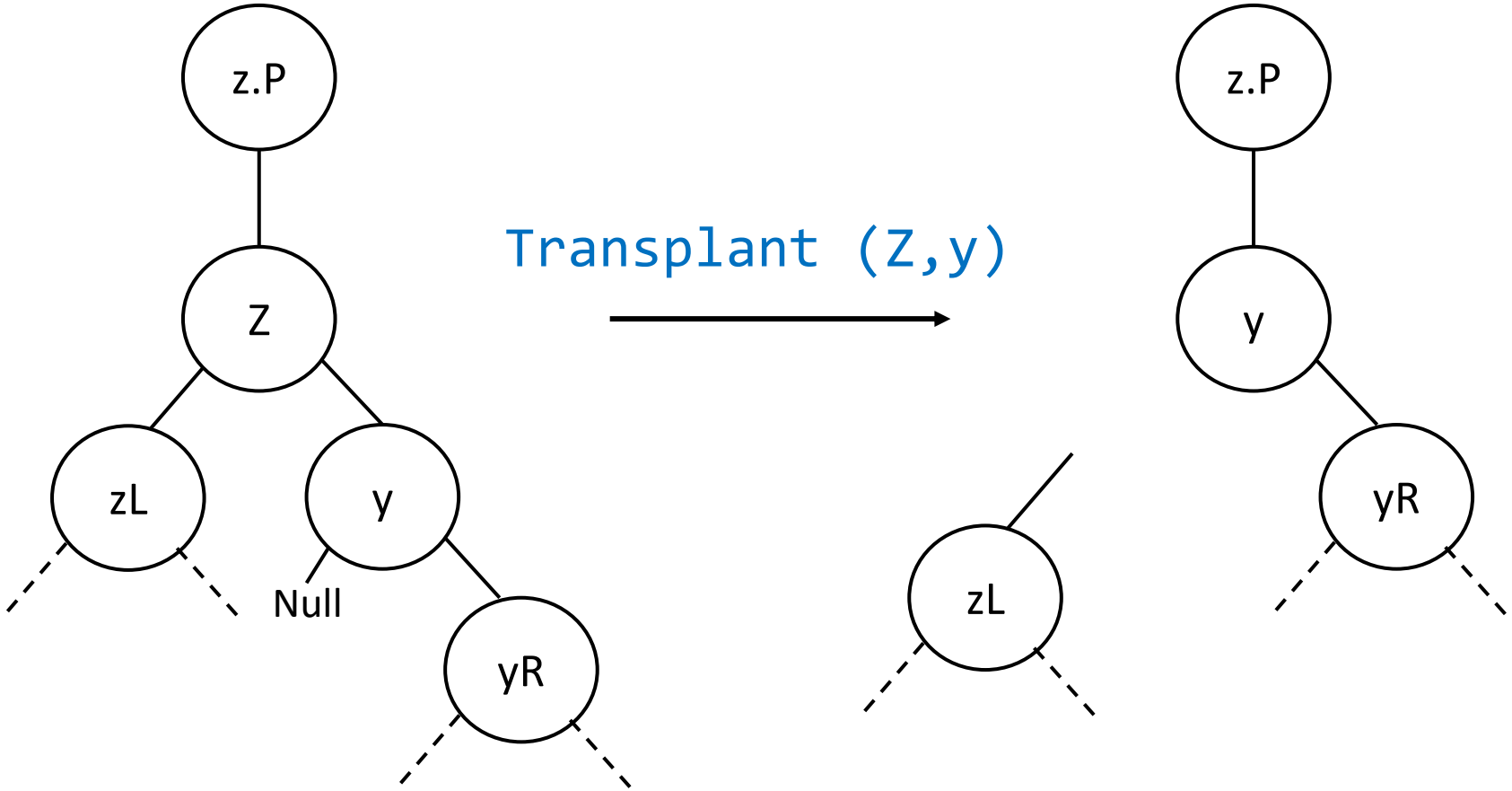


# Case-C

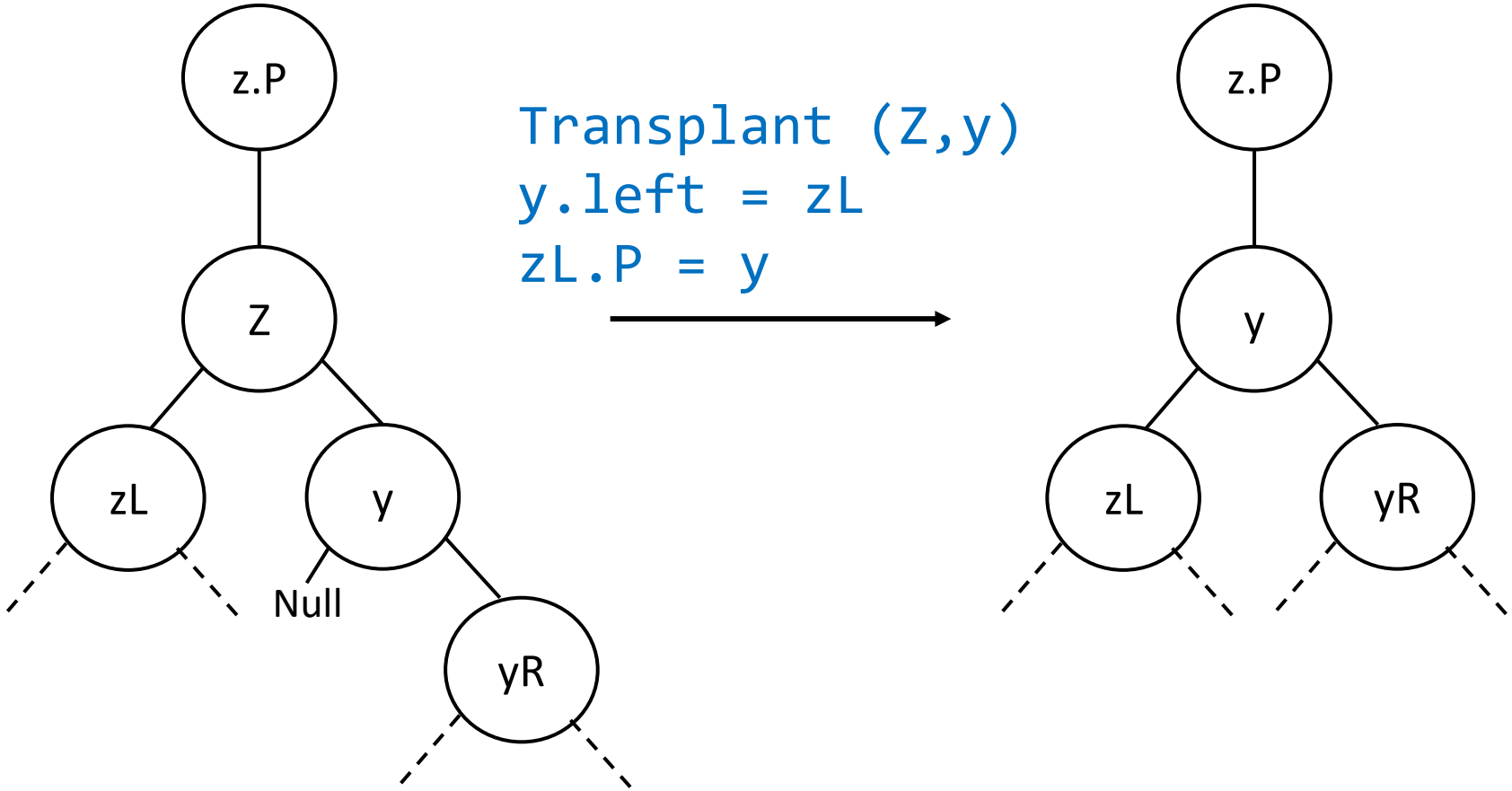



Transplant (Z,y)

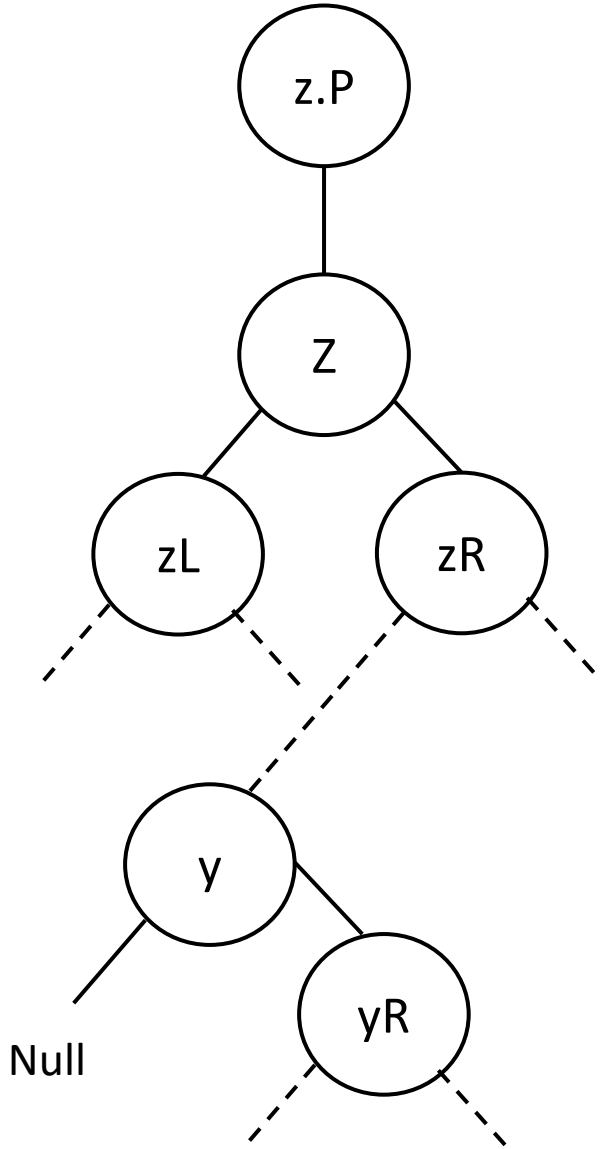
# Case-C



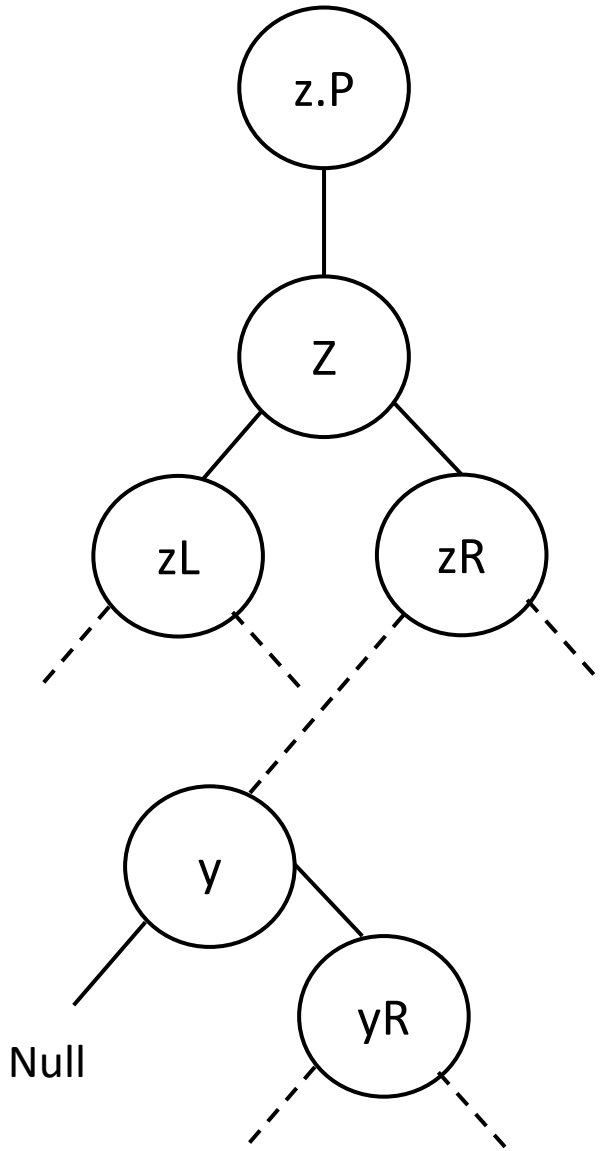
# Case-C



Case-D

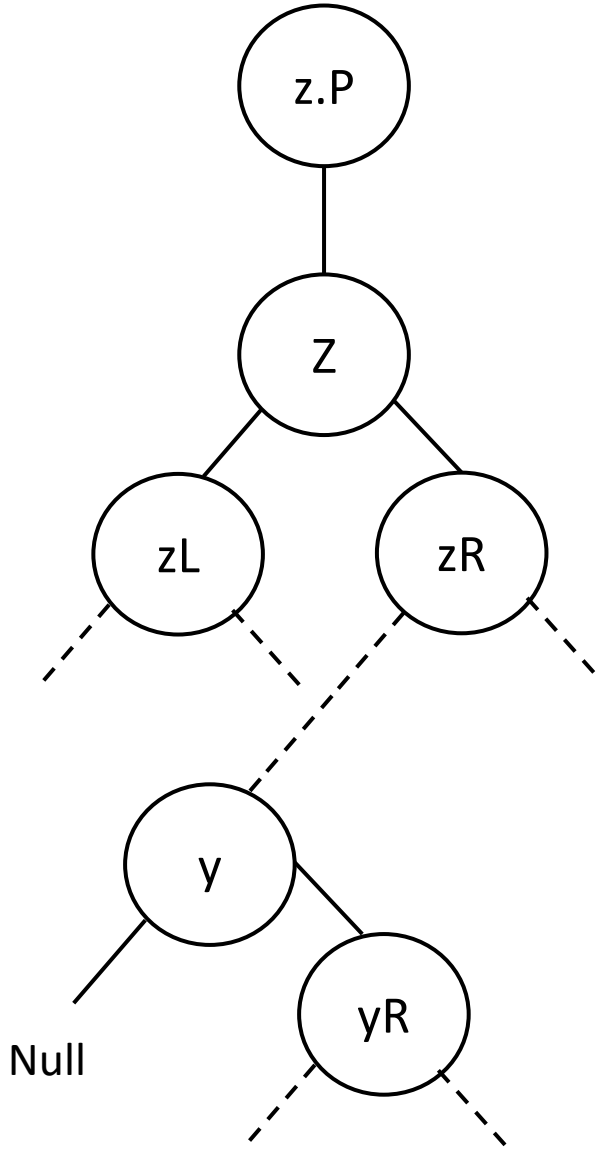


Case-D




*Plan:*
*y should come above, replace Z, catch zL & zR*

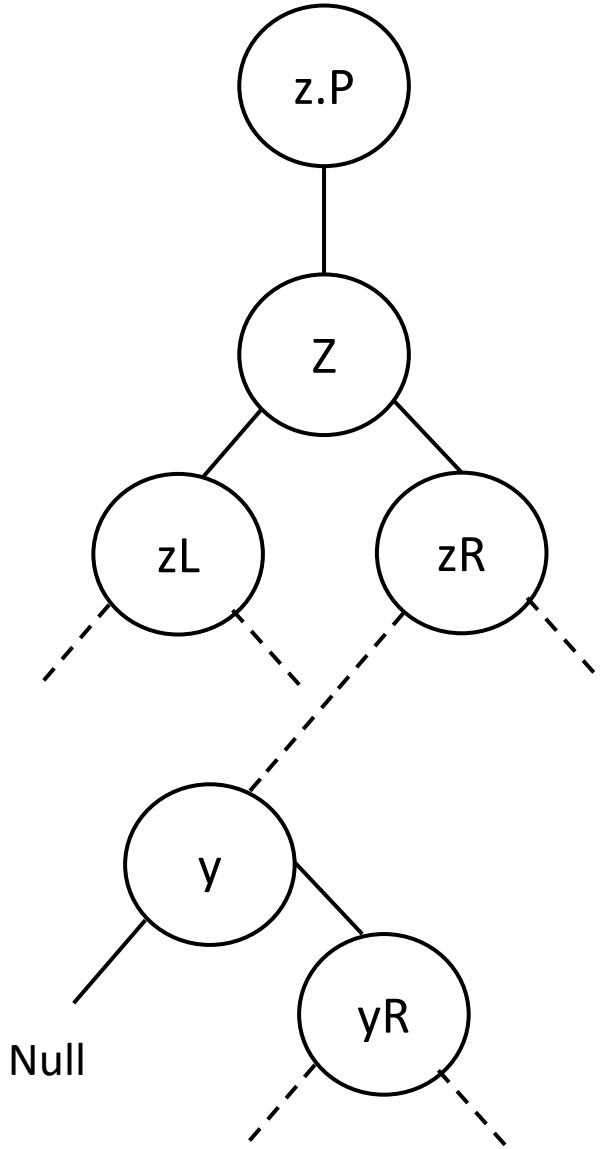
Case-D




*Plan:*
*y should come above, replace Z, catch zL & zR*

*To enable y to catch zL, zR ---*
*First y should be free!!!*

Case-D

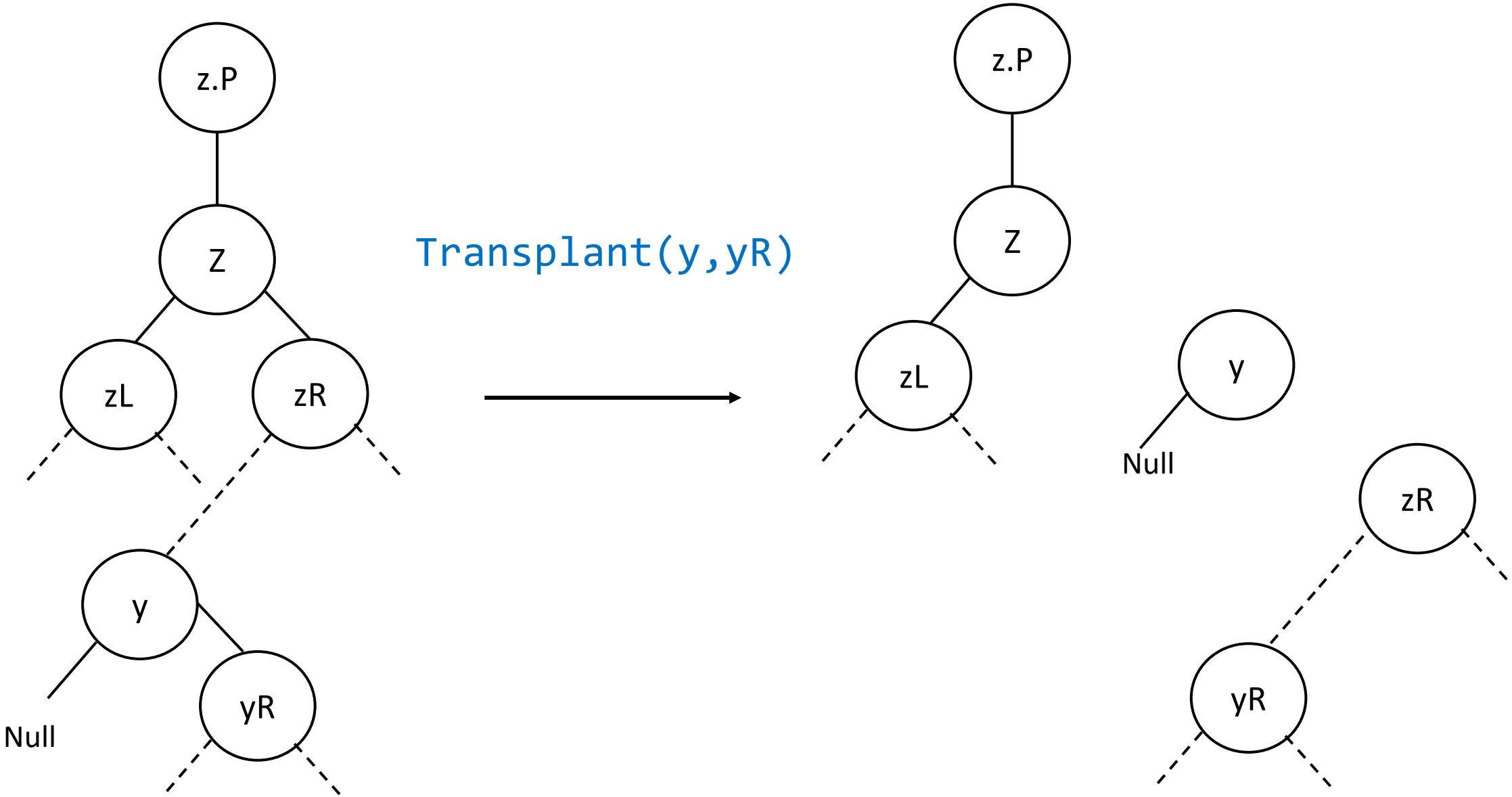



*Plan:*
*y should come above, replace Z, catch zL & zR*

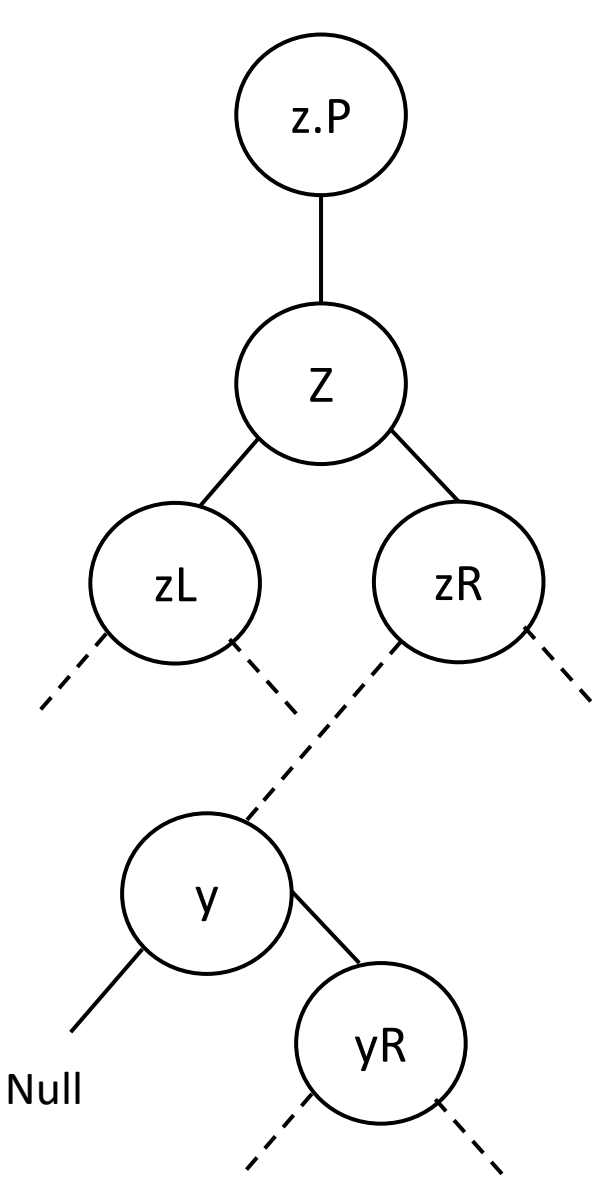
*To enable y to catch zL, zR ---*
*First y should be free!!!*

*yR becomes orphan*
*So, yR should be connected with y.p*

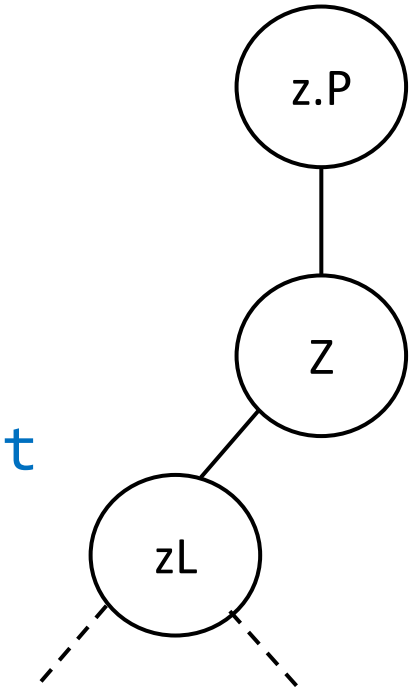
Case-D

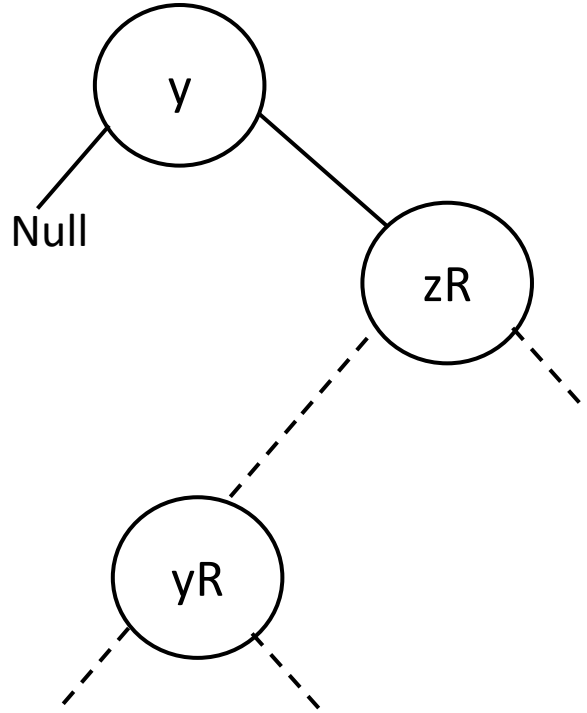


Case-D

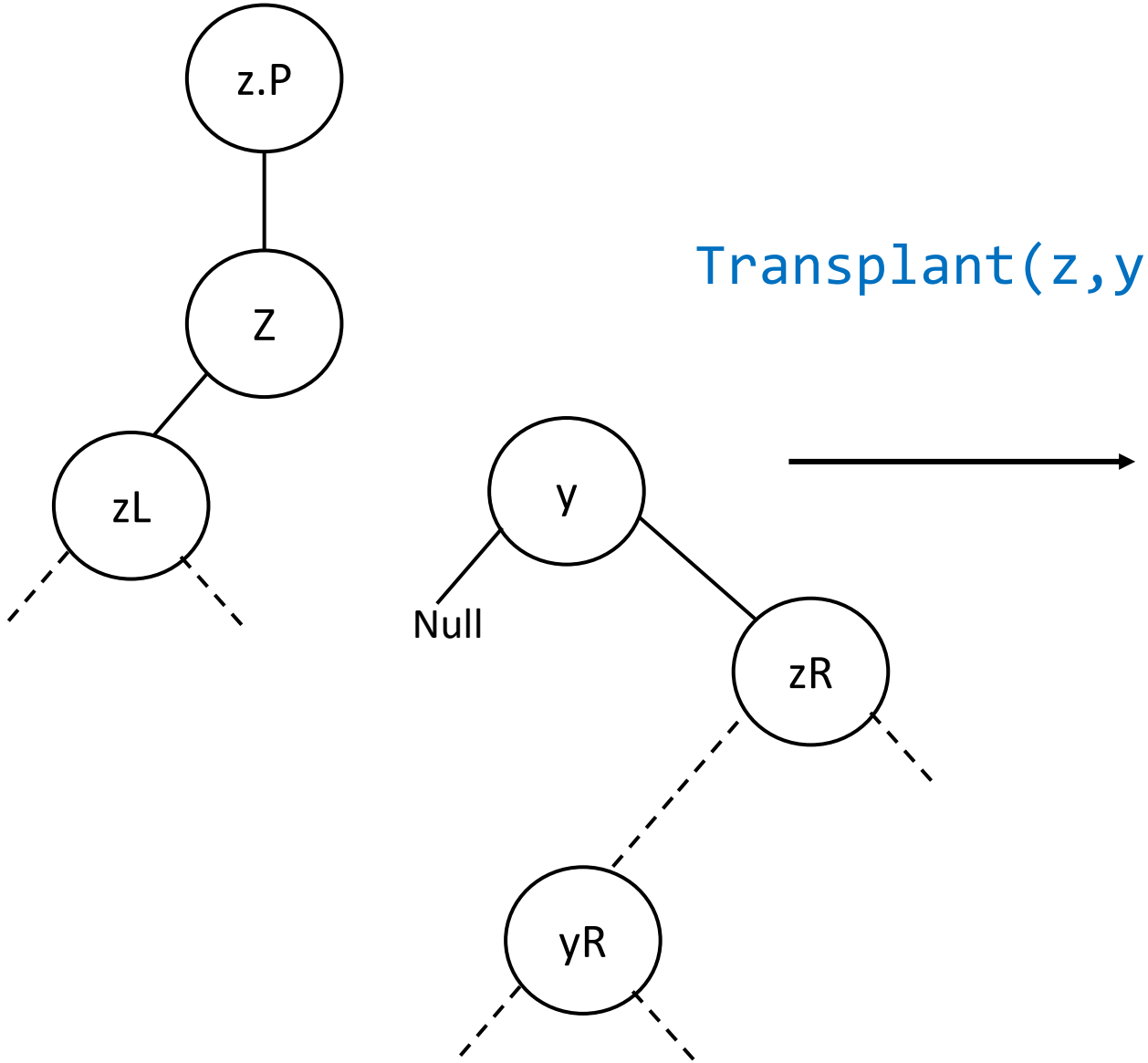



```
Transplant(y,yR)
y.right = z.right
z.right.p = y
```

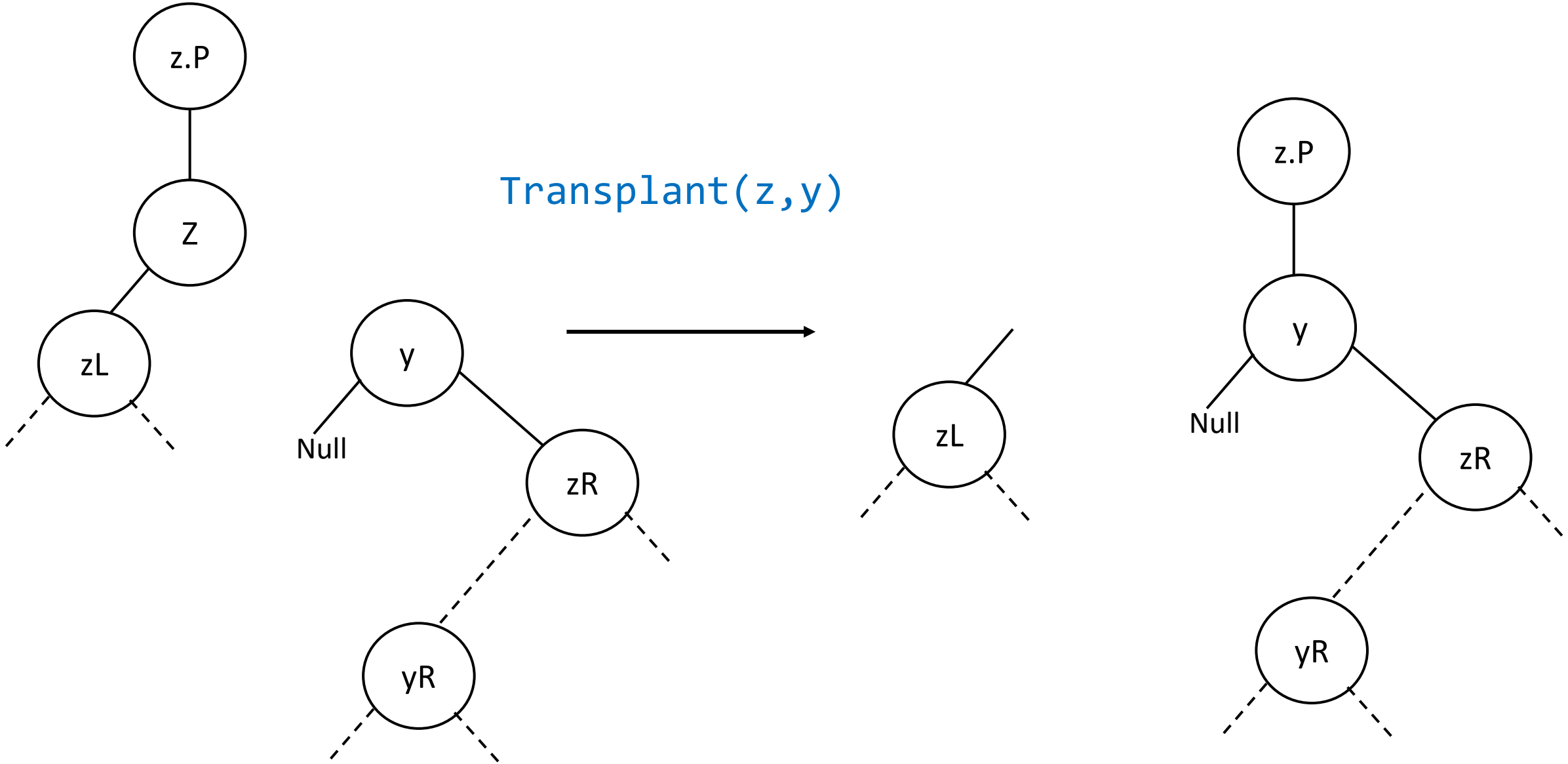


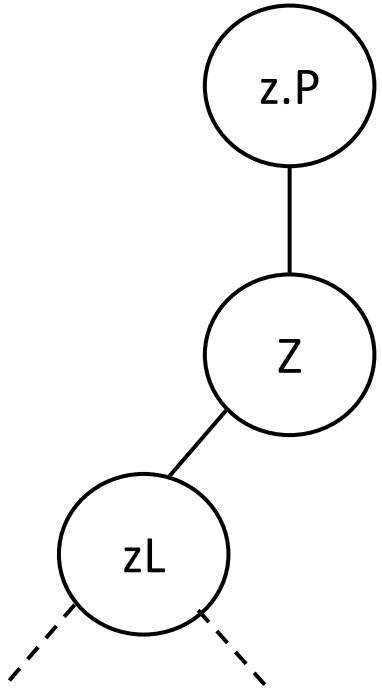
Case-D



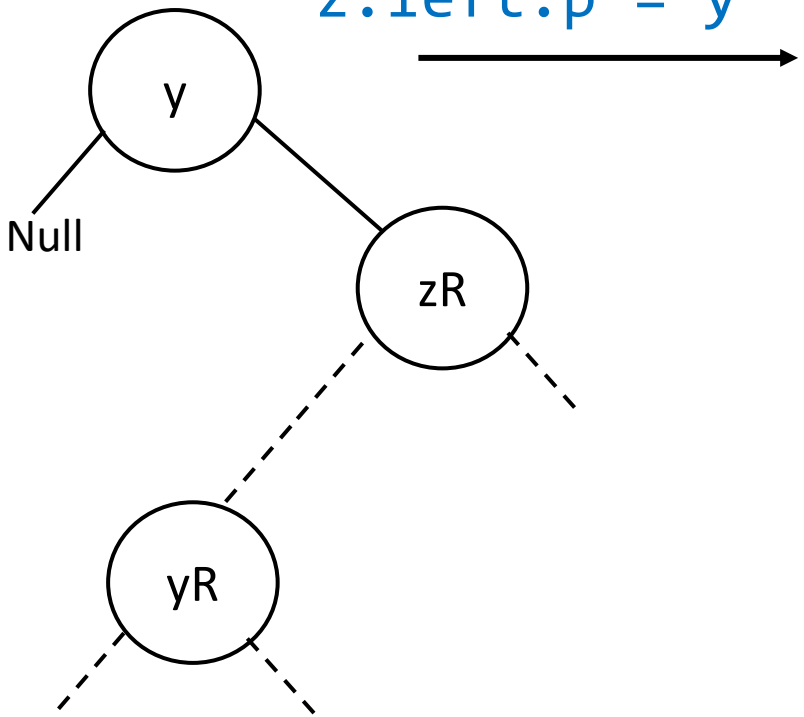
Case-D

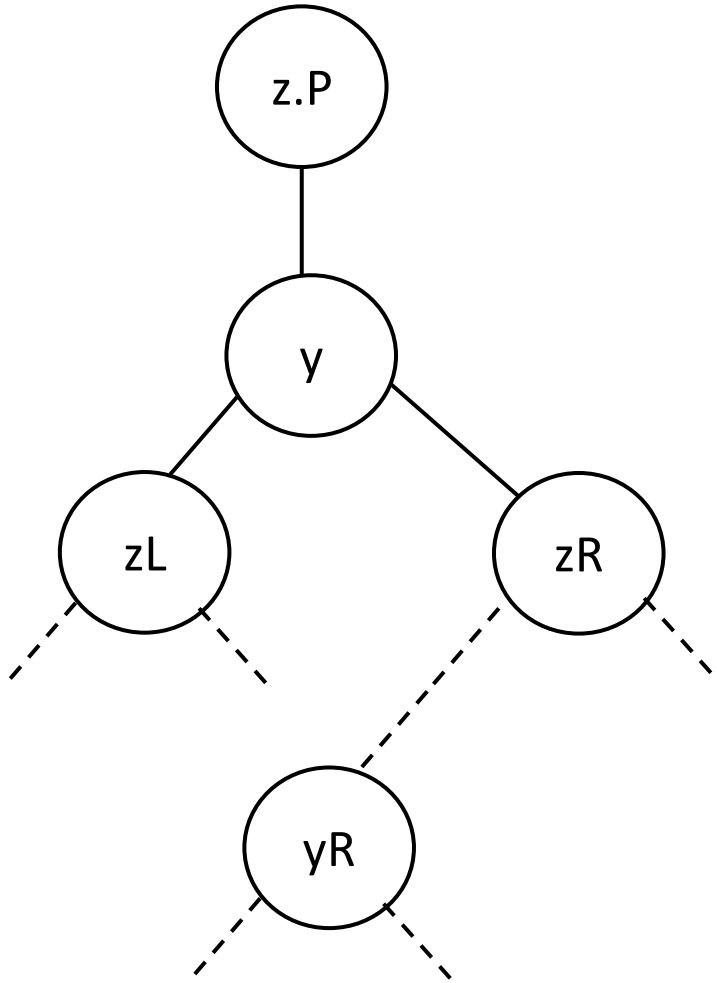


Case-D




```
Transplant(z,y)
y.left = z.left
z.left.p = y
```

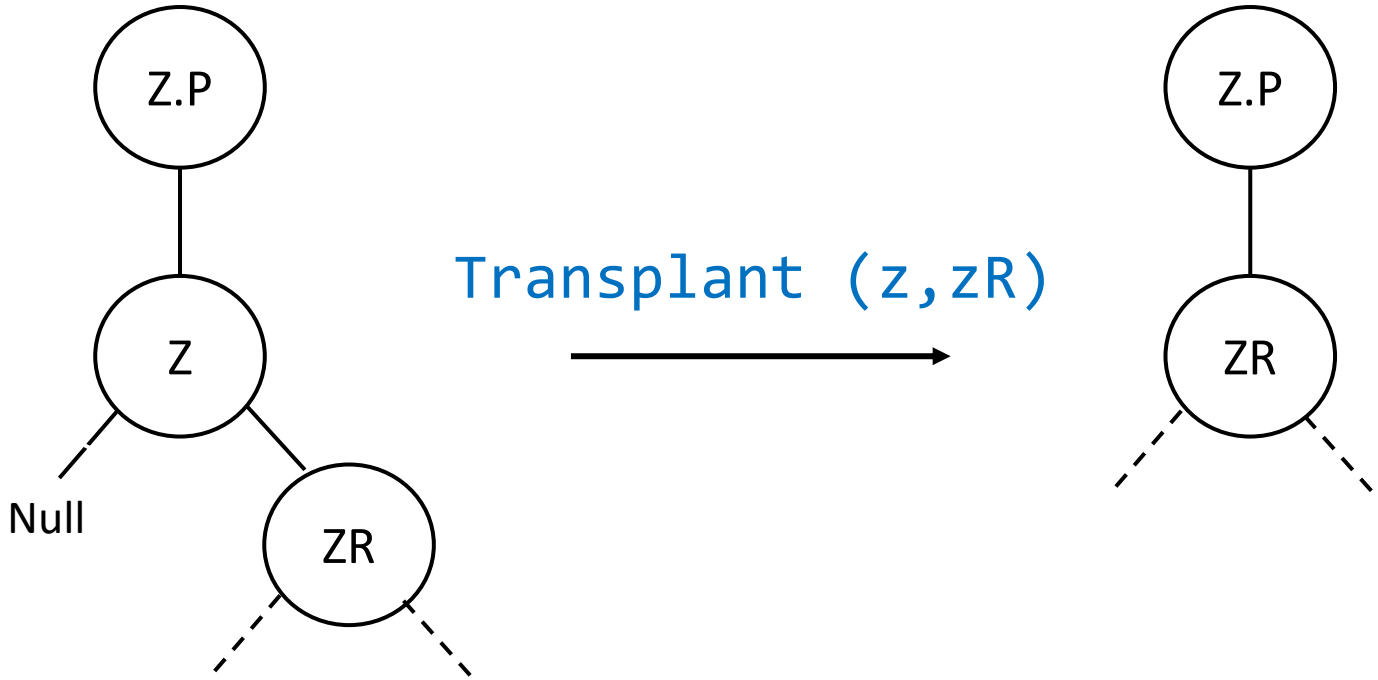


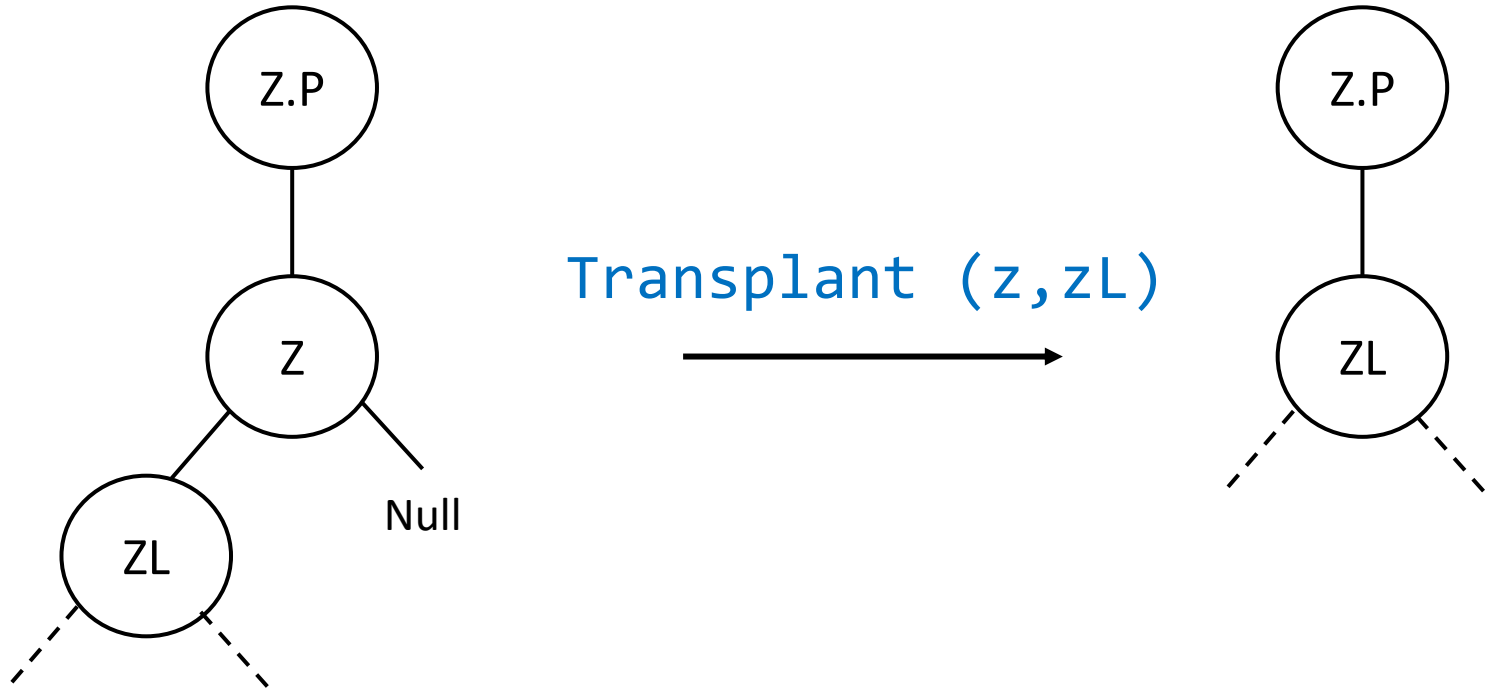
```
Tree_delete(int key):
```

```
Tree_delete(int key):
      Node* z = search(key)
```

```
Tree_delete(int key):
      Node* z = search(key)
      if z.left == Null:
            transplant(z, z.right)
```

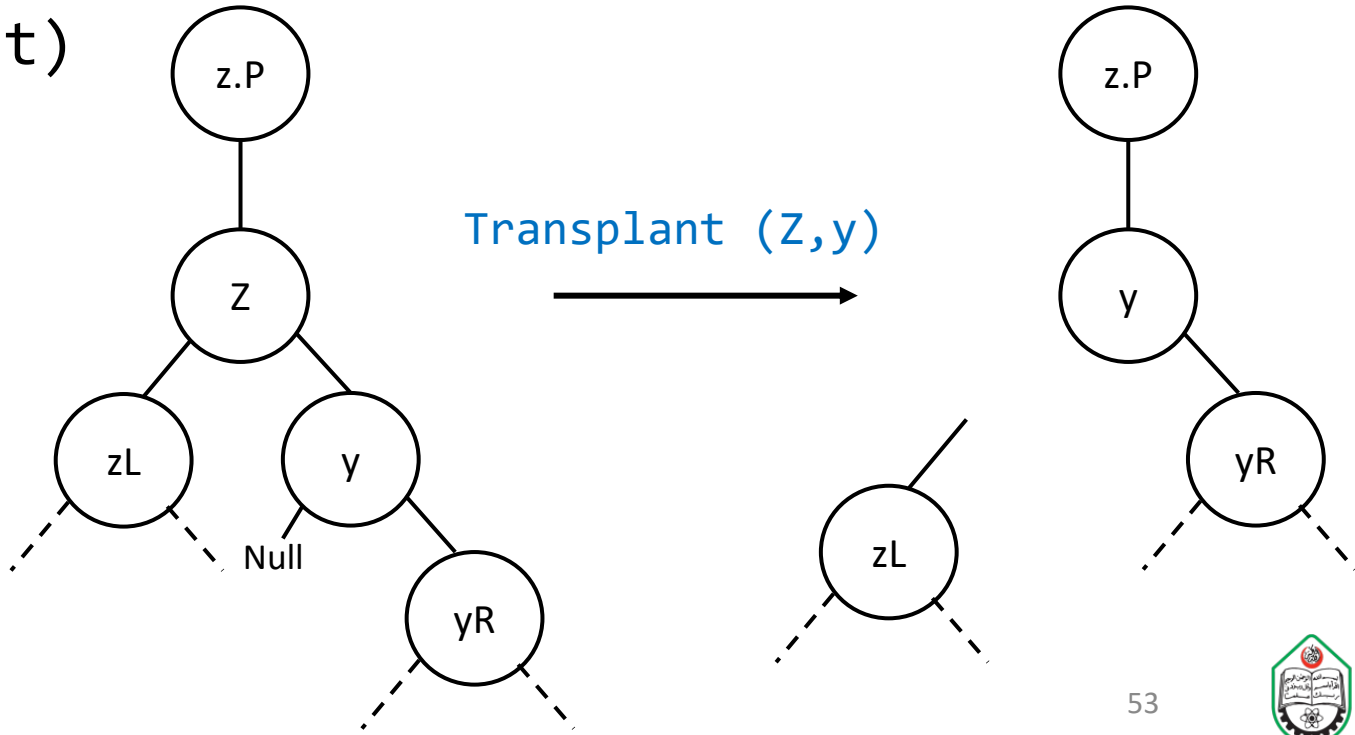


```
Tree_delete(int key):
      Node* z = search(key)
      if z.left == Null:
            transplant(z, z.right)
      else if z.right == Null:
            transplant(z, z.left)
```

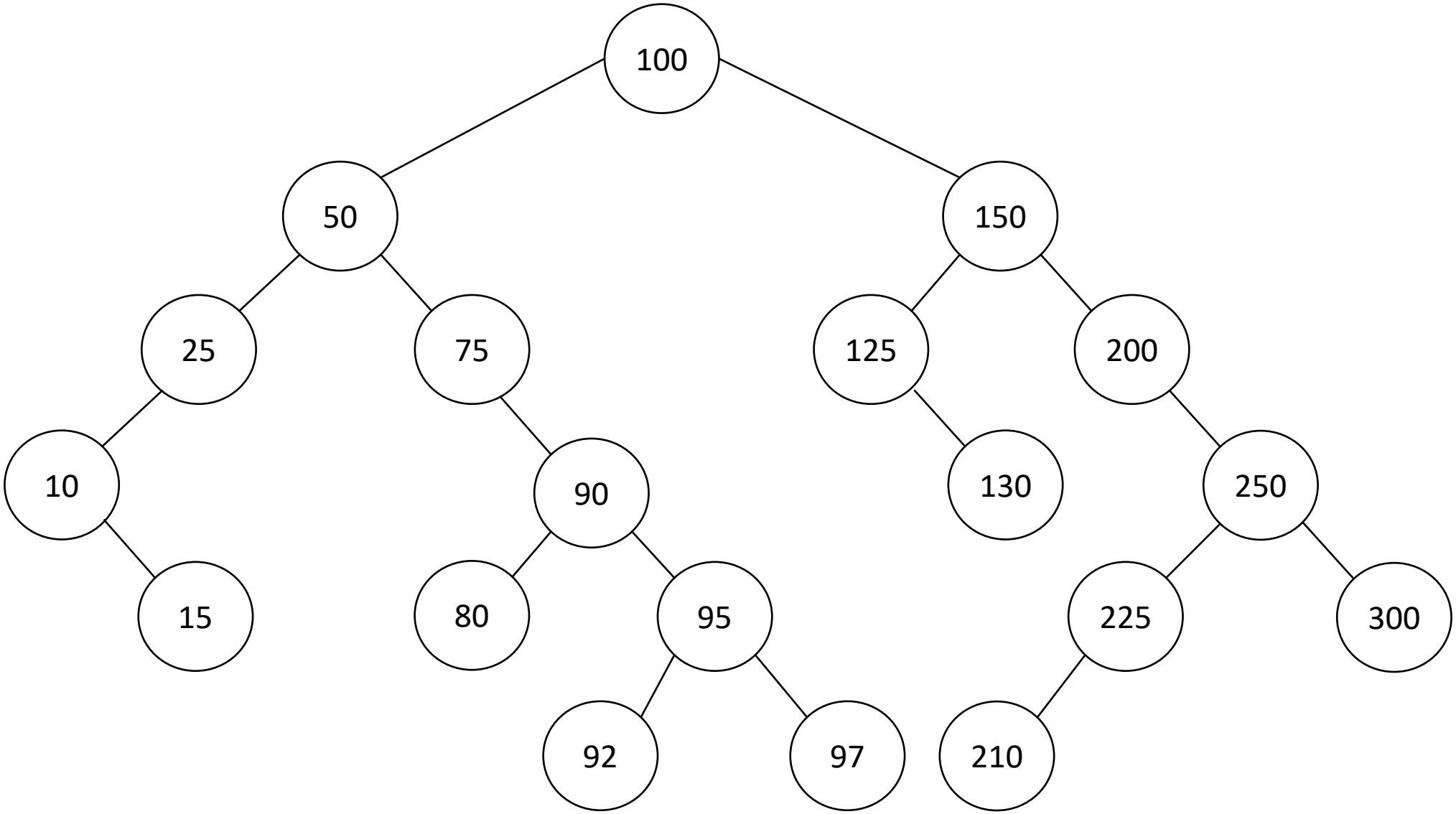


```
Tree_delete(int key):
      Node* z = search(key)
      if z.left == Null:
            transplant(z, z.right)
      else if z.right == Null:
            transplant(z, z.left)
      else
            y == minimum(z.right)

                  transplant(z,y)
                  y.left = z.left
                  z.left.p = y
```

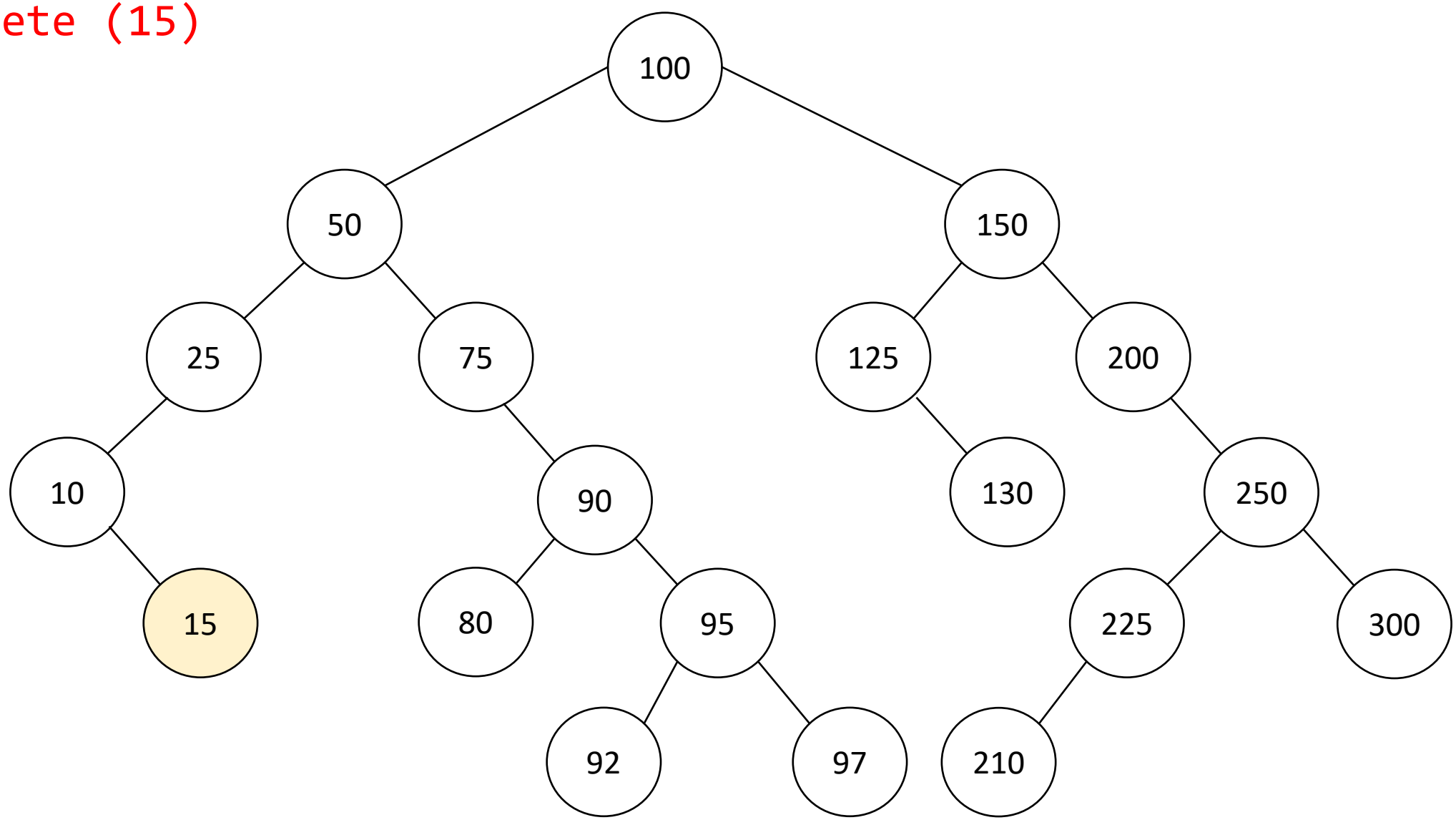
*(Assuming y.p == z)*



```
Tree_delete(int key):
      Node* z = search(key)
      if z.left == Null:
            transplant(z, z.right)
      else if z.right == Null:
            transplant(z, z.left)
      else
            y == minimum(z.right)
            if y.p != z:
                  transplant(y, y.right)
                  y.right = z.right
                  z.right.p = y
            transplant(z,y)
            y.left = z.left
            z.left.p = y
```

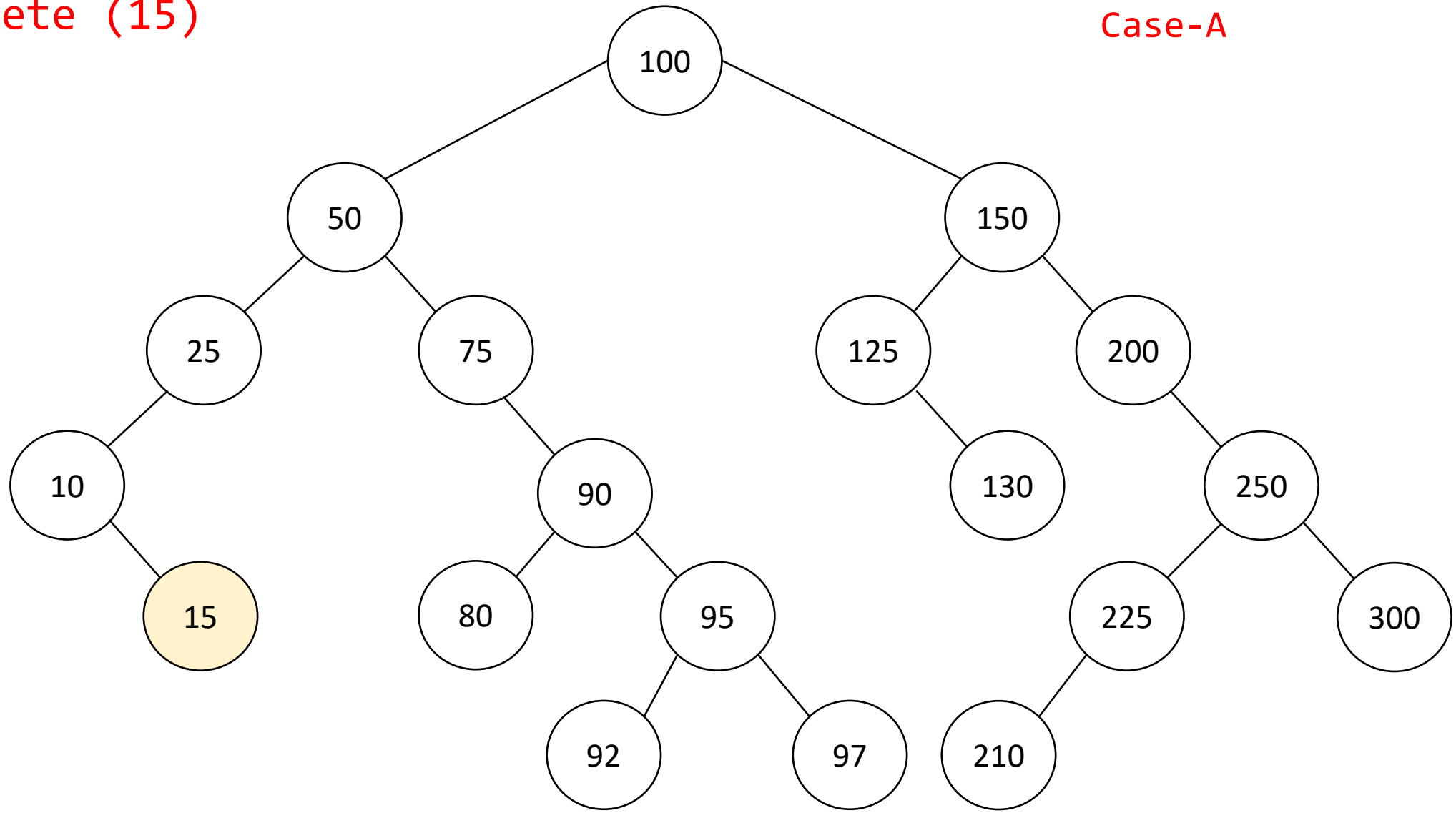

100
50
150
25
75
125
200
10
90
130
250
15
80
95
225
300
92
97
210

# Delete (15)



# Delete (15)

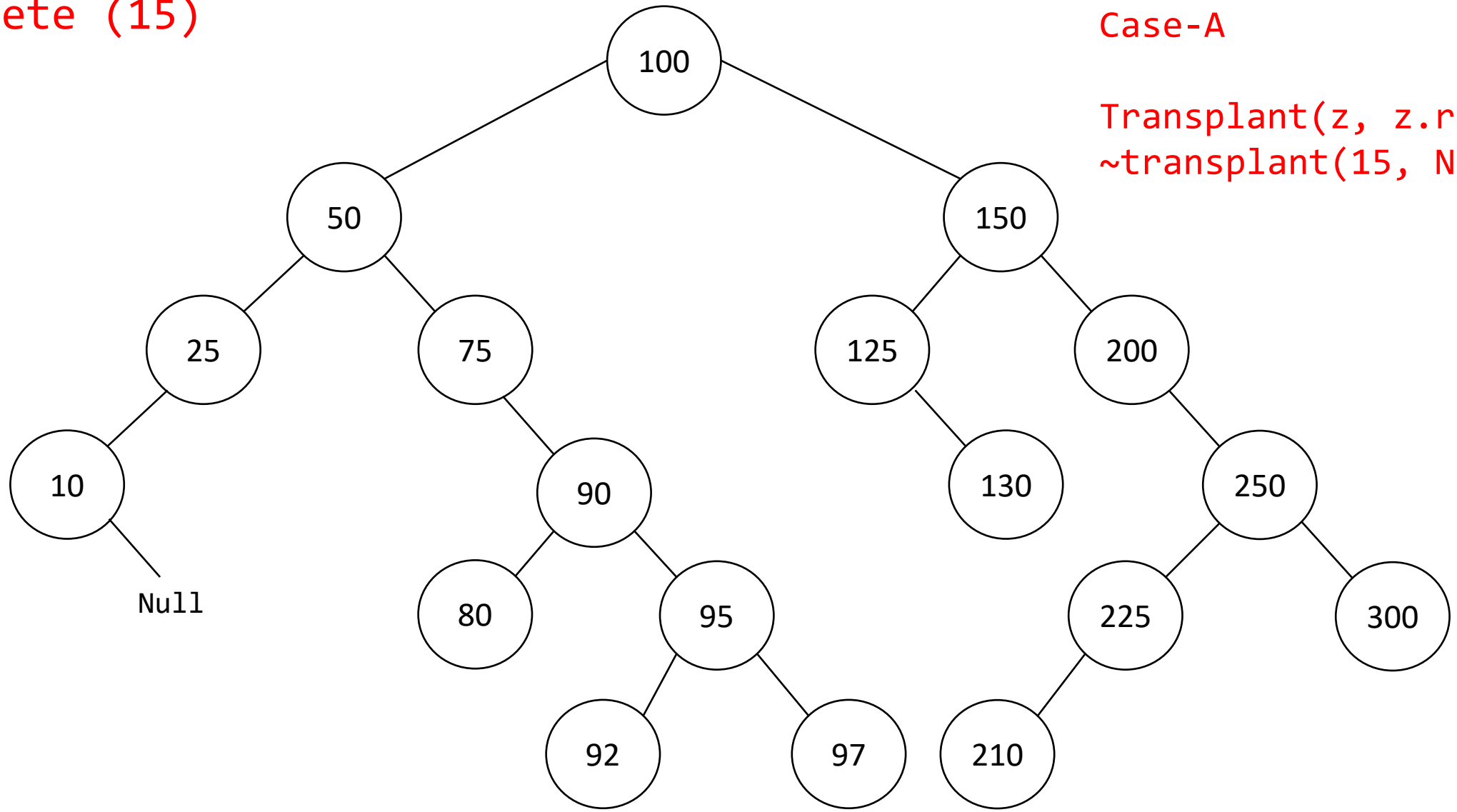
z=15, z.left=Null
Case-A



# Delete (15)

z=15, z.left=Null
Case-A

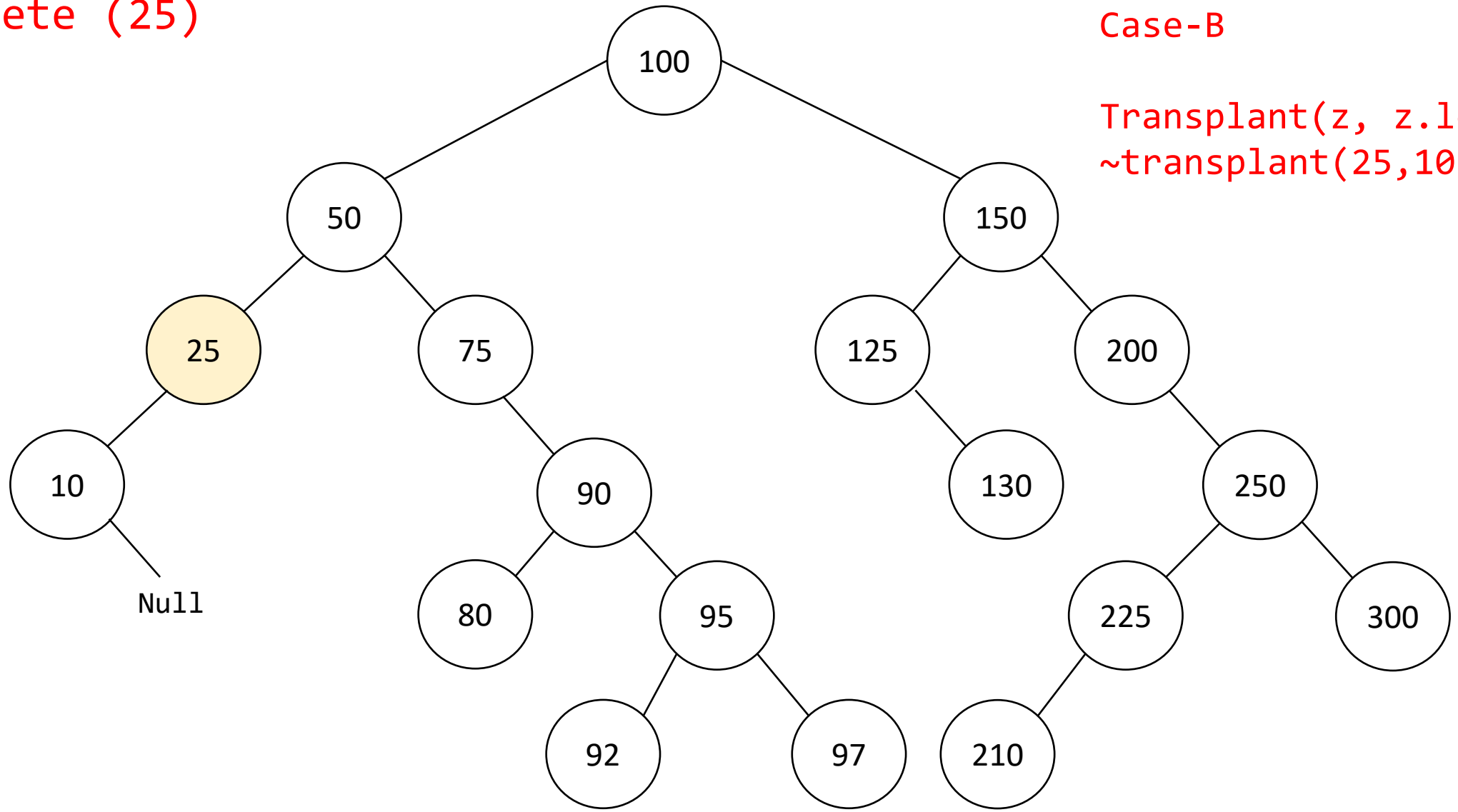
Transplant(z, z.right)
~transplant(15, Null)



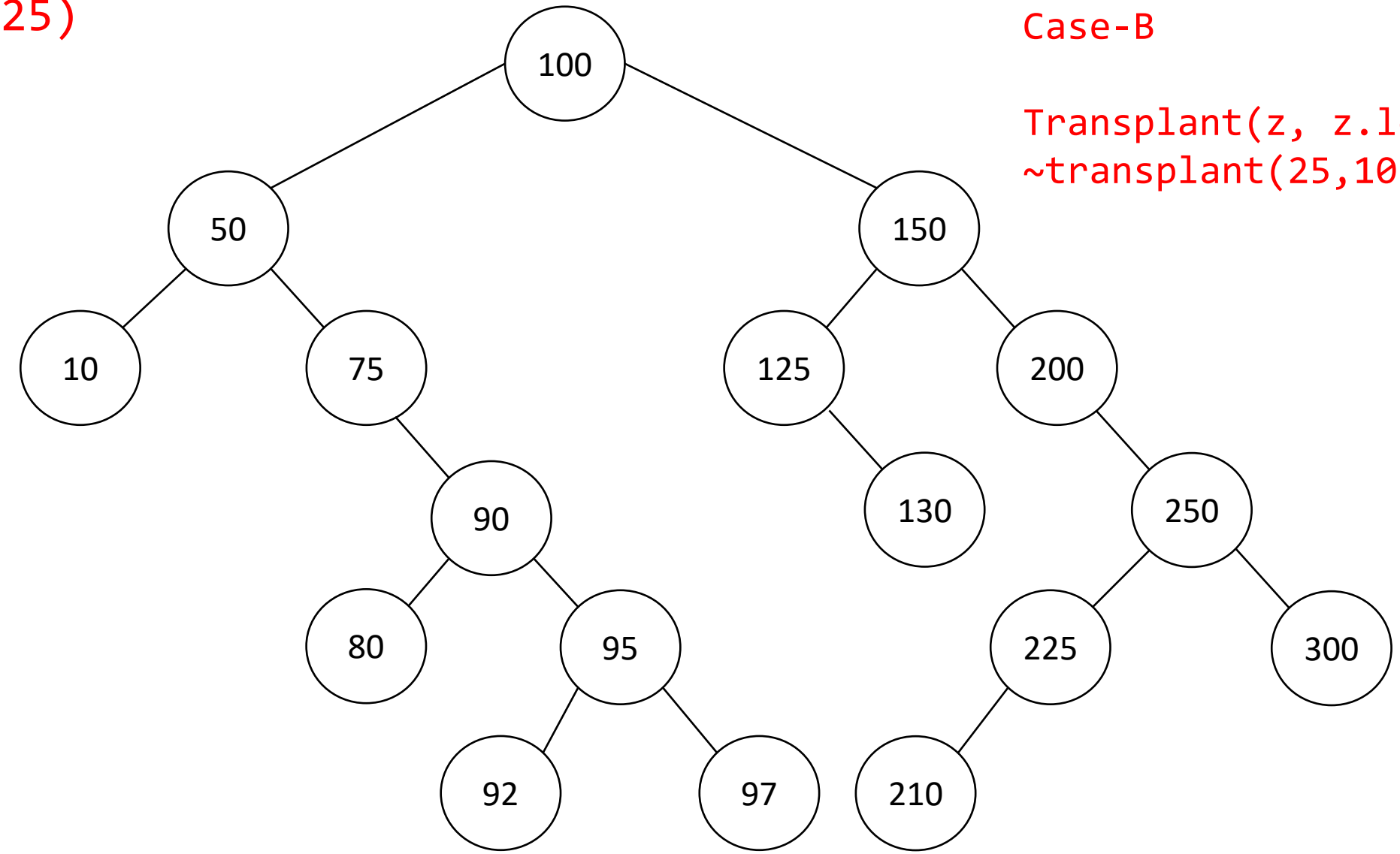
# Delete (25)

z=25, z.right=Null
Case-B

Transplant(z, z.left)
~transplant(25,10)



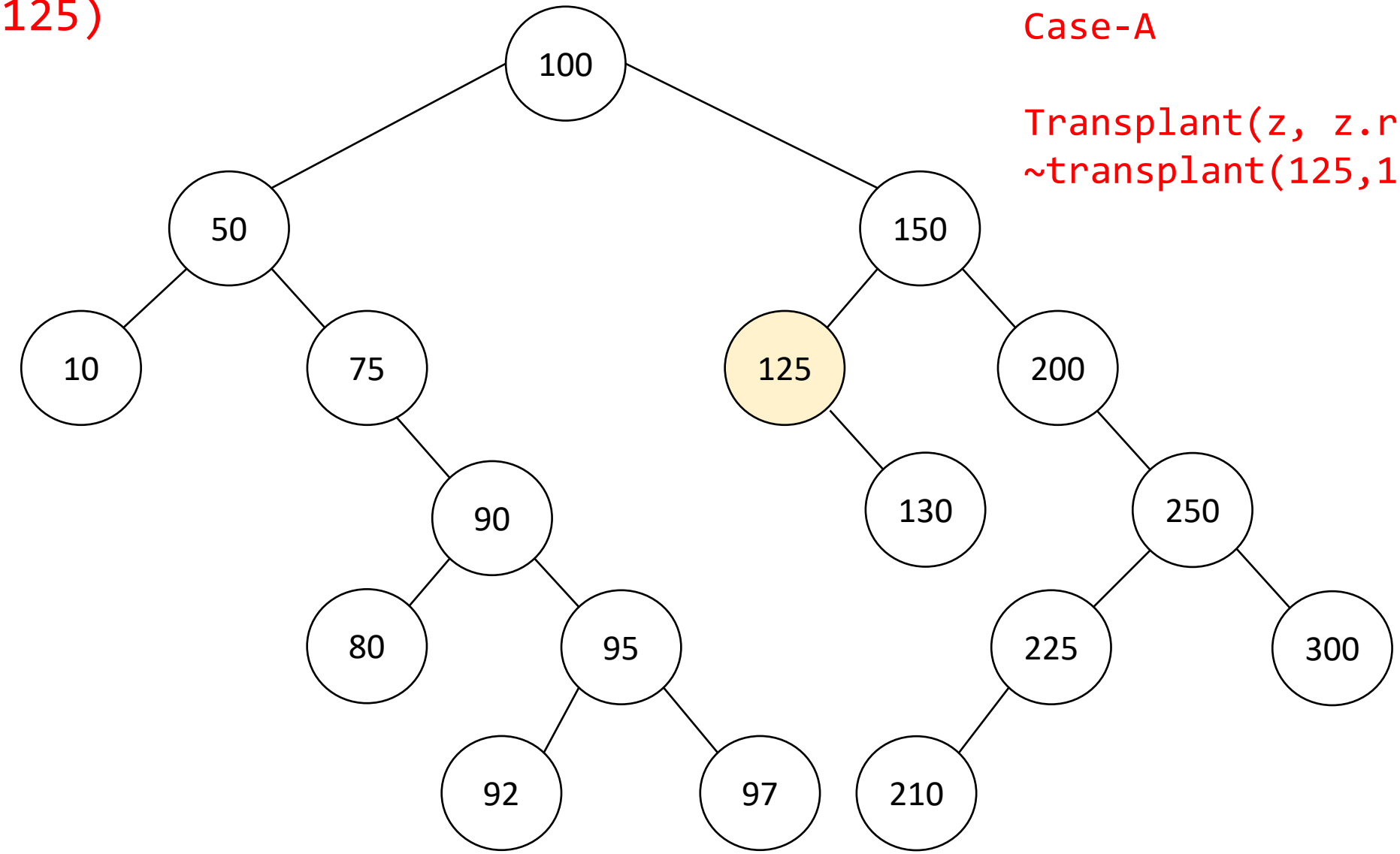
# Delete (25)




z=25, z.right=Null
Case-B

Transplant(z, z.left)
~transplant(25,10)

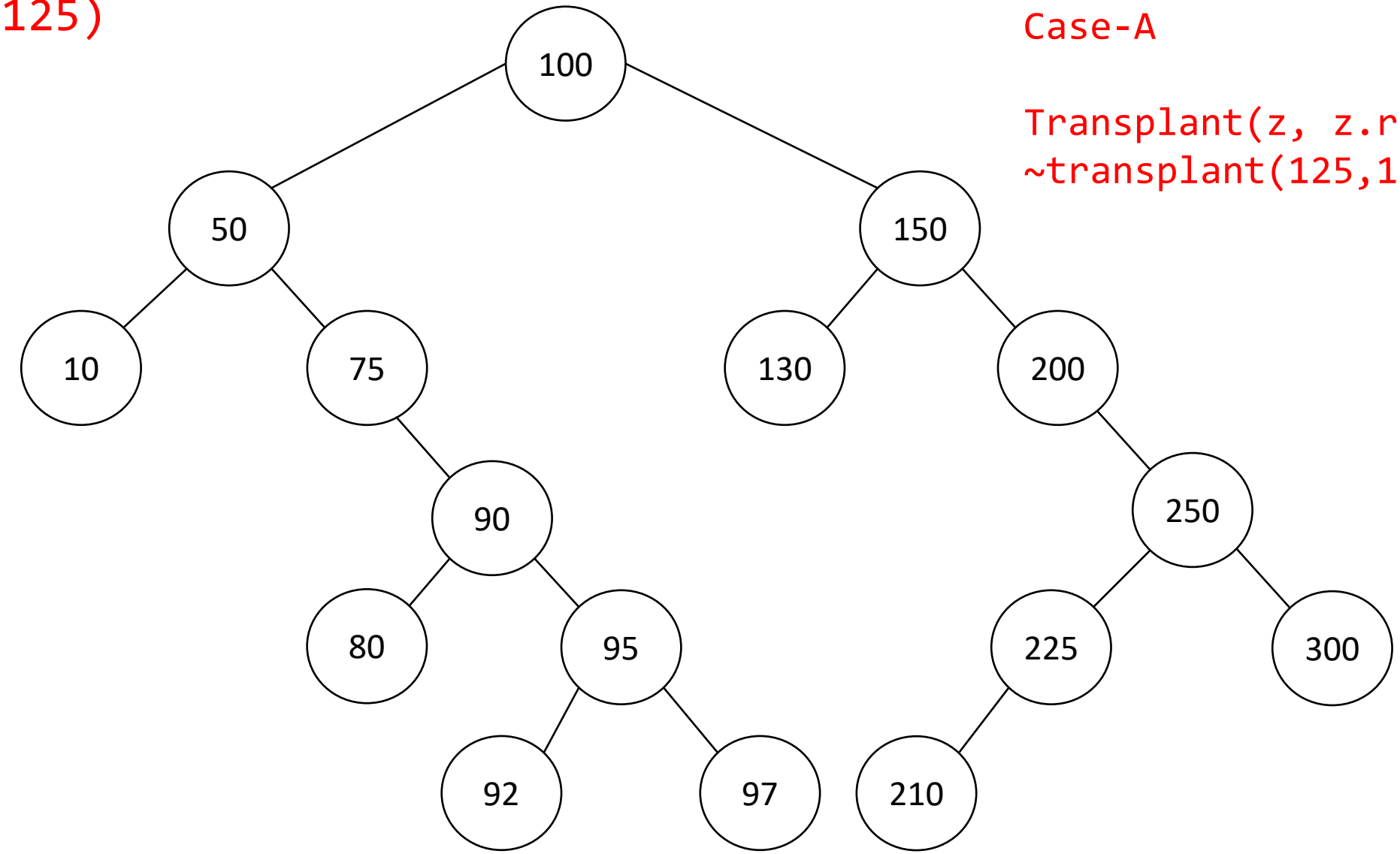
# Delete (125)




z=125, z.left=Null
Case-A

Transplant(z, z.right)
~transplant(125,130)

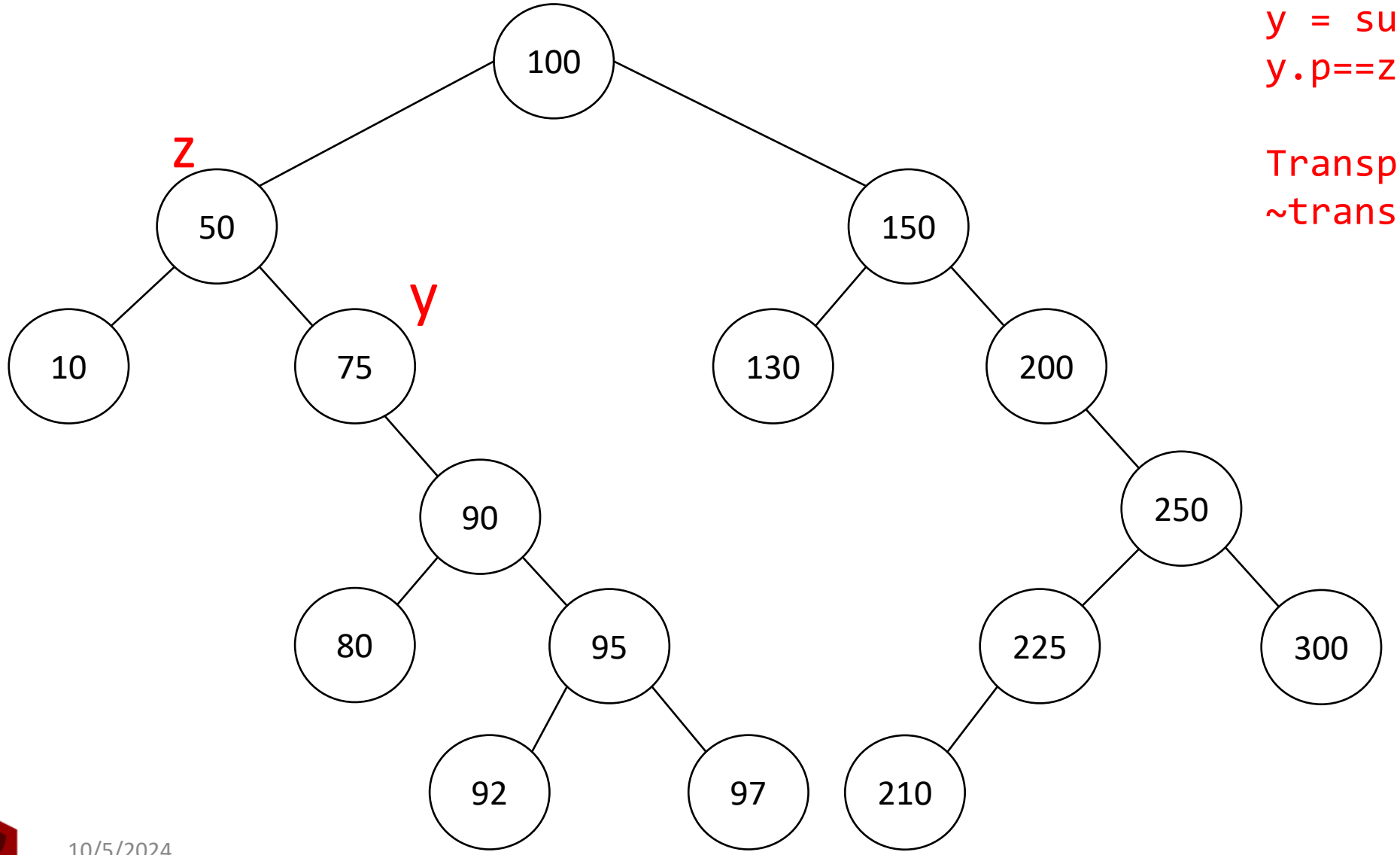
# Delete (125)




z=125, z.left=Null
Case-A

Transplant(z, z.right)
~transplant(125,130)

# Delete (50)

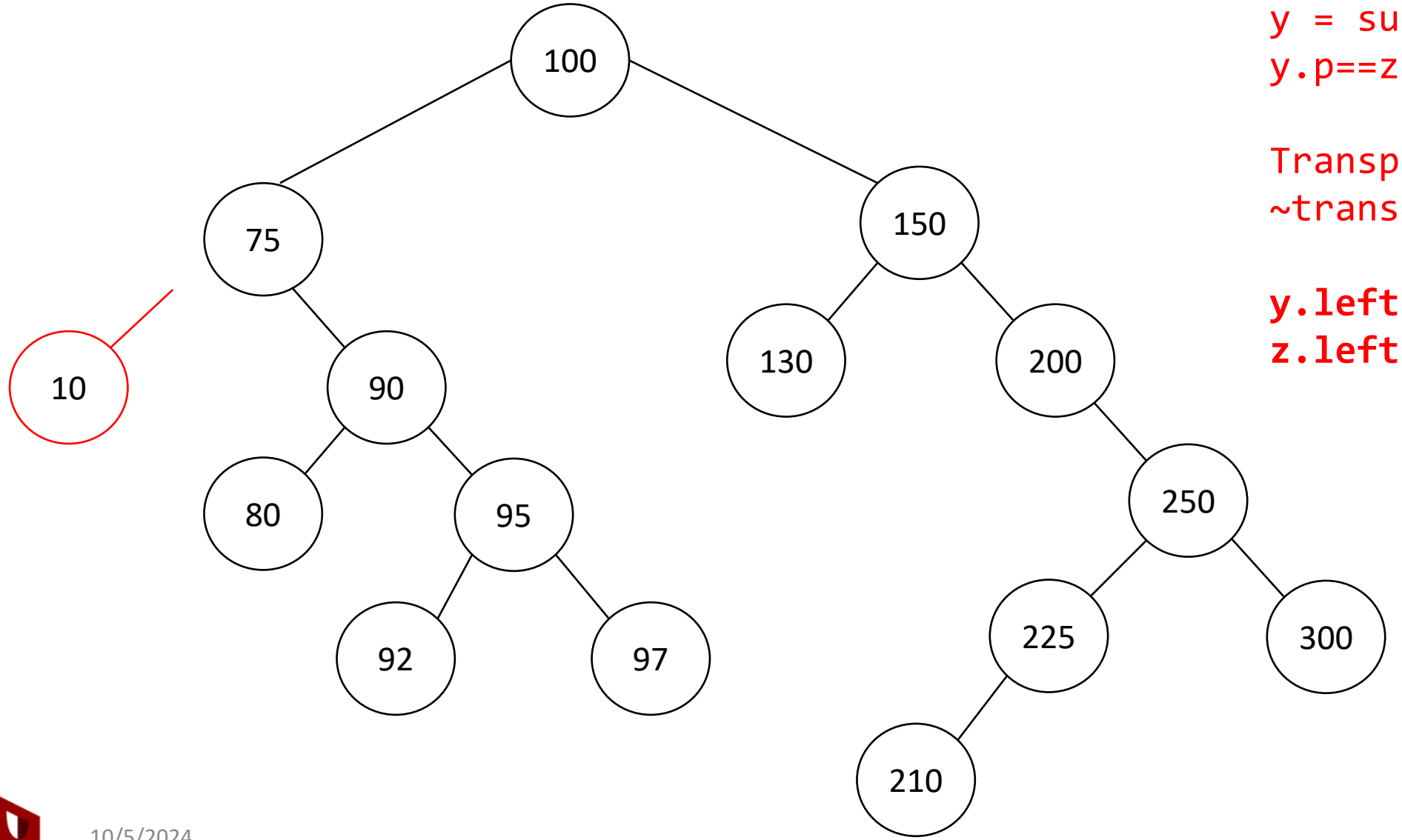



z=50, z.left!=Null,
z.right!=Null

y = successor(50) = 75
y.p==z, case-C

Transplant(z, y)
~transplant(50,75)

# Delete (50)



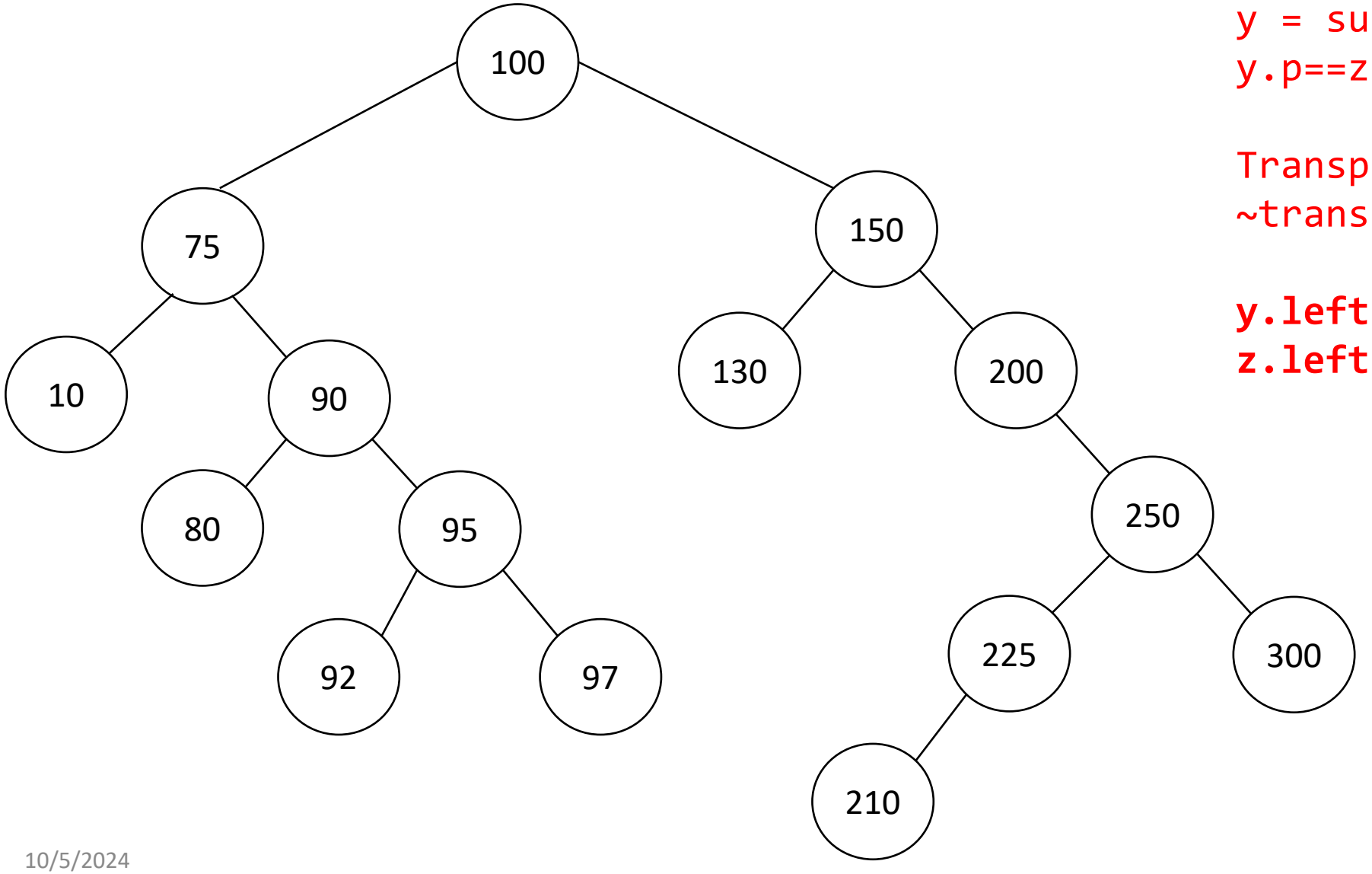

z=50, z.left!=Null,
z.right!=Null

y = successor(50) = 75
y.p==z, case-C

Transplant(z, y)
~transplant(50,75)

**y.left = z.left**
**z.left.p = y**

# Delete (50)



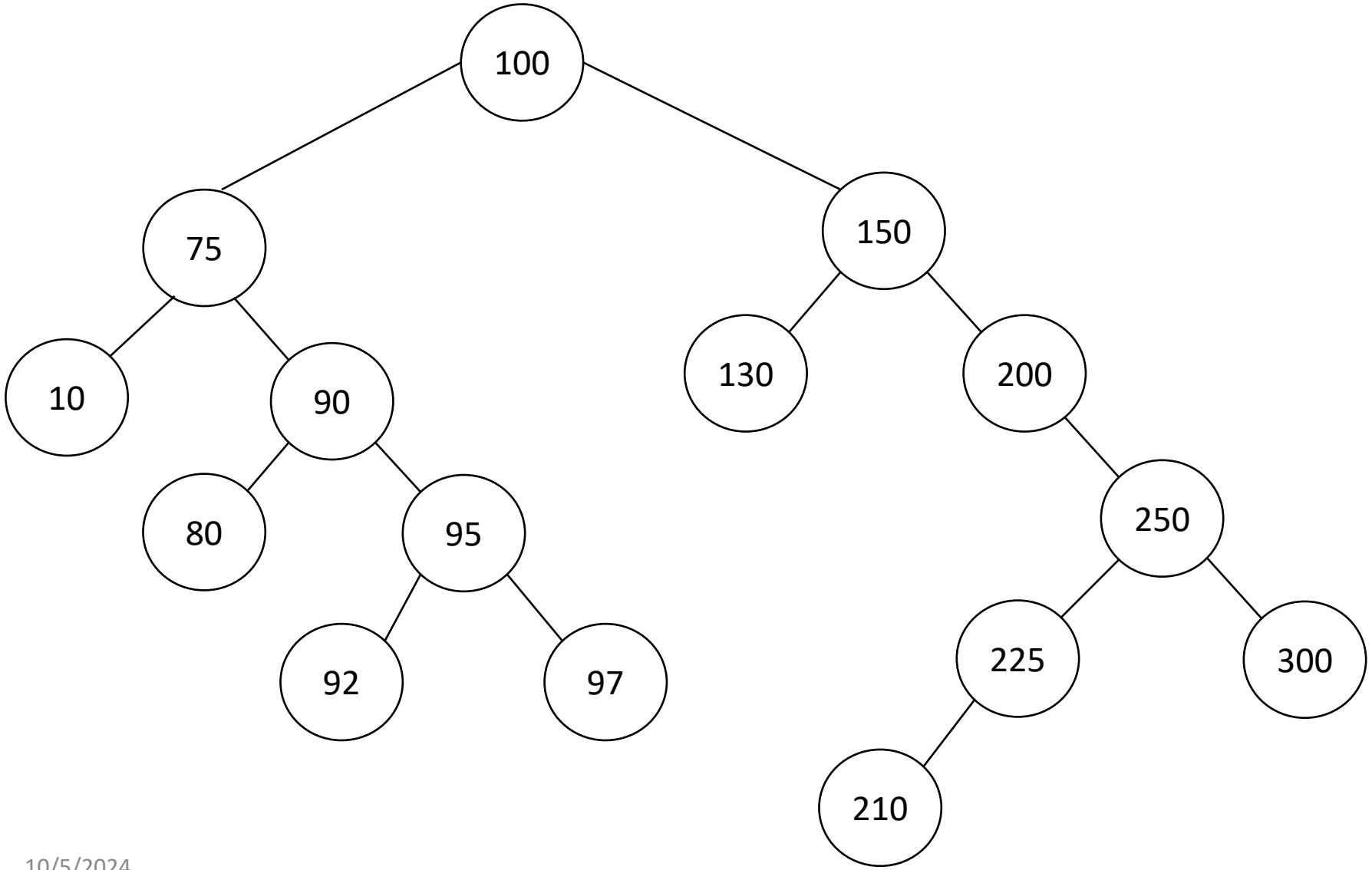

z=50, z.left!=Null,
z.right!=Null

y = successor(50) = 75
y.p==z, case-C

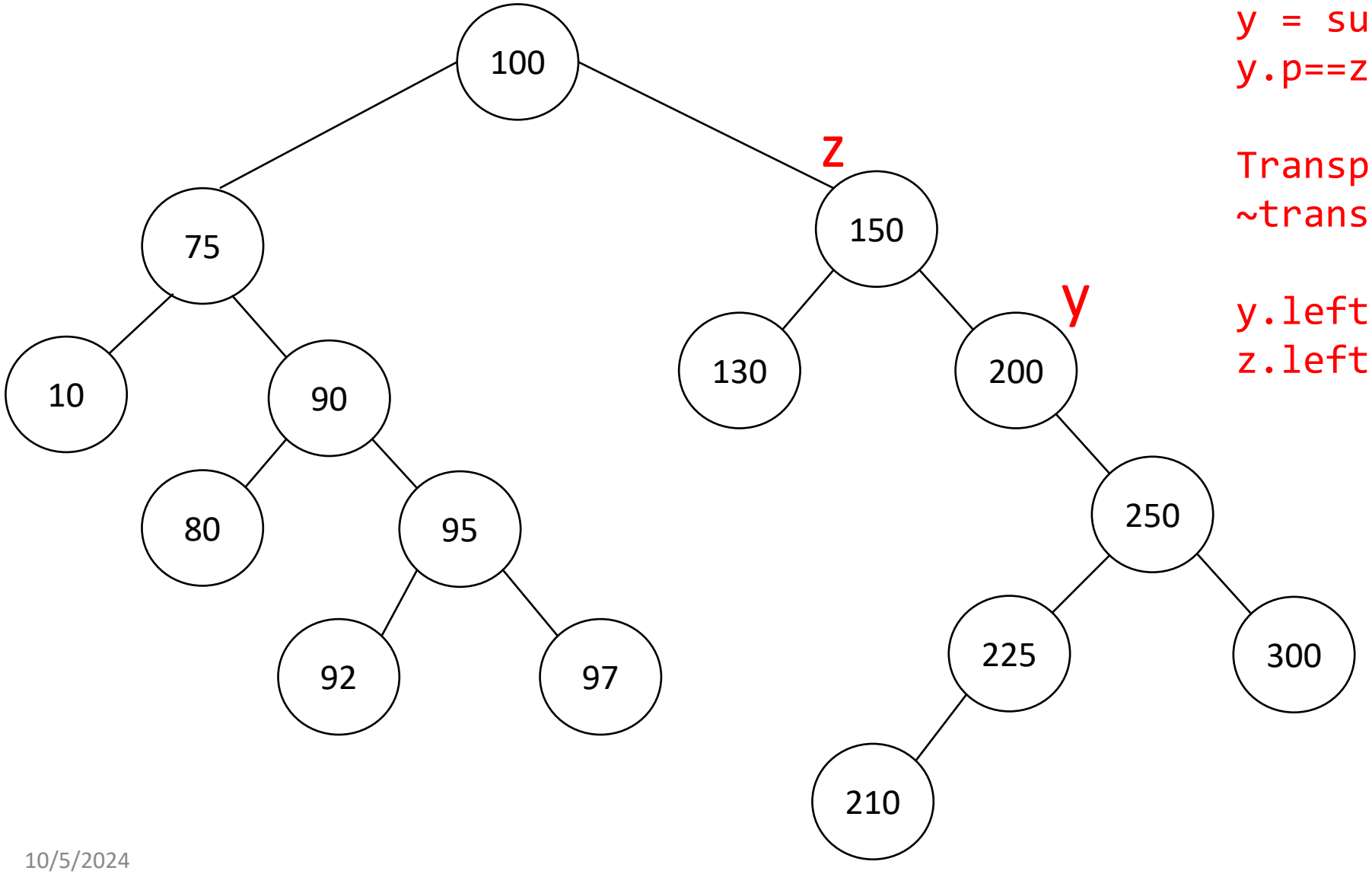
Transplant(z, y)
~transplant(50,75)

**y.left = z.left**
**z.left.p = y**

# Delete (150)



# Delete (150)



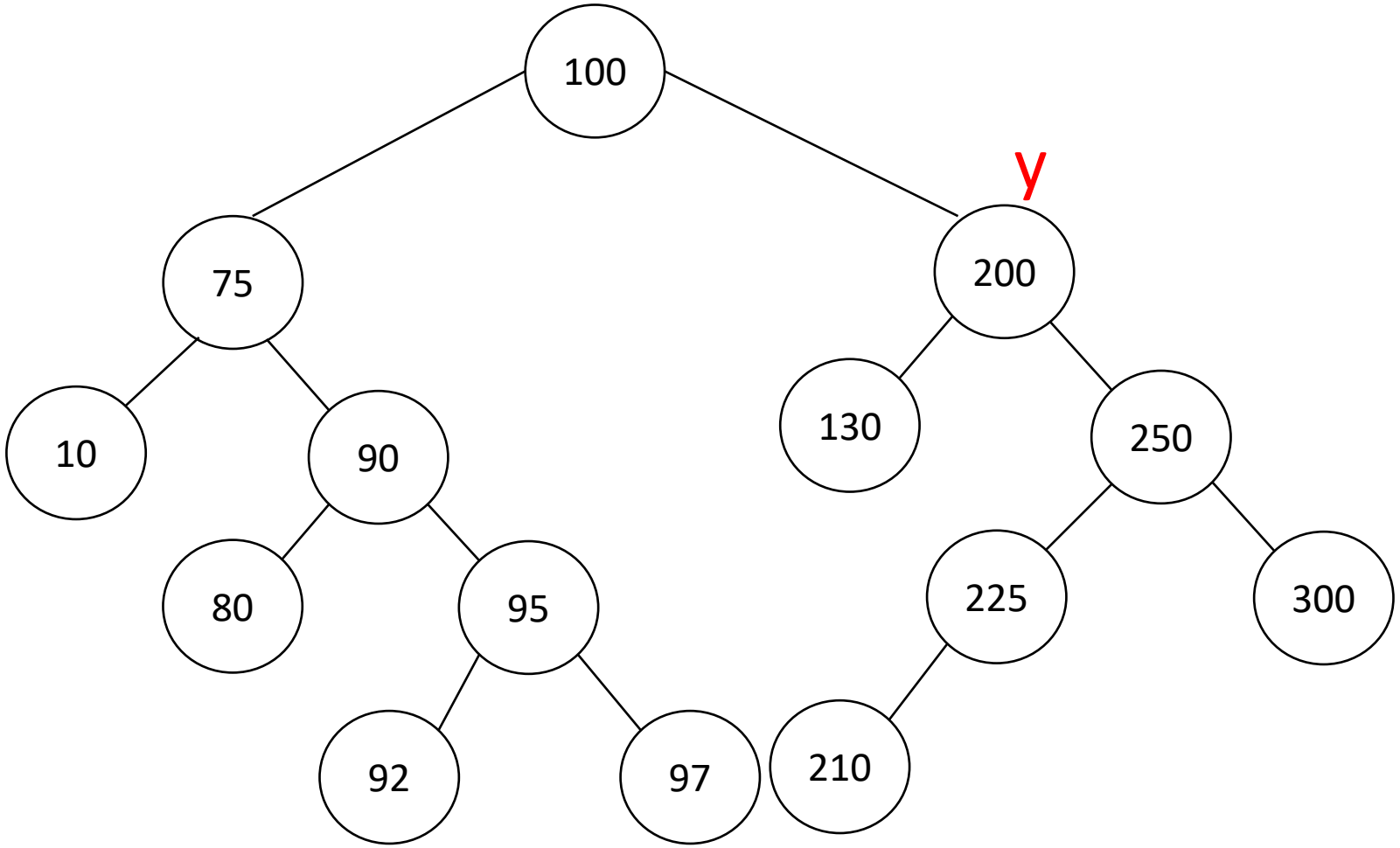

z=150, z.left!=Null,
z.right!=Null

y = successor(150)=200
y.p==z, case-C

Transplant(z, y)
~transplant(150,200)

y.left = z.left
z.left.p = y

# Delete (150)



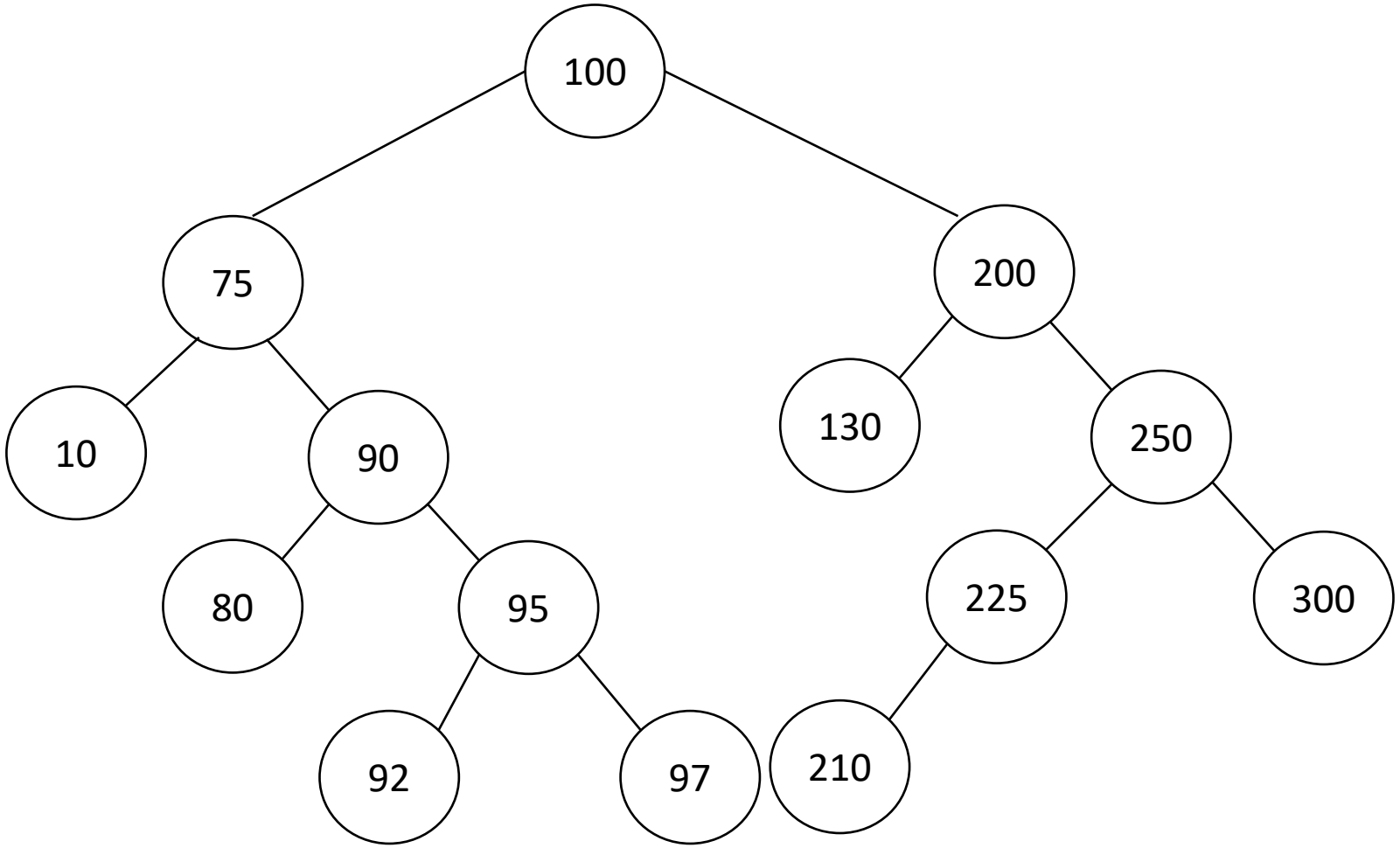

z=150, z.left!=Null,
z.right!=Null

y = successor(150)=200
y.p==z, case-C

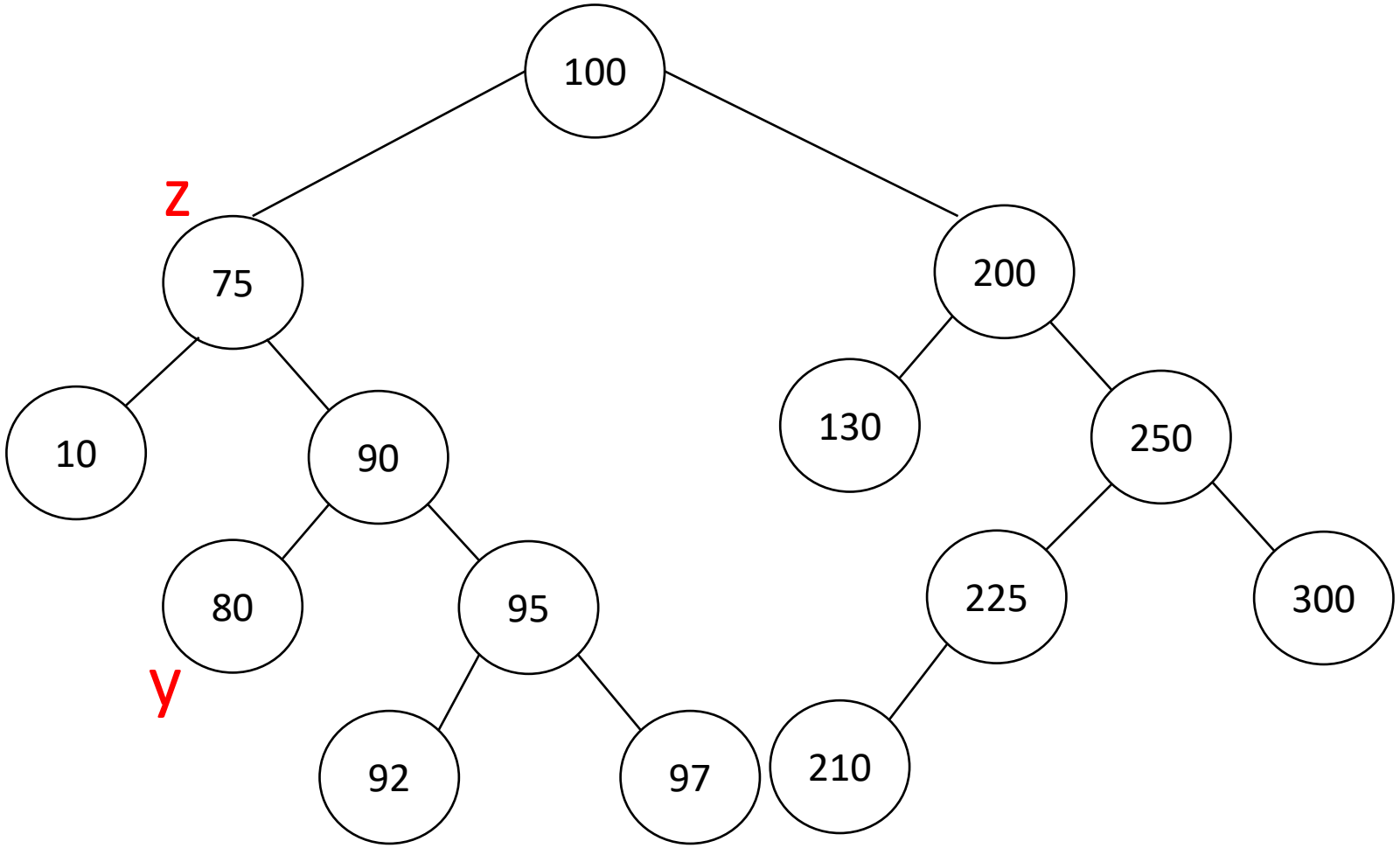
Transplant(z, y)
~transplant(150,200)

y.left = z.left
z.left.p = y

# Delete(75)



Delete(75)

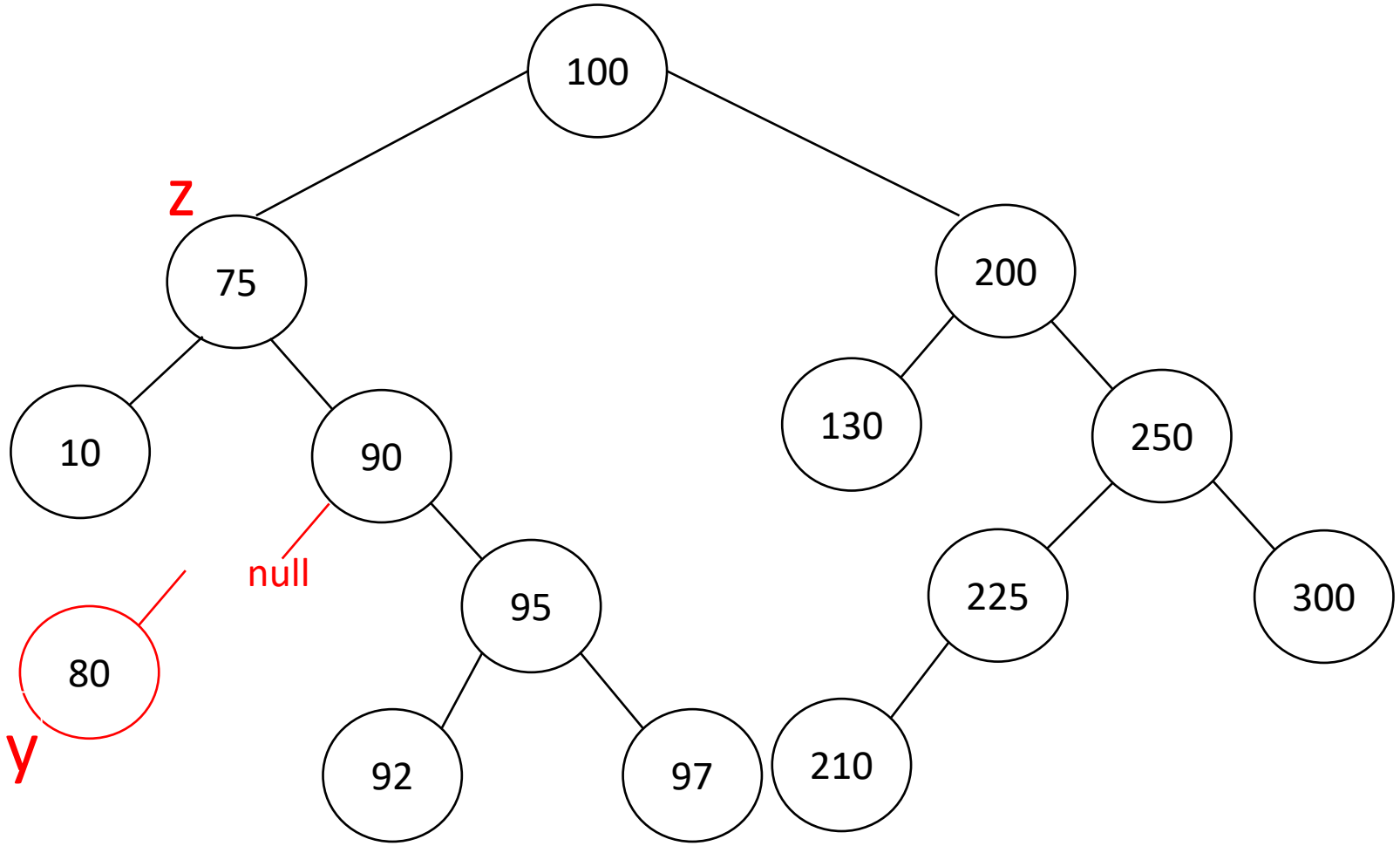



z=75, z.left!=Null,
z.right!=Null

y = successor(75)=80
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(80, Null)

# Delete(75)

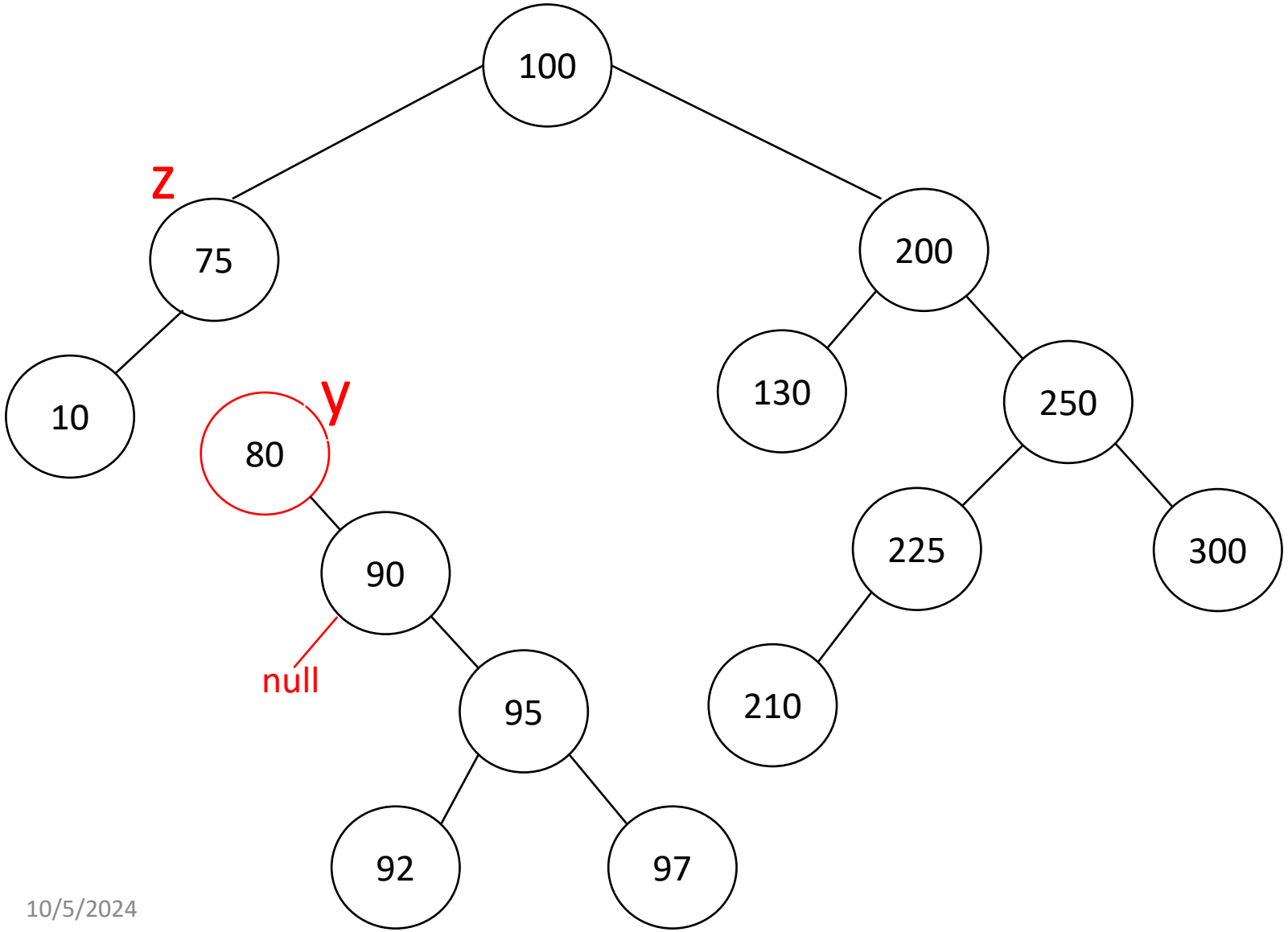



z=75, z.left!=Null,
z.right!=Null

y = successor(75)=80
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(80, Null)

# Delete(75)



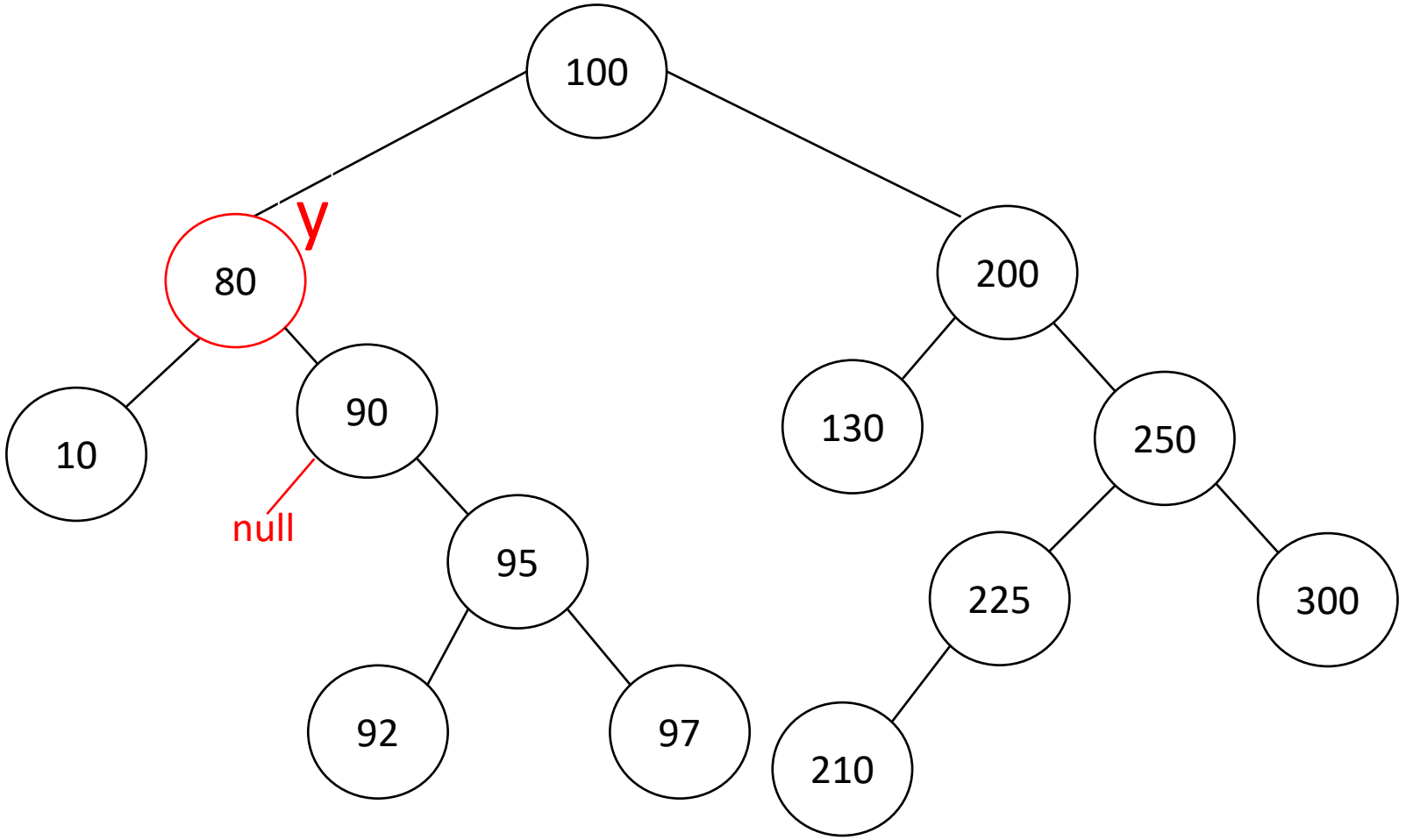

z=75, z.left!=Null,
z.right!=Null

y = successor(75)=80
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(80, Null)

y.right = z.right
z.right.p = y

# Delete(75)




z=75, z.left!=Null,
z.right!=Null

y = successor(75)=80
**y.p!=z, case-D**

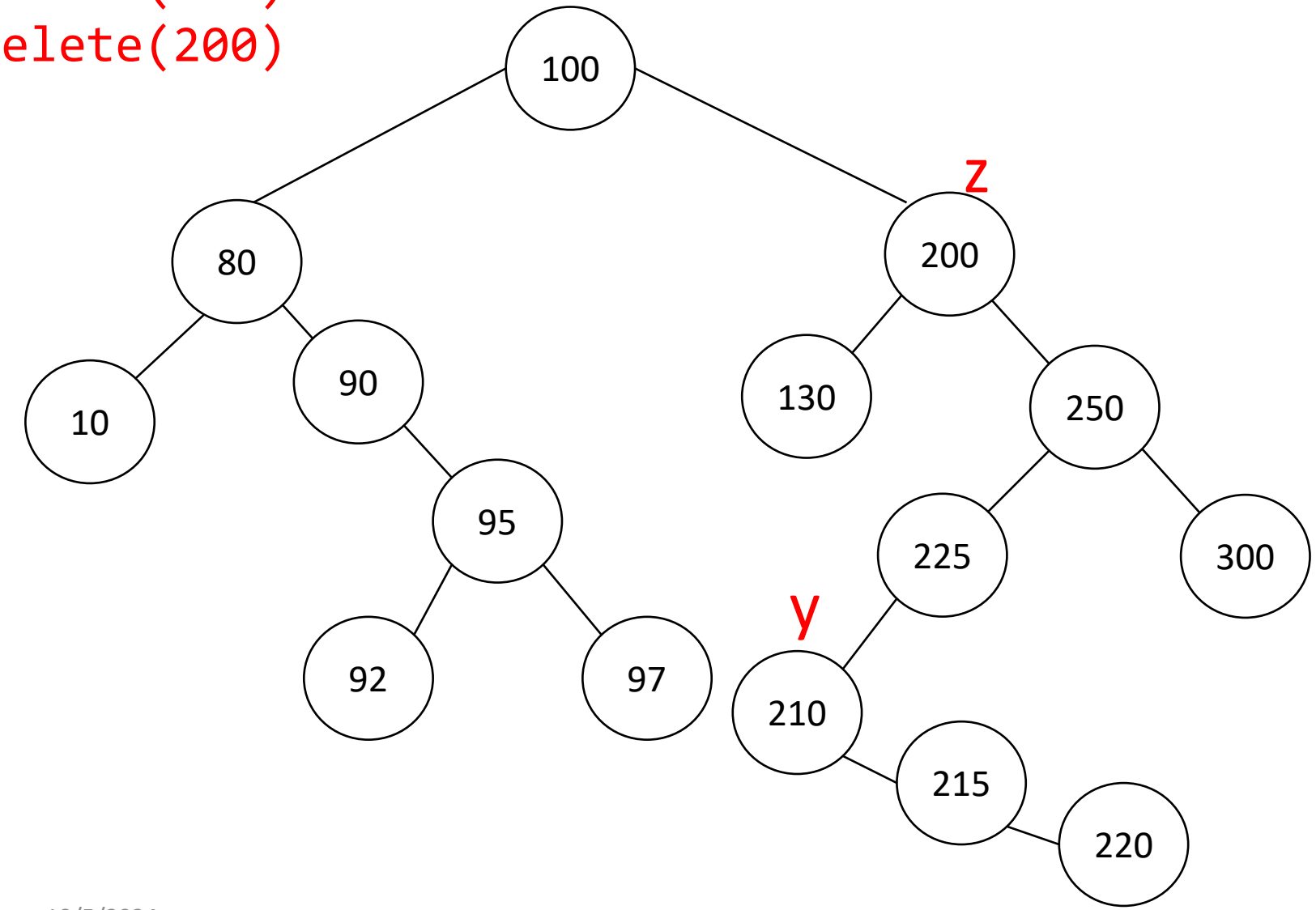
Transplant(y, y.right)
~Transplant(80, Null)

y.right = z.right
z.right.p = y

Transplant(z,y)
~transplant(75,80)

y.left = z.left
z.left.p=y

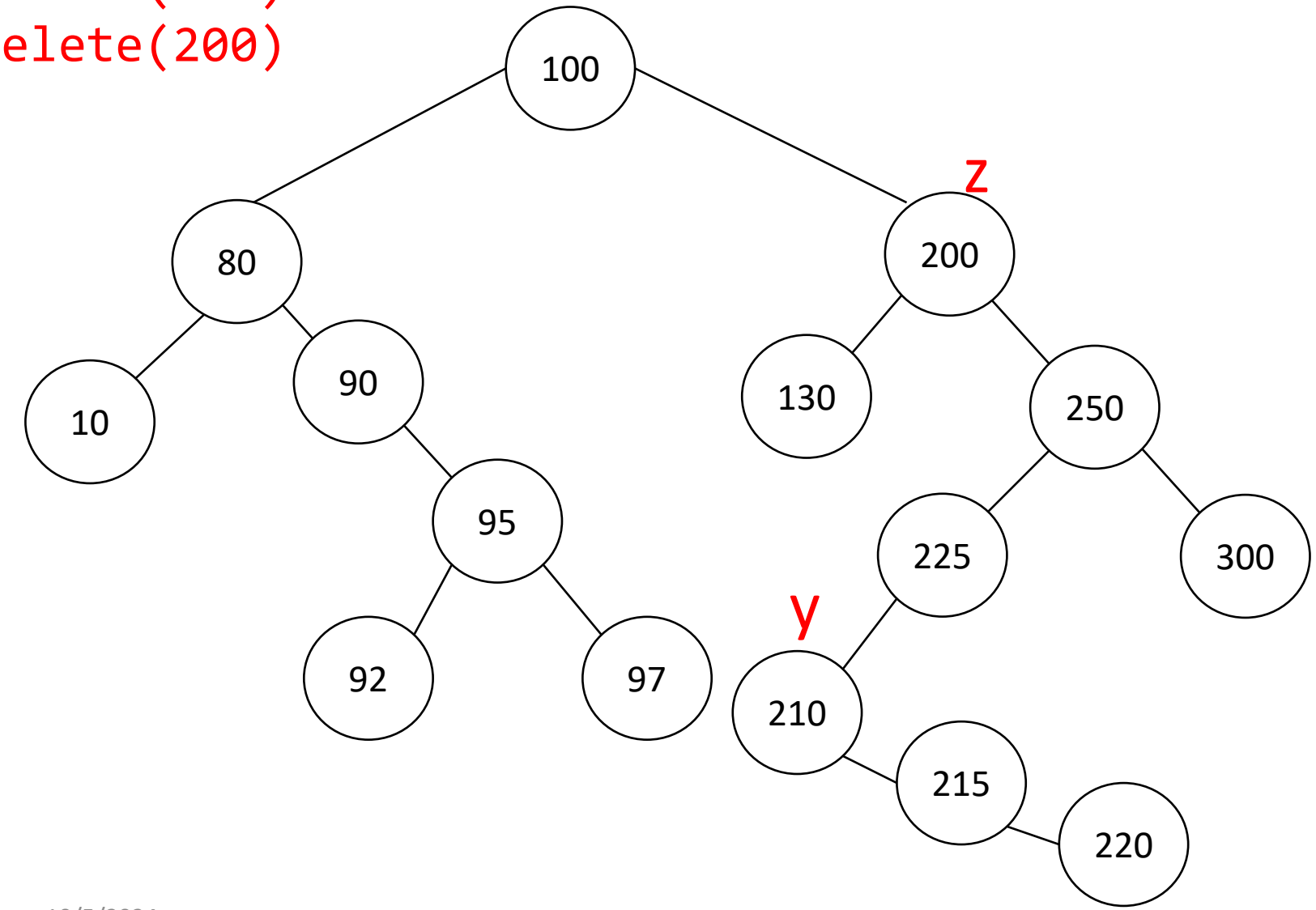
Insert(215)
Insert(220)
Delete(200)




z=200, z.left!=Null,
z.right!=Null

y = successor(200)=210
**y.p!=z, case-D**

Insert(215)
Insert(220)
Delete(200)

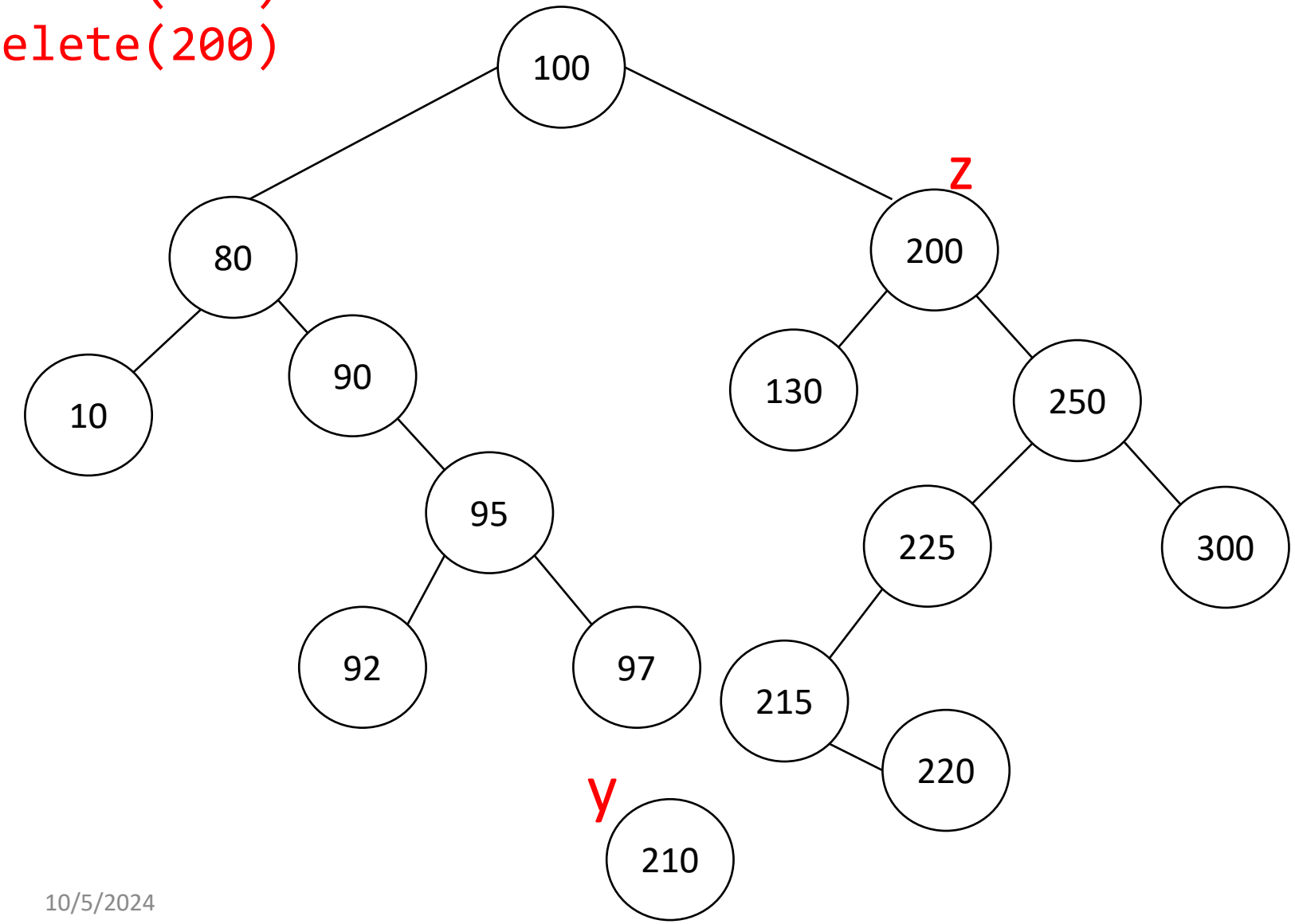



z=200, z.left!=Null, z.right!=Null

y = successor(200)=210
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(210, 215)

Insert(215)
Insert(220)
Delete(200)

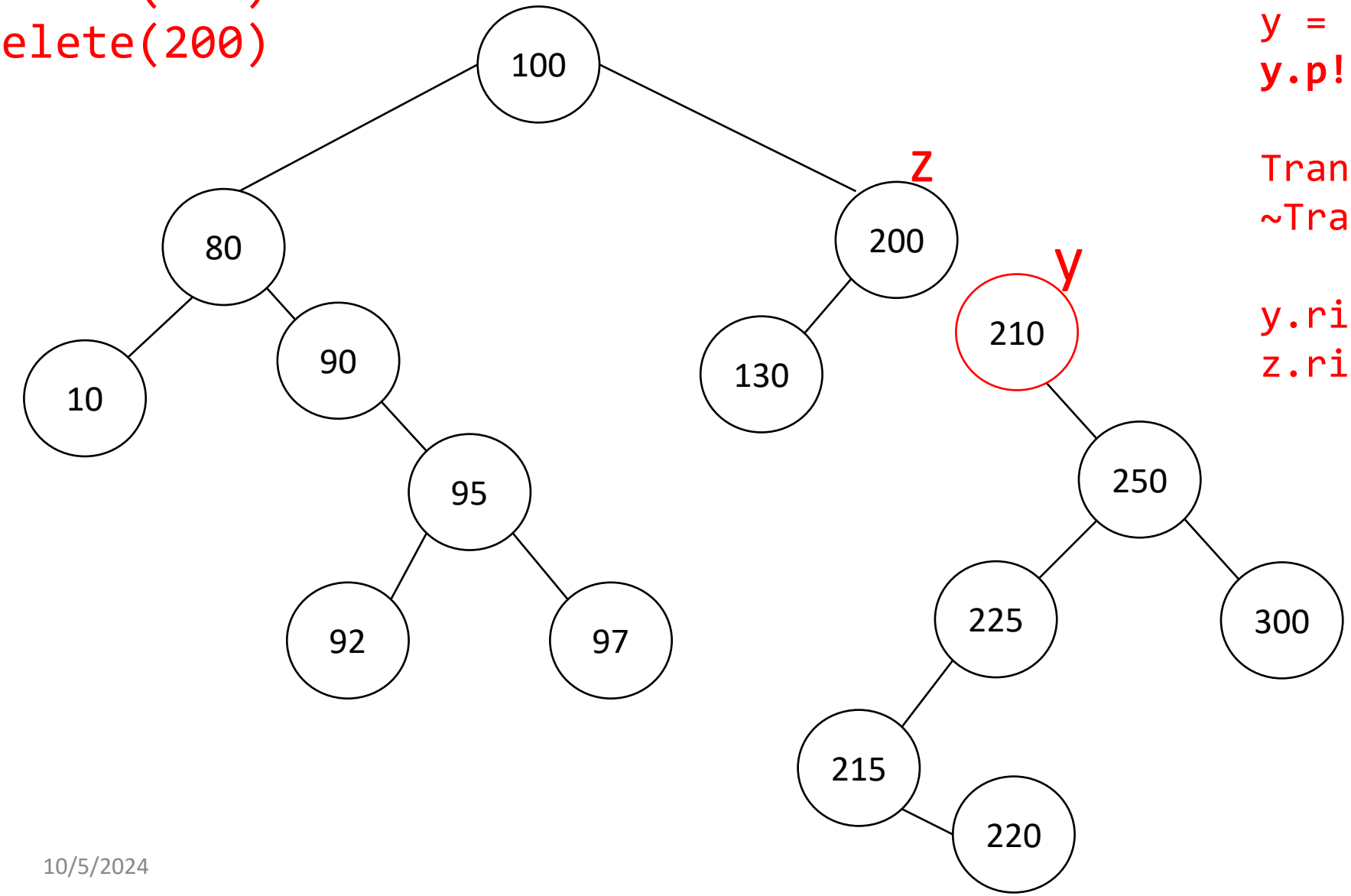



z=200, z.left!=Null,
z.right!=Null

y = successor(200)=210
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(210, 215)

Insert(215)
Insert(220)
Delete(200)



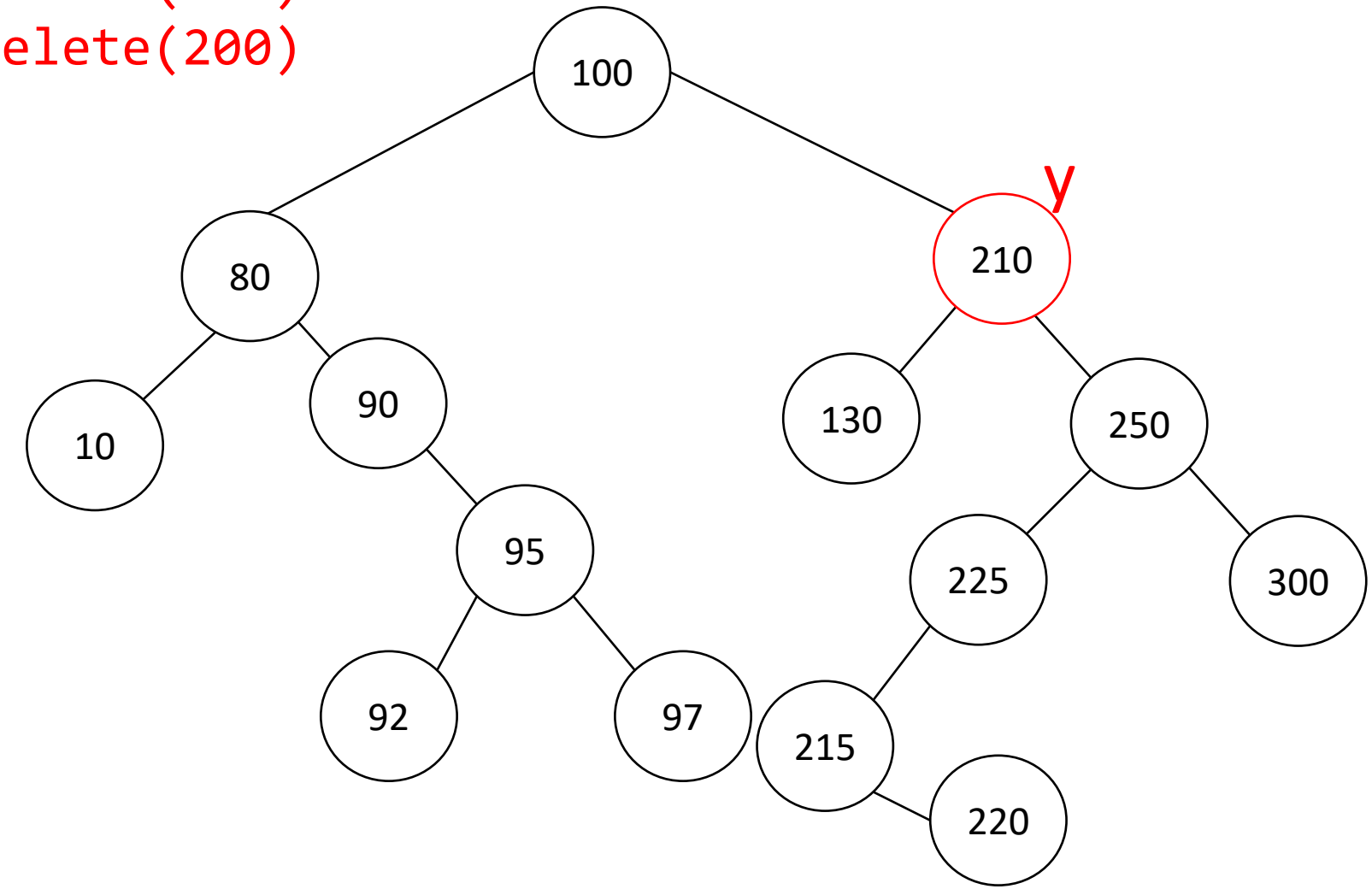

z=200, z.left!=Null,
z.right!=Null

y = successor(200)=210
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(210, 215)

y.right = z.right
z.right.p = y

Insert(215)
Insert(220)
Delete(200)




z=200, z.left!=Null, z.right!=Null

y = successor(200)=210
**y.p!=z, case-D**

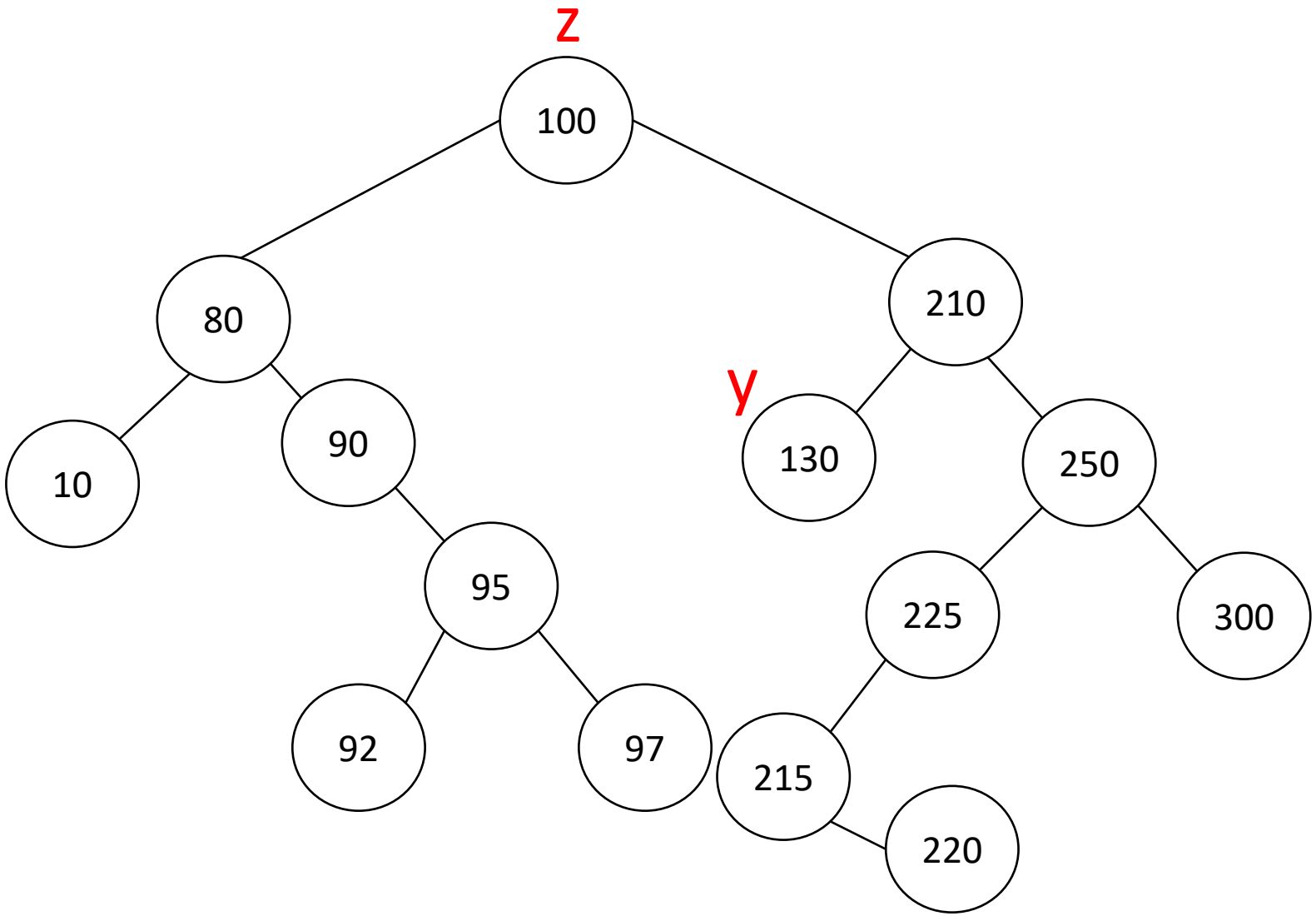
Transplant(y, y.right)
~Transplant(210, 215)

y.right = z.right
z.right.p = y

Transplant(z,y)
~transplant(200,210)

y.left = z.left
z.left.p=y

Delete(200)

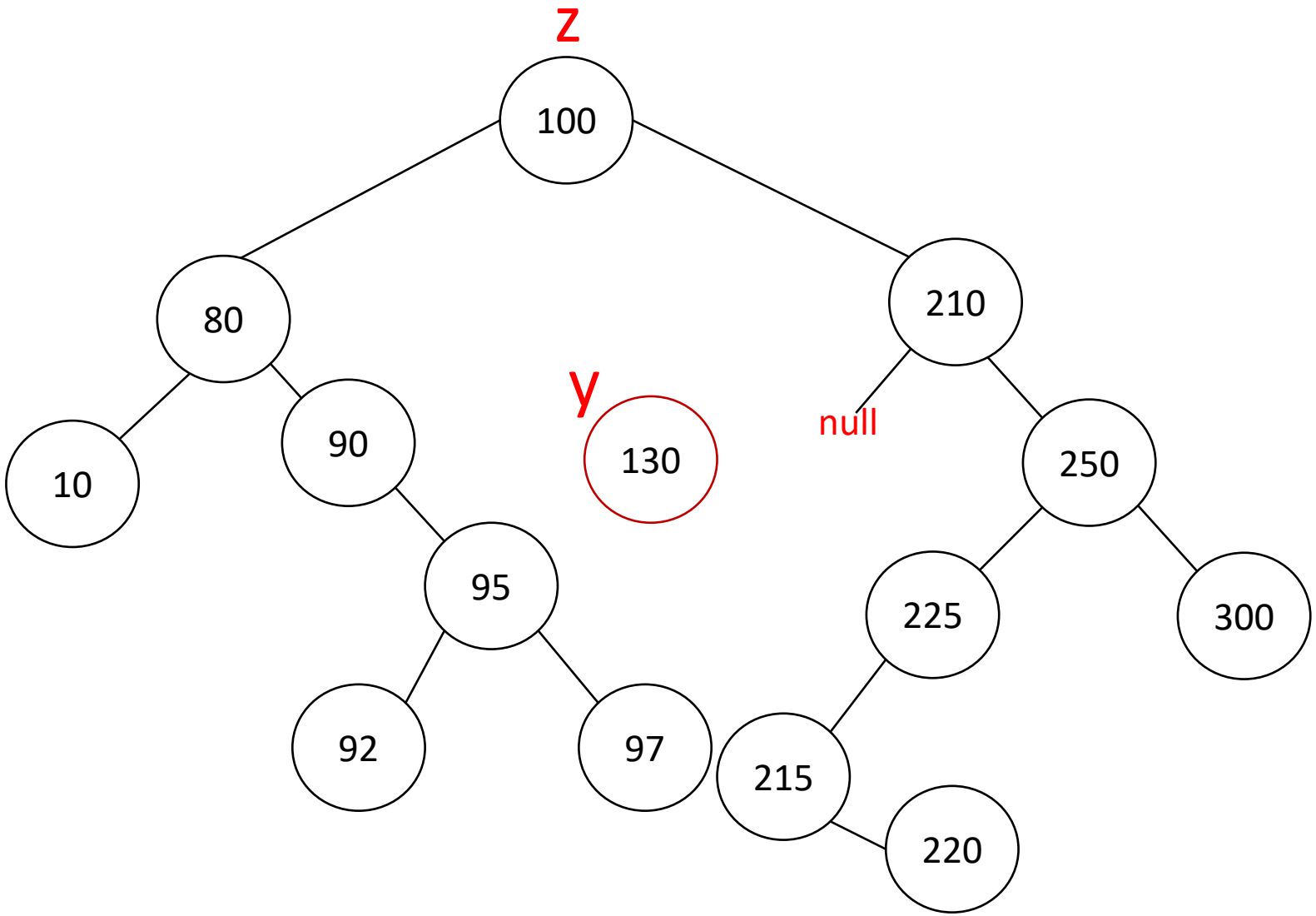



z=100, z.left!=Null, z.right!=Null

y = successor(100)=130

**y.p!=z, case-D**

Delete(200)

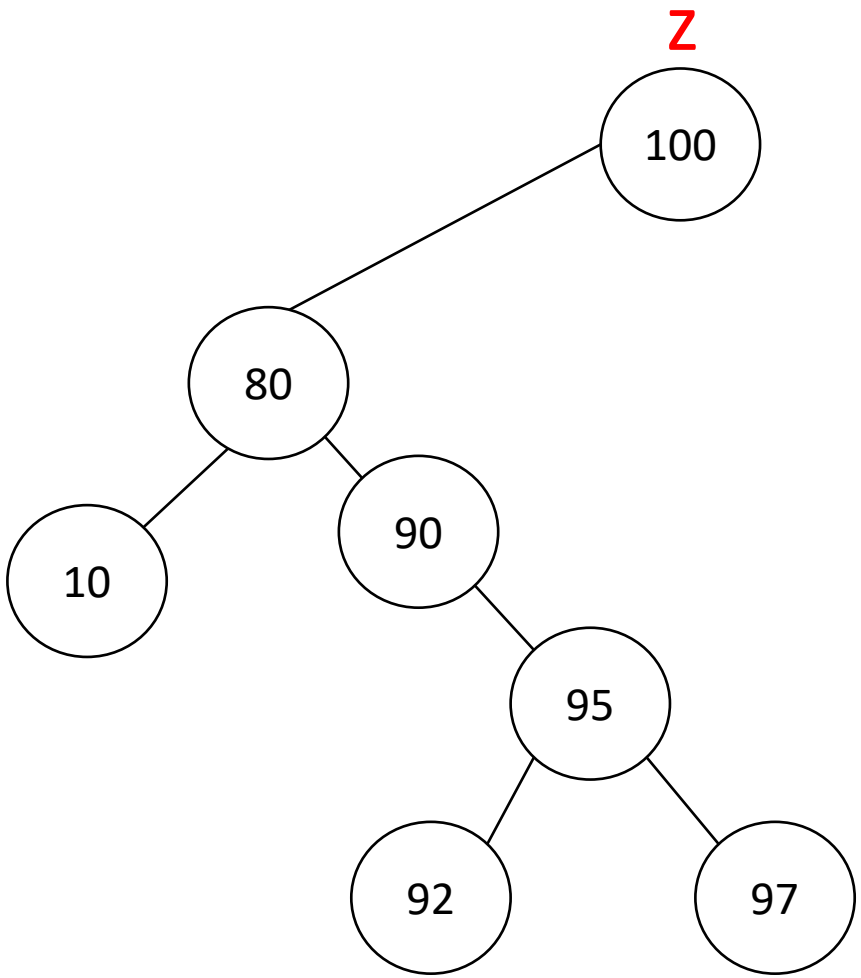


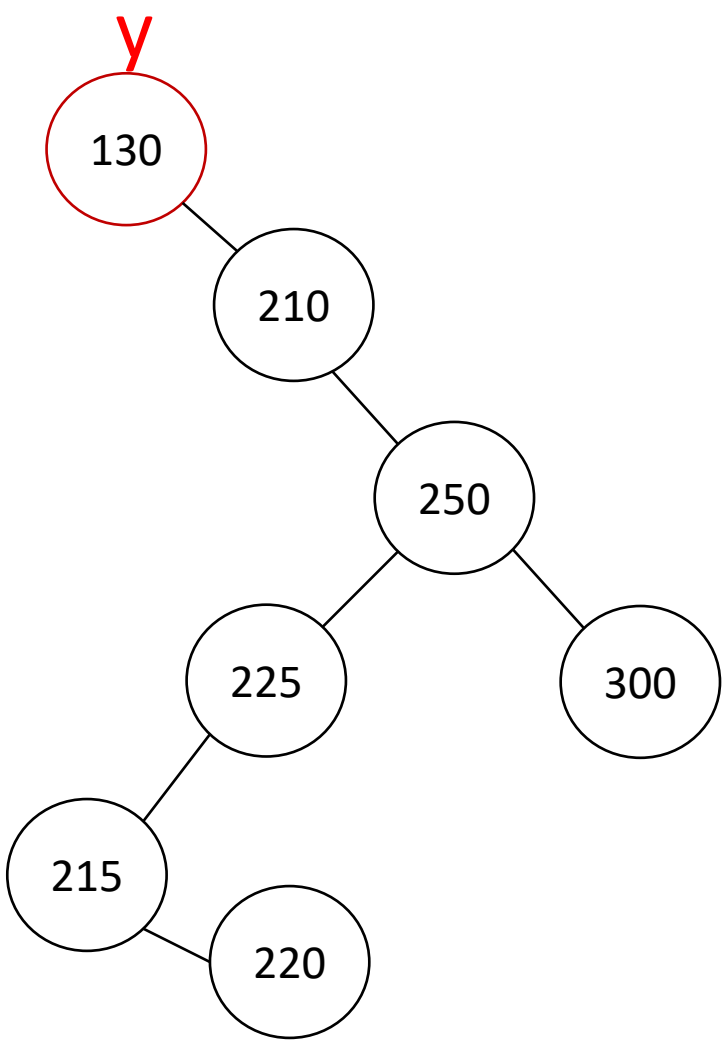

z=100, z.left!=Null,
z.right!=Null

y = successor(100)=130
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(130, Null)

Delete(200)



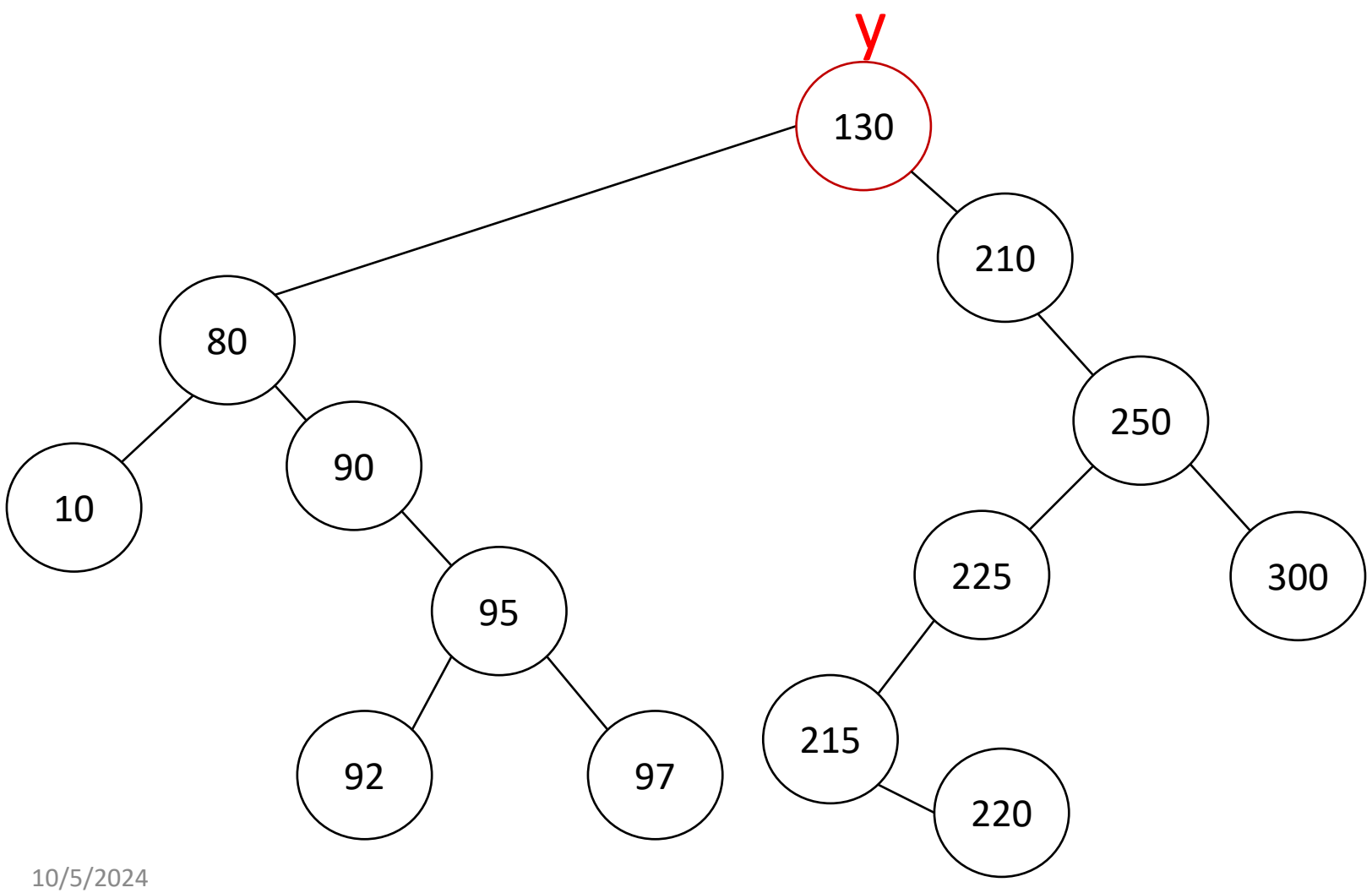

z=100, z.left!=Null,
z.right!=Null

y = successor(100)=130
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(130, Null)

y.right = z.right
z.right.p = y

# Delete(200)




z=100, z.left!=Null,
z.right!=Null

y = successor(100)=130
**y.p!=z, case-D**

Transplant(y, y.right)
~Transplant(130, Null)

y.right = z.right
z.right.p = y

Transplant(z,y)
~transplant(200,210)

y.left = z.left
z.left.p=y

# References

1. Chapter 12, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. 2009. Introduction to Algorithms, Third Edition (3rd. ed.). The MIT Press.

# Thank you